Archives of Computational Methods in Engineering
https://doi.org/10.1007/s11831-026-10716-2

REVIEW

Integrating Quantum Computing into Intelligent Transportation Systems: A Comprehensive Review of Emerging Trends, Open Challenges, and Future Research Directions

Pretom Sarkar[1] · Mahbub Hassan[2] · Ali Hanafiah Rambe[3] · Muzammil Jusoh[4]

Received: 14 April 2026 / Revised: 16 June 2026 / Accepted: 28 June 2026
© The Author(s) under exclusive licence to International Center for Numerical Methods in Engineering (CIMNE) 2026

Abstract

Quantum computing is increasingly proposed as a pathway for addressing computationally demanding problems in intelligent transportation systems, yet the field still lacks a rigorous, evidence-grounded assessment of its scientific progress, methodological maturity, and translational readiness. This paper presents an integrated bibliometric and qualitative systematic review of 108 Scopus-indexed peer-reviewed publications from 2014 to 2025, covering applications in traffic optimization, vehicle routing, demand forecasting, vehicular communication security, and autonomous vehicle intelligence. The study combines bibliometric performance analysis and science mapping with a structured qualitative synthesis of 27 representative studies evaluated through a four-dimension framework covering baseline fairness, problem scale realism, reproducibility, and deployment practicality. The bibliometric results show rapidly accelerating publication activity, geographically concentrated output led by the United States, China, India, and the United Kingdom, pronounced institutional and venue fragmentation, and very limited direct participation by transportation agencies and infrastructure operators. Science mapping identifies four dominant thematic clusters: quantum optimization, quantum machine learning, post-quantum cryptography, and quantum communication. The qualitative synthesis shows that the field is expanding across promising application domains, but empirical maturity remains limited. Existing studies do not yet demonstrate rigorous quantum computational advantage under fair classical benchmarking, reproducible experimental conditions, and operationally realistic problem scales, while quantum-inspired classical metaheuristics may inflate perceptions of field maturity when not separated from genuine quantum computing implementations. Post-quantum cryptography emerges as the most deployment-ready quantum-related domain because it is a classical quantum-resistant security pathway, occupying Technology Readiness Levels 6–8. Quantum annealing remains less mature at Technology Readiness Levels 3–4, gate-based QAOA and quantum machine learning remain largely exploratory at Technology Readiness Levels 2–3, and quantum communication occupies Technology Readiness Levels 4–6 for fixed or controlled links. To the best of the authors' knowledge, this review provides the first comprehensive bibliometric mapping of the quantum-ITS intersection, an original multidimensional critical evaluation framework, and a prioritized roadmap organized across immediate, strategic, and long-term horizons. The findings offer evidence-based guidance for researchers, transportation agencies, and funding bodies seeking to distinguish near-term security priorities from longer-term quantum computing opportunities in intelligent transportation systems.

✉ Ali Hanafiah Rambe
ali3@usu.ac.id

Pretom Sarkar
pretom.ce.20220103020@aust.edu

Mahbub Hassan
mahbub.hassan@ieee.org

Muzammil Jusoh
muzammil@unimap.edu.my

[1] Department of Civil Engineering, Ahsanullah University of Science and Technology (AUST), Dhaka 1208, Bangladesh

[2] Department of Civil Engineering, Faculty of Engineering, Chulalongkorn University, Pathumwan, Bangkok 10330, Thailand

[3] Department of Electrical Engineering, Universitas Sumatera Utara, Medan 20155, Indonesia

[4] Centre of Excellence for Advanced Computing (AdvComp), Faculty of Electronic Engineering Technology, Universiti Malaysia Perlis, Perlis, Malaysia

Published online: 10 July 2026

1 Introduction

Modern intelligent transportation systems (ITS) have evolved into large-scale cyber-physical mobility networks in which connected vehicles, adaptive roadside infrastructure, and traffic management platforms continuously exchange and process high-velocity data at metropolitan scale [1–3]. This integration of advanced computation, sensing, and vehicle-to-everything (V2X) communication enables coordinated autonomous operation, real-time route optimization, adaptive signal control, and predictive demand management [4–6]. It also concentrates computational and security challenges whose difficulty increases with network scale, vehicle density, and operational complexity.

Three challenge domains motivate interest in post-classical computing approaches for ITS. First, traffic signal coordination and network-level flow control involve combinatorial optimization problems whose feasible solution spaces grow rapidly with the number of controlled intersections. Tailored classical methods, including max-pressure control and deep reinforcement learning, achieve strong empirical performance by exploiting problem structure, but they do not eliminate the possibility that post-classical paradigms may be useful for selected hard subclasses [7–9]. Second, vehicle routing with time windows, stochastic demand, and real-time multimodal constraints remains computationally demanding, even though modern vehicle-routing reviews and exact branch-price-and-cut methods show that advanced classical solvers handle many structured benchmark variants effectively [10, 11]. Third, V2X security architectures relying on RSA and elliptic-curve cryptography are vulnerable in principle to Shor's polynomial-time algorithms for integer factorization and discrete logarithms [12]. The harvest-now-decrypt-later threat, where encrypted communications collected today may be decrypted once fault-tolerant quantum computers become available, makes cryptographic migration a present planning issue rather than a distant theoretical concern [13]. Beyond optimization and security, traffic prediction and autonomous-vehicle perception also impose increasing training and inference demands as models learn complex spatiotemporal dependencies from network-wide sensor data [14–16].

Quantum computing, which exploits superposition, entanglement, and quantum interference to process information in ways fundamentally different from classical binary architectures, has therefore attracted attention as a candidate framework for selected ITS problems [17, 18]. The Quantum Approximate Optimization Algorithm (QAOA) [19, 20] and quantum annealing [21, 22] are structurally compatible with combinatorial traffic coordination and routing formulations, while quantum machine learning offers alternative feature-space representations and kernel methods for transportation intelligence applications [23]. On the security side, post-quantum cryptographic (PQC) schemes are classical algorithms designed to remain secure against attackers equipped with large-scale quantum computers. Peer-reviewed PQC and migration studies show that quantum-resistant public-key alternatives are maturing for conventional computing environments, including resource-constrained IoT systems; therefore, PQC offers a near-term security migration pathway that does not require quantum infrastructure [24–26]. This distinction is essential: PQC is a classical response to the quantum threat, whereas noisy intermediate-scale quantum (NISQ) devices and future fault-tolerant quantum platforms represent different technological pathways with substantially lower near-term deployment readiness [18]. High-profile demonstrations of quantum computational advantage on specialized tasks [27] have intensified interdisciplinary interest, but the distance between theoretical promise, laboratory-scale demonstration, NISQ-era feasibility, and operational ITS deployment remains insufficiently clarified.

Despite rapid publication growth at the quantum-ITS intersection, the field still lacks an evidence-grounded characterization of its intellectual structure, methodological quality, and translational maturity. Existing domain surveys have discussed possible quantum computing applications in ITS [28], and bibliometric work has begun to map quantum computing in adjacent application domains [29]. However, an integrated bibliometric and critical-evidence synthesis of quantum-ITS research remains absent. Five gaps motivate the present review. First, rapid quantum hardware development makes it difficult to distinguish promising transportation applications from speculative claims that are not benchmarked against strong classical methods. Second, the field spans quantum physics, computer science, transportation engineering, and operations research, communities that often use different evaluation standards and few shared benchmarks. Third, the absence of standardized datasets, problem instances, and reporting protocols prevents meaningful cross-study comparison. Fourth, many studies labeled as quantum-ITS research use quantum-inspired classical metaheuristics without quantum hardware, entanglement, or interference, thereby conflating metaphorical algorithm design with genuine quantum computation. Fifth, the geographic, institutional, and funding structures shaping the field remain poorly characterized, and the limited participation of transportation agencies signals a disconnect between quantum algorithm development and operational validation. An integrated bibliometric and qualitative synthesis is therefore needed: bibliometric analysis can map publication trends, citation flows, and collaboration networks [30, 31], while structured qualitative evaluation can assess the

evidential quality of quantum advantage and deployment claims [32].

This study is guided by the following research objectives:

- **RO1:** To map the temporal evolution, geographic distribution, institutional concentration, and collaboration structure of quantum computing research in ITS through systematic bibliometric analysis of the 2014–2025 literature.
- **RO2:** To identify dominant thematic clusters, quantum computing paradigms, and ITS application domains through science mapping, and to characterize the field's intellectual organization and thematic trajectory.
- **RO3:** To critically evaluate the methodological rigor, evidential basis, and practical validity of quantum advantage claims across optimization, security, machine learning, and simulation applications in representative publications.
- **RO4:** To synthesize persistent technical, methodological, infrastructural, and institutional barriers impeding quantum-ITS deployment, and to propose a prioritized research roadmap grounded in demonstrated maturity and operational need.

These objectives are operationalized through the following research questions:

- **RQ1:** How has the quantum-ITS research landscape evolved in terms of publication output, citation impact, geographic concentration, and collaboration patterns, and what does this structural evolution reveal about the field's maturity and institutional dynamics?
- **RQ2:** What thematic clusters characterize the intellectual structure of quantum-ITS research, which quantum computing paradigms and ITS application domains dominate the literature, and how have research emphases shifted over the 2014–2025 period?
- **RQ3:** What quantum computing techniques and algorithms have been applied to ITS challenges, what evidence supports or refutes quantum computational advantage in these contexts, and what methodological weaknesses limit the validity of reported performance claims?
- **RQ4:** What technical, methodological, and institutional challenges currently impede the translation of quantum-ITS research into operational deployment, and what research directions offer the most productive pathways toward bridging this gap?

To address these questions, this study integrates bibliometric analysis with structured qualitative synthesis in a single critical review framework. Five contributions distinguish the work. First, to the best of the authors' knowledge, it provides the first comprehensive bibliometric mapping of the quantum-ITS intersection, covering publication trends, citation patterns, geographic distribution, and co-authorship structure across 108 publications from 2014 to 2025. This mapping reveals pronounced institutional and venue fragmentation and the near-absence of transportation agencies and infrastructure operators as direct contributors. Second, it applies science mapping through keyword co-occurrence, bibliographic coupling, and strategic thematic analysis to identify dominant clusters, structural coverage gaps, and emerging-versus-mature themes. Third, it conducts a qualitative synthesis of 27 representative publications using a four-dimension evaluation framework covering baseline fairness, problem scale realism, reproducibility, and deployment practicality. This synthesis establishes the review's central analytical finding: within the examined evidence base, claims of quantum computational advantage remain weak under fair classical benchmarking, reproducible experimental conditions, and operationally realistic problem scales, while the labeling of quantum-inspired classical metaheuristics as quantum computing inflates apparent field maturity. Fourth, it provides a technology readiness assessment distinguishing post-quantum cryptography (TRL 6–8), quantum annealing optimization (TRL 3–4), gate-based QAOA and quantum machine learning (TRL 2–3), and quantum communication (TRL 4–6). Fifth, it proposes a prioritized three-tier research roadmap spanning 2025–2027, 2026–2032, and 2028–2040+, aligning stakeholder responsibilities, success metrics, and investment priorities with demonstrated maturity rather than speculative hardware trajectories.

The remainder of this paper is organized as follows. Section 2 establishes the technical foundations through a four-dimensional ITS-quantum taxonomy, an analysis of ITS computational demands, and a qualified assessment of quantum hardware and algorithmic capabilities. Section 3 describes the systematic search strategy, PRISMA screening procedure, data extraction protocol, and bibliometric analysis tools. Section 4 presents the bibliometric performance analysis, science mapping results, and qualitative synthesis. Section 5 synthesizes findings across the research questions, provides maturity assessments, and develops stakeholder recommendations. Section 6 delineates open technical and institutional challenges and presents the strategic research roadmap. Section 7 summarizes the principal findings, theoretical and practical implications, and future research directions.

2 Background

This section establishes the technical and conceptual foundations needed to interpret the bibliometric and qualitative findings presented in Sects. 4 and 5. It introduces a four-dimensional taxonomy of quantum computing applications in intelligent transportation systems, examines the transportation demands that motivate quantum investigation, summarizes the quantum principles and hardware architectures relevant to ITS applications, and clarifies the gap between theoretical alignment and current operational feasibility. This foundation is necessary for distinguishing productive research directions from speculative extrapolation.

2.1 Taxonomy of Quantum Computing Applications in ITS

The intersection of quantum computing and intelligent transportation systems spans heterogeneous research directions: combinatorial optimization for traffic management, quantum-enhanced machine learning for prediction and perception, quantum communication for V2X security, and post-quantum cryptographic protocols for connected vehicle authentication. These directions cannot be meaningfully compared, evaluated, or synthesized without a unifying classificatory structure. Figure 1 presents the four-dimensional taxonomic hierarchy employed throughout this review to organize the 108-paper corpus. The four dimensions are quantum computing paradigm, ITS application domain, quantum algorithm or technique, and implementation maturity level; each publication is classified along all four axes during the data-extraction phase described in Sect. 3.

Table 1 presents the formal taxonomy with descriptions and representative sub-categories for each dimension. Three features of this structure warrant emphasis before the subsequent subsections develop the conceptual foundations underlying each dimension.

The implementation maturity dimension is the most analytically consequential axis of the taxonomy because it determines whether quantum performance claims are hypothetical constructs, simulated approximations under idealized conditions, or empirically grounded results subject to realistic hardware constraints. The bibliometric and qualitative mapping in Sect. 4 indicates that much of the reviewed corpus remains at the two lower maturity levels, theoretical framework and classical simulation, a pattern with direct implications for interpreting quantum advantage claims and calibrating investment priorities.

A second feature deserving attention is the explicit category of quantum-inspired classical algorithms. This class does not constitute a quantum computing paradigm in any substantive sense: algorithms within it do not exploit superposition, entanglement, or quantum interference, and reported performance improvements derive entirely from metaheuristic design rather than quantum mechanics. Their systematic labeling as quantum-ITS research inflates

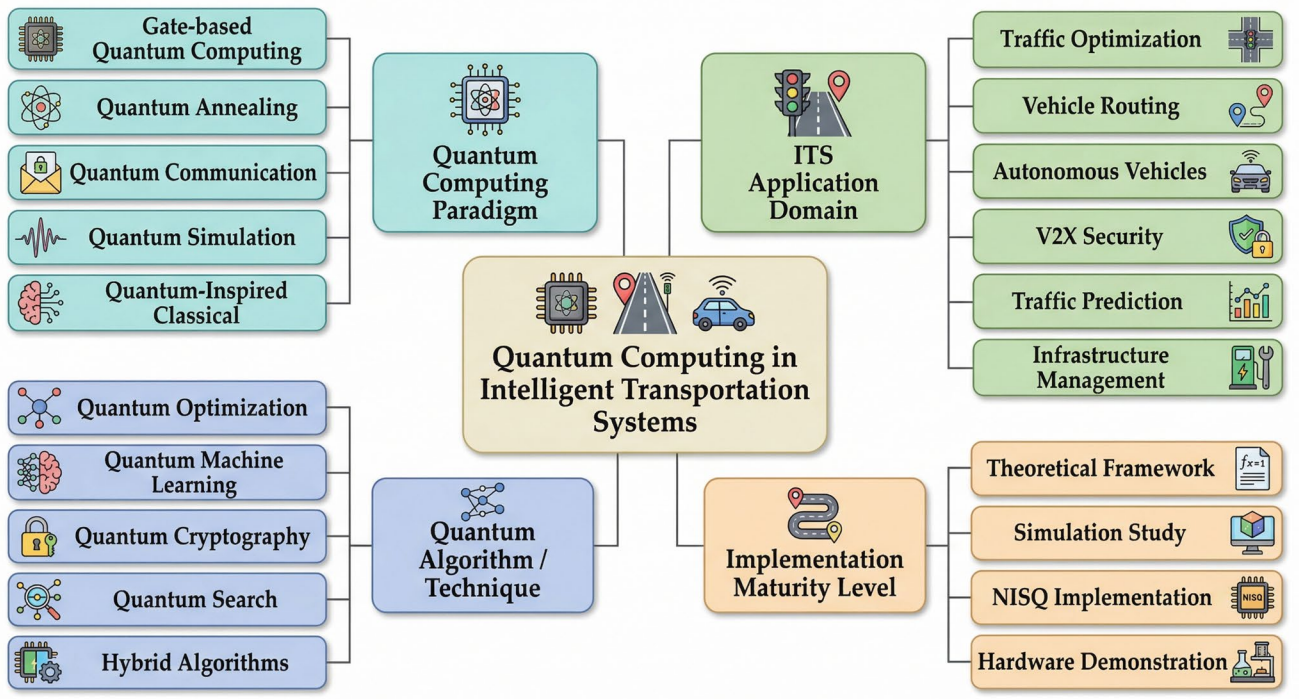


Fig. 1 Four-dimensional taxonomy of quantum computing applications in intelligent transportation systems

Table 1 Four-dimensional taxonomy of quantum computing applications in intelligent transportation systems

Dimension	Category	Description and sub-categories
Quantum Computing Paradigm	Gate-based Quantum Computing	Universal quantum computers operating through quantum gate circuits; encompasses variational quantum algorithms, quantum machine-learning circuits, and hybrid quantum–classical workflows.
	Quantum Annealing	Adiabatic quantum optimization encoding combinatorial objectives into physical Ising Hamiltonians; specialized hardware, including D-Wave systems, solves QUBO and Ising-model formulations.
	Quantum Communication	Quantum key distribution (QKD) protocols and quantum-channel vehicular authentication for V2X network security.
	Quantum Simulation	Analog and digital quantum simulation of transportation dynamics, traffic-flow phenomena, and multi-agent coordination scenarios.
	Quantum-Inspired Classical	Metaheuristics borrowing quantum metaphors, such as Q-bit representation and rotation-gate-style updates, while executing entirely on classical processors without genuine quantum hardware or quantum effects.
ITS Application Domain	Traffic Optimization	Signal timing, intersection coordination, network-level flow control, and congestion management.
	Vehicle Routing	Static and dynamic routing, multi-vehicle coordination, logistics optimization, and fleet dispatch.
	Autonomous Vehicles	Perception enhancement, decision-making, path planning, and cooperative driving coordination.
	V2X Security	Cryptographic protocol design, authentication, intrusion detection, and privacy preservation for vehicular communication networks.
	Traffic Prediction	Short- and long-horizon demand forecasting, congestion prediction, and travel-time estimation.
	Infrastructure Management	Maintenance scheduling, resource allocation, parking optimization, and charging infrastructure planning.
Quantum Algorithm / Technique	Quantum Optimization	QAOA, VQE, quantum annealing, and hybrid quantum–classical decomposition for NP-hard combinatorial problems.
	Quantum Machine Learning	Quantum neural networks, quantum kernel methods, variational classifiers, quantum feature maps, and quantum reinforcement learning.
	Quantum Cryptography	QKD protocols such as BB84, E91, and CV-QKD, quantum random number generation, and post-quantum cryptographic schemes for vehicular network security applications; PQC is treated as a classical quantum-resistant security pathway rather than a quantum-hardware implementation.
	Quantum Search	Grover's algorithm, amplitude amplification, and quantum pattern matching for large-scale transportation queries.
	Hybrid Algorithms	Quantum–classical decomposition, variational hybrid methods, and quantum-assisted subroutine integration.
Implementation Level	Theoretical Framework	Mathematical formulations, complexity analysis, algorithm design, and theoretical performance bounds.
	Simulation Study	Classical simulation of quantum circuits; correctness validation on small instances constrained by exponential simulation cost.
	NISQ Implementation	Near-term execution on noisy intermediate-scale devices; variational and noise-adaptive approaches.
	Hardware Demonstration	Experimental validation on physical quantum processors with real or realistic ITS datasets.

apparent field maturity and is a principal source of the terminological confusion that Sects. 4 and 5 characterize in detail.

Third, post-quantum cryptography occupies a structurally distinctive position in the taxonomy. It is not a quantum computing technology but a classical response to the quantum threat to public-key cryptographic infrastructure. Its deployment maturity is categorically distinct from, and substantially exceeds, that of NISQ-era quantum hardware, a distinction the security sub-literature has not consistently enforced.

2.2 ITS Computational Demands Motivating Quantum Investigation

Building on the motivation established in Sect. 1, this subsection distinguishes three transportation demand categories that make post-classical computing relevant to ITS: combinatorial optimization, machine learning at transportation scale, and cryptographic transition. The strength of the quantum motivation differs across these domains, and each requires a different evidentiary threshold before practical advantage can be claimed.

Combinatorial complexity in traffic optimization and routing Urban traffic signal coordination across n intersections,

each with k discrete timing configurations, generates a solution space of $\mathcal{O}(k^n)$ feasible configurations whose exhaustive enumeration is computationally intractable for any realistic metropolitan deployment [7]. Even obtaining provably near-optimal solutions requires integer programming formulations that scale poorly with network size. Tailored methods such as max-pressure controllers [8] and deep reinforcement learning agents [9] achieve strong empirical performance through problem-structure exploitation, but the computational difficulty of the general coordination problem leaves open whether post-classical paradigms could yield meaningful gains for specific hard sub-problems. Critically, NP-hardness of the general formulation does not imply that instances arising from real transportation networks resist efficient classical solution: problem structure, spatial locality, and temporal decomposability create openings for specialized algorithms that generic hardness arguments do not preclude. Quantum optimization must therefore demonstrate advantage over specifically optimized solvers, a considerably more demanding target than the toy-scale comparisons that characterize most of the reviewed literature.

Vehicle routing problems with time windows present combinatorial complexity that grows rapidly with the number of stops [33], and real-world extensions incorporating dynamic travel times, stochastic demand, and heterogeneous fleets compound this further. State-of-the-art metaheuristics and exact solvers, including the Lin-Kernighan-Helsgaun algorithm family [34] and hybrid genetic search methods [35], reach near-optimal solutions on structured instances of hundreds of customers. These results establish a performance threshold that quantum routing methods must exceed at operationally realistic scales, well beyond the three-to-four-stop instances evaluated in current quantum-ITS literature.

Machine learning at transportation scale Machine learning has become structurally integral to ITS, with deep spatio-temporal architectures, including graph neural networks and Transformer-based models, processing months of high-resolution multi-sensor data to learn complex network-wide dependency structures [14, 15]. These tasks already rely on highly optimized classical hardware and software stacks. Quantum machine learning has been proposed as a possible route to alternative feature-space representations and kernel-based classifiers [23, 36], but this remains a research hypothesis rather than an established ITS capability. Current NISQ-era algorithms remain constrained by data encoding, repeated measurement, noise, and limited circuit depth [37]. These constraints are especially important for transportation tasks involving large datasets, continuous sensor values, and strict latency requirements. Autonomous vehicle perception systems process high-dimensional multimodal sensor inputs under real-time latency constraints [16], and traffic incident detection or anomaly identification often faces severe class imbalance between normal and anomalous events [38]; these settings are sometimes cited as domains where quantum kernel classifiers could, in principle, offer richer feature-space mappings [36]. In each domain, the theoretical quantum opportunity is bounded by encoding overhead and measurement-shot complexity constraints whose practical implications are analyzed in Sect. 2.4.

Security imperatives and the cryptographic transition The security dimension is structurally distinct from optimization and prediction because the threat is already present rather than prospective. V2X communications secured by RSA and elliptic-curve Diffie–Hellman are exposed to the harvest-now-decrypt-later model, in which traffic captured today is archived for decryption once fault-tolerant quantum computers exist; migration therefore cannot wait for quantum hardware to mature [12, 13]. Peer-reviewed transition and IoT-security studies show that lattice-based and other quantum-resistant public-key schemes have advanced through standardization and migration pathways suited to conventional and resource-constrained environments, making cryptographic migration actionable on existing infrastructure [24–26]. This combination of an operational threat and deployable defenses makes post-quantum cryptography the most deployment-ready quantum-related area in ITS, a conclusion the TRL assessment in Sect. 5 confirms quantitatively.

2.3 Quantum Computing Foundations Relevant to ITS

Quantum computing derives its computational properties from three quantum mechanical principles that collectively enable information processing distinct in kind from classical binary architectures [17, 39].

Superposition. A single qubit described by $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ exists as a linear combination of both computational basis states, where the complex probability amplitudes satisfy $|\alpha|^2 + |\beta|^2 = 1$ [39]. An n-qubit register spans a 2^n-dimensional Hilbert space, but measurement returns classical outcomes rather than direct access to all amplitudes. Superposition therefore does not imply classical parallelism; its computational value depends on whether an algorithm can use interference to amplify useful outcomes and suppress unhelpful ones. This distinction bears directly on realistic assessments of quantum benefit for ITS applications.

Entanglement. Entanglement creates correlations between qubits whose joint quantum state cannot be decomposed into a product of individual qubit states [40]. The

Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ is a canonical two-qubit example: measurements of the two qubits in the same computational basis produce perfectly correlated outcomes. Entanglement enables selected quantum algorithms to encode and manipulate global problem structure in ways that may be expensive to represent classically, making it central to advantage claims associated with quantum linear-systems solvers and quantum principal component analysis [41, 42]. However, entanglement alone does not imply practical advantage for ITS; the advantage also depends on data access, circuit depth, noise tolerance, and benchmark design.

Quantum interference. Quantum algorithms are fundamentally exercises in interference engineering: gate sequences are arranged so that probability amplitudes corresponding to undesirable outcomes cancel while amplitudes corresponding to target outcomes reinforce [17]. This phase-based computation differs from classical probabilistic search and underlies canonical speedups, including Grover's $\mathcal{O}(\sqrt{N})$ unstructured search [43] and Shor's polynomial-time algorithms for integer factorization and discrete logarithms [12]. The same mechanism is fragile in current hardware: decoherence, gate errors, and finite measurement budgets limit the depth and scale of circuits that can be executed reliably.

2.4 Quantum Hardware, Algorithms, and Practical Constraints

Hardware platforms. Four principal physical implementations are relevant to ITS application contexts, each with distinct performance profiles. Superconducting quantum processors, the platform underlying IBM Quantum, Google Sycamore, and Rigetti systems, implement qubits as Josephson junctions operating at millikelvin temperatures, with fast gate operation but coherence and gate-error constraints that limit practical NISQ circuit depth [27, 44, 45]. The requirement for dilution-refrigerator infrastructure is a fundamental constraint incompatible with on-board vehicular installation.

Trapped-ion processors offer long coherence times and high-fidelity gates, but slower gate operations and scaling challenges constrain throughput for large-scale optimization workloads [46]. Quantum annealers, including D-Wave systems with more than 5,000 physical qubits, provide a natural formulation pathway for ITS combinatorial problems through QUBO and Ising encodings [21, 22, 47]. Limited hardware connectivity nevertheless requires minor embedding of logical problem variables onto physical qubit chains, and embedding overhead or chain-breaking errors can reduce effective problem size and degrade solution quality [48–50]. Neutral-atom platforms in optical tweezer arrays have demonstrated programmable Rydberg-interaction systems with hundreds of atoms, offering flexible graph structures that may eventually reduce embedding burdens for network-structured optimization problems [51, 52].

Quantum algorithms for ITS problems. The Quantum Approximate Optimization Algorithm targets combinatorial optimization through alternating layers of problem and mixing Hamiltonian evolution, with circuit parameters optimized classically to maximize objective expectation values [19, 20]. In the standard depth-p formulation, QAOA uses $2p$ global variational parameters, and circuit depth increases with p for a fixed problem encoding. Deeper QAOA circuits are therefore difficult to execute reliably on current NISQ hardware, especially when transportation formulations require many variables and constraints. Theoretical performance guarantees also remain limited: symmetry and locality constraints can prevent constant-depth QAOA from exceeding strong classical approximation algorithms, and for certain MaxCut instances the classical Goemans-Williamson algorithm can outperform QAOA at any constant circuit depth [53]. Quantum annealing seeks low-energy configurations of the Ising Hamiltonian encoding optimization objectives through the adiabatic theorem [21, 22]. Finite anneal times, hardware noise, and small spectral gaps for hard instances prevent guaranteed global-optimum recovery, and no study in the reviewed literature has yet established competitive advantage over classical simulated annealing or commercial integer-programming solvers on ITS problem instances.

Quantum machine learning encompasses algorithms whose advantage claims require careful qualification [23]. Quantum principal component analysis and quantum linear-systems algorithms can offer asymptotic speedups under strong data-access assumptions, but these assumptions often shift the cost to state preparation or QRAM-like input models [54, 55]. Parameterized quantum circuits face trainability limitations from barren plateaus, where gradients may vanish exponentially with qubit count or circuit depth under broad conditions [56, 57]. Quantum kernel methods avoid some deep-circuit training difficulties by mapping classical data into quantum feature spaces [36], but their superiority over classical kernels on realistic transportation datasets remains an open empirical question. For ITS, the unresolved bottleneck is not only model expressiveness but also end-to-end data loading, repeated measurement, latency, and reproducibility [37, 58].

The NISQ constraint in context. Current quantum hardware is still best understood through the NISQ framework: devices may contain many noisy physical qubits, but they generally lack the error-corrected logical qubits required for reliable execution of deep, large-scale algorithms [18, 37]. Fault-tolerant quantum computation would suppress

decoherence through quantum error correction, but surface-code approaches require substantially more physical than logical qubits, with overheads commonly reaching hundreds to thousands, or more, depending on physical error rates, target logical error rates, and circuit requirements [59]. This gap is not a minor implementation detail; it defines the difference between algorithms that are theoretically attractive and systems that can meet the problem sizes, precision requirements, and latency budgets of operational ITS.

2.5 Critical Alignment Between ITS Needs and Real Quantum Capability

The theoretical alignment between quantum computing capabilities and ITS computational demands is genuine: the combinatorial structure of traffic signal coordination maps naturally to QUBO formulations, the high-dimensional data of autonomous vehicle perception connects to quantum kernel expressiveness, and the quantum threat to V2X cryptography creates an urgent security mandate. Where the literature falls short is in translating these structural correspondences into credible evidence of practical advantage, and the nature of that shortfall differs across application domains.

Quantum optimization research is constrained primarily by scale and baseline selection. Early demonstrations of QUBO formulation for traffic-flow rerouting on D-Wave hardware [60] confirm technical feasibility but evaluate instances orders of magnitude smaller than metropolitan deployments and compare against elementary classical alternatives rather than commercial solvers. This methodological gap, rather than fundamental infeasibility, is the primary reason the existing literature cannot support confident claims of quantum computational advantage. Hybrid quantum-classical decomposition offers a pragmatic pathway forward, but the quantum contribution within such workflows must be isolated analytically from classical engineering when evaluating advantage claims.

Quantum machine learning faces a different set of constraints rooted in algorithmic fundamentals. The barren-plateau problem, data-encoding information loss, and measurement-shot latency continue to constrain near-term realization of quantum ML advantages for ITS tasks [37, 55–57]. Quantum kernel methods are theoretically more tractable, but their superiority over classical counterparts, including support vector machines with domain-specific transportation features on spatiotemporal datasets, has not yet been demonstrated empirically.

Quantum key distribution provides information-theoretically secure key exchange in principle and is most credible for fixed or well-controlled infrastructure links, but practical QKD systems still face implementation, range, network-integration, and cost constraints that make near-term vehicular deployment substantially less mature than PQC [61]. Post-quantum cryptography addresses the same security objective on existing hardware using quantum-resistant public-key algorithms [24–26], making it the operationally viable choice for connected-vehicle security in the current decade. The two approaches differ by at least one generation of hardware maturity, and conflating them in ITS security discussions produces systematically misleading readiness assessments.

This domain-differentiated assessment motivates the critical evaluation framework applied in Sect. 4, which tests advantage claims against baseline fairness, problem-scale realism, reproducibility, and deployment practicality. Post-quantum cryptography represents an immediately actionable priority. Quantum annealing and gate-based QAOA offer medium-term research directions contingent on hardware scaling and improved algorithmic guarantees. Quantum machine learning for ITS remains in an exploratory phase defined by unresolved theoretical barriers rather than solvable engineering challenges. Quantifying the distance between current hardware capabilities and the demands of operational transportation systems, and identifying the conditions under which that distance may be reduced, is a primary analytical purpose of this review.

3 Methodology

This section describes the systematic approach used to identify, select, analyze, and synthesize the peer-reviewed literature on quantum computing applications in intelligent transportation systems. The study adopts a hybrid design that integrates quantitative bibliometric analysis with structured qualitative content synthesis, an approach increasingly adopted for evidence-grounded critical reviews in rapidly evolving interdisciplinary engineering fields [62–64].

The two components are complementary. Bibliometric analysis provides reproducible quantification of field structure, collaboration networks, thematic evolution, and citation dynamics across a consistently evaluated corpus, but it cannot assess the methodological rigor or internal validity of individual studies. No citation or keyword-based procedure can determine whether a reported quantum speedup reflects a genuine empirical advantage or an artifact of weak experimental design, such as benchmarking against suboptimal classical baselines, evaluating laboratory-scale instances far smaller than operational systems, or presenting simulator-only results as hardware validation. Qualitative content synthesis supplies precisely this study-level critical assessment. Together, the two methods yield field-level

structural insight and study-level methodological evaluation that neither approach could provide alone [64].

Figure 2 summarizes the integrated workflow, from source identification and search design through PRISMA-compliant screening, data extraction, and bibliometric analysis to qualitative synthesis and integrated evidence reporting. The subsections below describe each component.

3.1 Data Source Selection and Search Strategy

Scopus was selected as the sole primary bibliographic database for four complementary reasons spanning quantum computing, transportation engineering, computer science, and operations research.

Disciplinary coverage Scopus is one of the principal databases used in bibliometric research and offers broad multidisciplinary coverage across the natural, physical, engineering, and computing sciences. Comparative studies indicate that Scopus generally indexes more journals than Web of Science, although coverage varies across fields, countries, languages, and document types [65, 66].

Metadata consistency Scopus provides internally standardized metadata suitable for consistent bibliometric analysis. Structured editorial curation of author identifiers, institutional affiliations, and citation counts supports reliable performance analysis and network construction. Because co-authorship, bibliographic-coupling, and keyword co-occurrence analyses depend on clean, internally coherent input, the choice and configuration of bibliometric software shape the science-mapping outputs that can be produced [67]. Merging records across multiple databases introduces heterogeneous metadata schemas, differing citation conventions, and duplicate inflation that can distort centrality measures, bibliographic-coupling boundaries, and temporal trends [64]. These risks are amplified in the present domain, where quantum-ITS work spans physics, computer science, and engineering outlets with divergent indexing conventions.

Single-database adequacy A companion transportation-AI bibliometric review on a structurally comparable topic found that Scopus retrieved approximately 5.8 times more relevant publications than Web of Science, and manual cross-verification indicated that most Web of Science records were already indexed in Scopus [68]. This supports the adequacy of single-database retrieval for the present class of review, without implying that the same ratio generalizes to other topics.

Methodological precedent Review designs of this kind have been applied in adjacent ITS domains, including machine learning in autonomous vehicles [69] and large language model applications in transportation [68], and in other emerging interdisciplinary engineering fields, including scientometric reviews of quantum computing for climate-change research [29] and 4D printing in computer engineering [70], together with systematic technology-assessment work in biomedical manufacturing [71]. Beyond review methodology, recent quantum-enhanced health-prediction research also illustrates how quantum machine-learning claims are increasingly evaluated in applied domains outside transportation, reinforcing the need for cautious evidence appraisal when translating quantum performance claims into domain-specific applications [72]. The restriction to Scopus is an acknowledged limitation whose scope is examined in Sect. 3.6.

The search strategy was developed through iterative refinement to maximize recall of substantively relevant quantum-ITS publications while limiting tangentially related retrieval. Two conceptual groups were combined with a Boolean `AND`: a quantum computing group capturing the relevant paradigms and techniques, and an ITS group spanning transportation application contexts. Within each group, synonymous and variant terms were combined with Boolean `OR`, and wildcard truncation captured morphological variants (*optimi*ation*, *transport**). Retrieval was restricted to title, abstract, and keyword fields (`TITLE-ABS-KEY`) to ensure substantive rather than incidental coverage. The complete query is given in Table 2.

January 2014 was selected as the lower temporal boundary to capture the field's full developmental arc. This window captures the period immediately preceding the earliest transportation-oriented quantum annealing demonstrations and spans the later consolidation of near-term quantum optimization research around QAOA and related variational methods [19, 20, 22, 60, 73]. Beginning retrieval in 2014 therefore covers both the formative period, in which foundational QUBO encodings and shared vocabulary were established, and the subsequent acceleration phase, supporting an accurate characterization of the field's intellectual trajectory. The search was executed on January 28, 2026, returning 312 records.

3.2 Study Screening and Eligibility Assessment

Screening followed the PRISMA 2020 reporting guidelines [32] through three sequential stages: automated

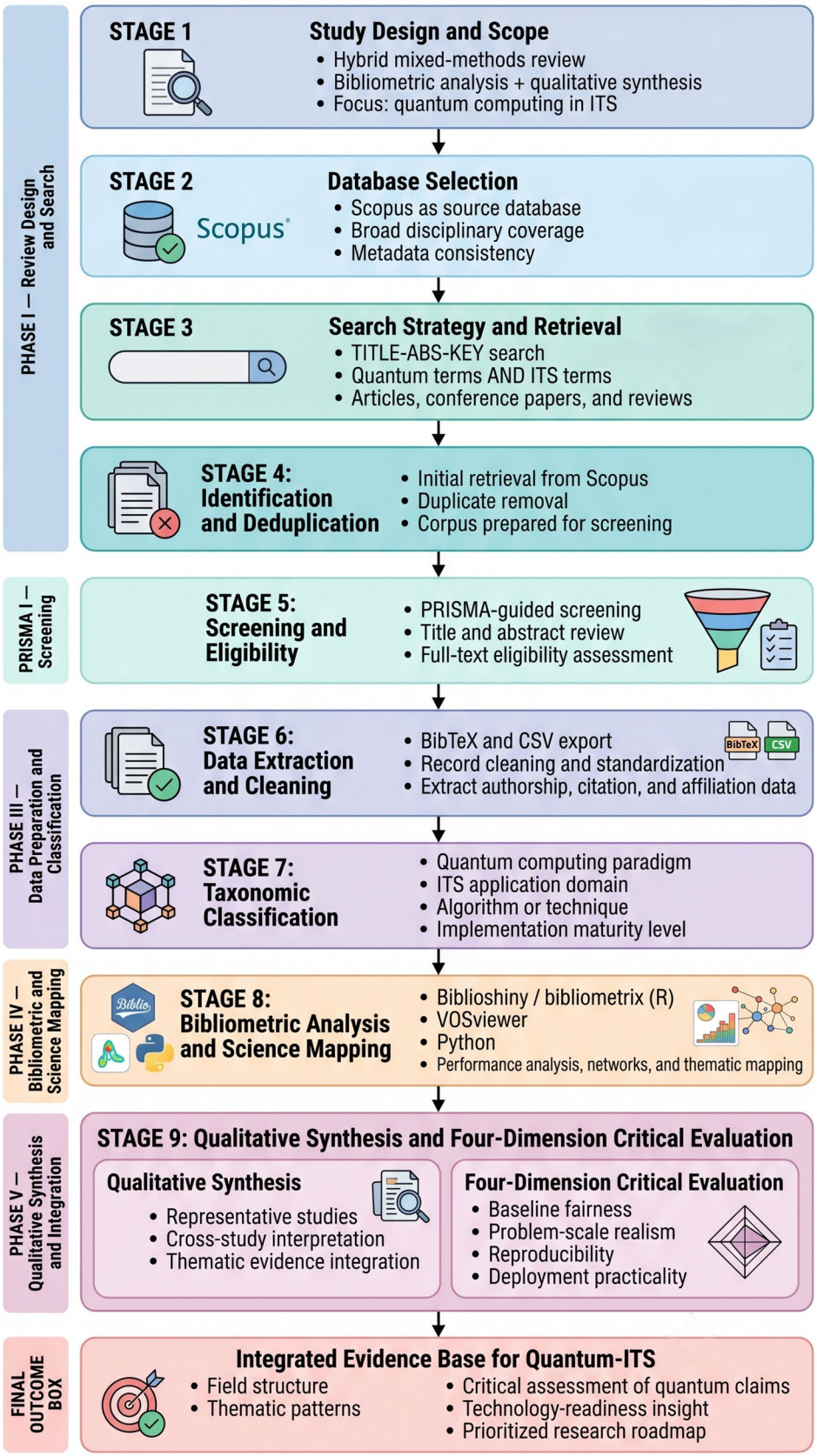


Fig. 2 Overview of the integrated methodological workflow

Table 2 Scopus search strategy and query design used for the literature retrieval process

Category	Specification
A. Search concept groups	
Quantum computing terms	"quantum computing" OR "quantum algorithm*" OR "quantum optimi*ation" OR "quantum annealing" OR "QAOA" OR "quantum machine learning" OR "quantum neural network*" OR "quantum cryptography" OR "quantum key distribution" OR "QKD" OR "quantum communication" OR "quantum simulation" OR "variational quantum" OR "hybrid quantum" OR "NISQ"
ITS terms	"intelligent transport*" OR "smart transport*" OR "connected vehicle*" OR "autonomous vehicle*" OR "traffic optimi*ation" OR"traffic management" OR "traffic signal*" OR "vehicle routing" OR"traffic prediction" OR "vehicular network*" OR "V2X" OR "V2V" OR"V2I" OR "VANET" OR "ITS" OR "parking optimi*ation" OR"fleet management"
Combined Boolean logic	(Quantum computing terms) AND (ITS terms)
B. Search filters and database settings	
Database	Scopus
Search field	TITLE-ABS-KEY (title, abstract, and author keywords)
Document types	Article, Conference Paper, and Review
Language	English
Time span	January 1, 2014–December 31, 2025
Search execution date	January 28, 2026
C. Retrieval outcome	
Initial records retrieved	$n = 312$

deduplication, title and abstract screening, and full-text eligibility assessment. Two independent reviewers conducted each stage, and disagreements were resolved through discussion until consensus was reached. Figure 3 presents the PRISMA 2020 flow diagram.

Automated deduplication removed 18 duplicate records, yielding 294 unique publications. Reviewers independently assessed each record against the eligibility criteria in Table 3, excluding 142 publications that did not substantively address both quantum computing and ITS, and advancing 152 records to full-text assessment. Full-text evaluation excluded a further 44 publications: 19 for insufficient integration of quantum and ITS content; 12 for purely theoretical quantum formulations with no ITS application; 8 for inaccessible full text despite institutional subscription access; and 5 for predatory or non-indexed venues. The final corpus comprised 108 publications.

Inter-reviewer agreement was quantified using Cohen's κ coefficient [74], the standard reliability measure for categorical screening decisions. Agreement was $\kappa = 0.87$ at the title and abstract stage and $\kappa = 0.91$ at full-text assessment, indicating high reviewer consistency without relying on an additional interpretive scale. The higher full-text agreement reflects the clearer evidentiary basis available once complete publication content was examined, relative to abstract screening, where quantum-ITS relevance was occasionally ambiguous for papers using quantum-inspired classical metaheuristics under quantum computing terminology.

3.3 Data Extraction and Analytical Classification

Bibliometric metadata and content-classification data were extracted from all 108 included publications through a combination of automated Scopus export and systematic manual extraction. Metadata were exported in BibTeX and CSV formats and processed through semi-automated cleaning to resolve author-name inconsistencies, institutional-affiliation variants, and citation-count discrepancies. Table 4 summarizes the extracted fields by category.

Content classification required independent manual review by two reviewers along the four taxonomic dimensions introduced in Sect. 2: quantum computing paradigm, ITS application domain, quantum algorithm or technique, and implementation maturity level. A classification distinction bearing directly on this review's central finding is the separation of quantum-inspired classical algorithms, which borrow quantum notation without involving quantum hardware, entanglement, or interference, from genuine quantum computing implementations on physical or cloud-accessible processors. This separation is maintained across all analytical phases, with classification agreement ($\kappa = 0.84$) reported alongside the other inter-rater statistics in Sect. 3.6.

3.4 Bibliometric Analysis Tools and Procedures

Three complementary software environments were used for the bibliometric analysis, coordinated as a single pipeline rather than as independent analyses. Table 5 summarizes the role and primary functions of each platform.

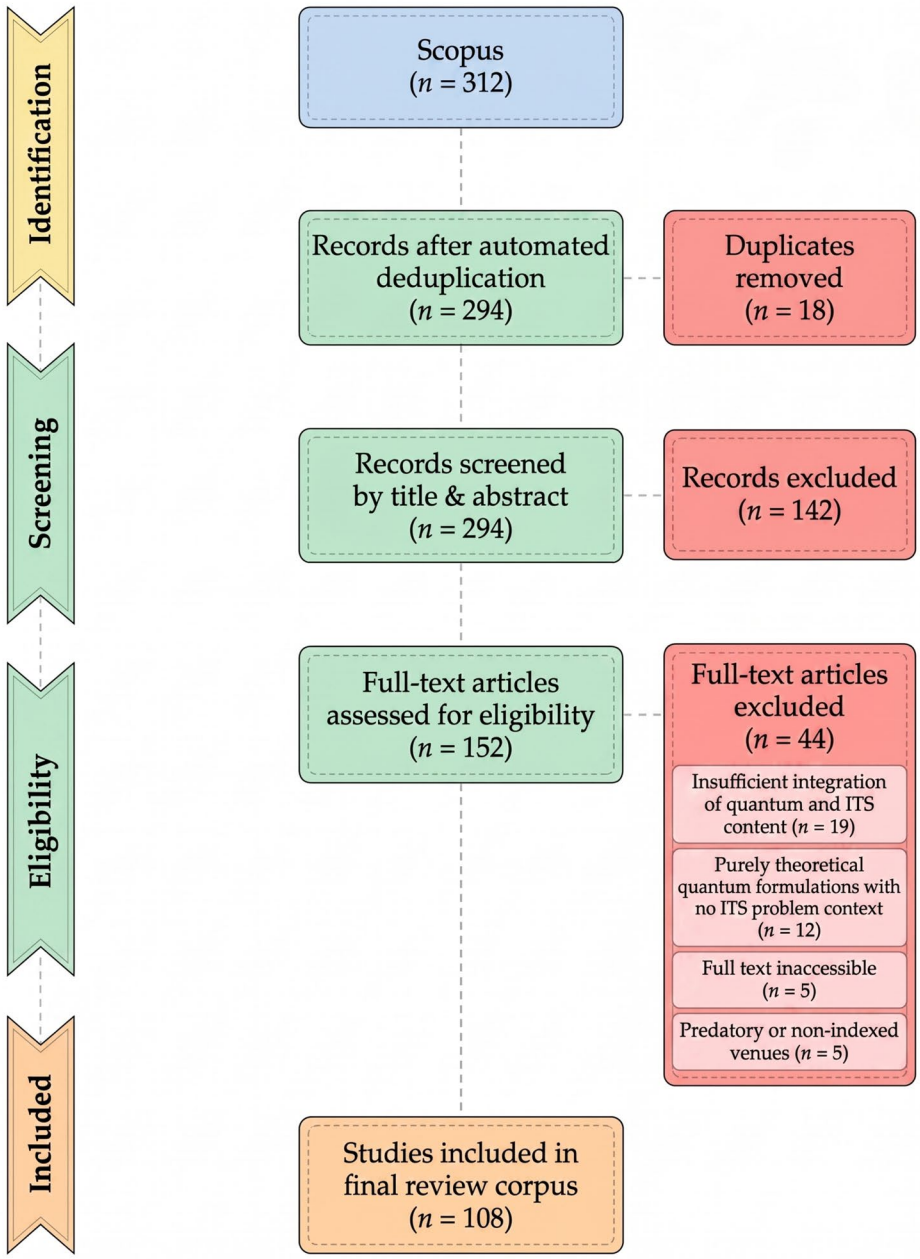


Fig. 3 PRISMA 2020 flow diagram tracing the systematic screening process from initial Scopus retrieval ($n = 312$) through automated deduplication, title and abstract screening, and full-text eligibility assessment to the final included corpus ($n = 108$)

Biblioshiny and `bibliometrix`. Biblioshiny, the web interface of the `bibliometrix` R package, served as the primary platform for performance analysis and science mapping [30]. It supported descriptive performance analysis, covering annual output, document-type distributions, source concentration, author productivity, and the most globally and locally cited publications, with mean total citations (MeanTC) and age-normalized mean citations per year (MeanTCpY) used to separate foundational impact from sustained annual influence. It also generated the core science-mapping outputs: co-authorship networks at the author, institutional, and country levels with degree, betweenness, and closeness centrality; bibliographic coupling and co-citation analysis; keyword co-occurrence networks from author-provided and Scopus-indexed fields; and strategic thematic mapping in centrality–density space.

VOSviewer. VOSviewer complemented these outputs with distance-based network visualization and modularity-based community detection for co-authorship, bibliographic coupling, co-citation, and keyword co-occurrence networks [31]. The resolution parameter was tuned to obtain thematically interpretable clusters, and text-mining normalized terms from titles and abstracts to canonical forms while

Table 3 Inclusion and exclusion criteria applied during the study selection process

Criterion	Inclusion	Exclusion
A. Topical relevance criteria		
Research focus	Studies in which quantum computing algorithms, techniques, or enabling technologies are substantively applied to intelligent transportation systems (ITS), transportation optimization, vehicular communications, or broader mobility-related problems	Studies containing only incidental references to quantum computing; purely classical computing approaches; or quantum-related studies in non-transportation domains
Quantum scope	Gate-based quantum computing, quantum annealing, quantum communication, quantum simulation, quantum machine learning, quantum optimization, quantum cryptography, and hybrid quantum–classical approaches	Classical metaheuristics without genuine quantum computing components; quantum physics studies without an explicit computational application
ITS domain relevance	Traffic optimization and signal control, vehicle routing, autonomous vehicles, V2X communication, traffic prediction, transportation cybersecurity, infrastructure management, parking optimization, and fleet management	Non-transportation applications or generic quantum algorithm studies in which transportation appears only as a minor illustrative example
B. Publication eligibility criteria		
Publication type	Peer-reviewed journal articles, conference papers indexed in Scopus, and review articles	Editorials, letters, book chapters, dissertations, preprints, andnon-peer-reviewed technical reports
Language	English	All other languages
Time period	January 1, 2014–December 31, 2025	Publications outside this time window
Accessibility	Full text accessible through institutional subscriptions or open-access sources	Full text not accessible after reasonable retrieval efforts
C. Quality screening criteria		
Publication quality	Publications appearing in indexed journals or established conference proceedings with clearly documented peer-review processes	Predatory journals, non-indexed venues, or outlets without verifiable peer-review procedures

Table 4 Data fields extracted for bibliometric analysis and qualitative content synthesis

Data category	Extracted fields
A. Bibliographic and source information	
Publication metadata	Title; abstract; author keywords; Scopus-indexed keywords; publication year; DOI; source title; volume; issue; page range; document type
Reference data	Full reference list for each included publication; cited source titles; cited publication years; reference count per document
Funding information	Funding agencies; grant identifiers; funding countries
B. Authorship and impact indicators	
Authorship information	Author names; institutional affiliations; countries of affiliation; corresponding author identifier; ORCID identifiers; Scopus author identifiers
Citation metrics	Total citations; annual citation counts; mean citations per year (MeanTCpY); author-level h-index and g-index; citing publication identifiers for local citation network construction
C. Analytical coding and content synthesis	
Content classification	Quantum computing paradigm (gate-based, annealing, communication, simulation, quantum-inspired classical); ITS application domain; quantum algorithm or technique; implementation maturity level (theoretical framework, simulation study, NISQ implementation, hardware demonstration)

filtering high-frequency generic terms lacking thematic discriminatory value.

Custom Python analyses. Custom Python scripts extended the analysis beyond the capabilities of the other two platforms. They performed temporal trend decomposition separating systematic growth from year-to-year fluctuation, produced choropleth maps of country-level output, and computed eigenvector centrality and PageRank to identify nodes whose influence propagates beyond direct connections. Community detection via the Louvain algorithm [75] was applied at resolution levels complementary to VOSviewer's modularity-based clustering, and probabilistic topic modeling based on Latent Dirichlet Allocation [76] extracted latent thematic structure from abstracts, with the topic count selected by minimizing perplexity and maximizing coherence.

The three platforms operate as an integrated framework. Biblioshiny establishes the quantitative foundation, including publication trends, citation dynamics, and Bradford's law source concentration, against which structural findings are interpreted. VOSviewer interrogates the same corpus through proximity-based visualization, so that convergence between the two platforms indicates a robust result; where they diverge, LDA topic modeling provides content-based adjudication, testing whether apparent sub-structure reflects a genuine thematic distinction or a terminological artifact. Python's network metrics and community detection extend coverage into geographic and recursive-influence dimensions not accessible through the other tools. Where

Table 5 Bibliometric analysis tools, analytical roles, and primary functions used in this study

Tool	Analytical role	Primary functions in this study
Biblioshiny / bibliometrix (R) [30]	Primary platform for performance analysis and science mapping	Annual publication trend analysis; source productivity and Bradford's law; author h-index and g-index; MeanTC and MeanTCpY citation metrics; co-authorship networks at the author, institution, and country levels; betweenness, degree, and closeness centrality; bibliographic coupling; co-citation analysis; keyword co-occurrence mapping; and strategic thematic mapping based on centrality–density quadrants.
VOSviewer [31]	Complementary platform for network visualization and community detection	High-resolution distance-based network visualization for co-authorship, bibliographic coupling, co-citation, and keyword co-occurrence networks; modularity-based community detection with resolution tuning; term normalization and thematic label extraction from abstract text; and thematic structure exploration through cluster-based mapping.
Python (`pandas`, `numpy`, `networkx`, `scikit-learn`)	Supplementary platform for specialized and advanced analyses	Temporal trend decomposition separating systematic growth from year-to-year fluctuation; choropleth visualization of country-level research output; eigenvector centrality and PageRank computation; Louvain community detection for collaboration networks [75]; and Latent Dirichlet Allocation topic modeling of publication abstracts [76].

feasible, the principal structural findings reported in Sect. 4 are cross-checked across at least two analytical approaches, strengthening the evidentiary basis of the conclusions without implying identical outputs from all tools [64].

3.5 Qualitative Synthesis Strategy

Bibliometric mapping characterizes the structural and relational properties of a corpus but cannot evaluate the methodological rigor, evidential quality, or practical validity of individual studies. A systematic qualitative synthesis was therefore conducted to address RQ3, specifically whether claimed quantum computational advantages over established classical solvers are substantiated by available evidence, a question no bibliometric procedure can answer alone [32].

The synthesis used structured thematic analysis adapted from systematic review methodology. For each major thematic cluster identified through science mapping, a representative sample was drawn through stratified purposive sampling satisfying five coverage criteria: *citation impact* (landmark papers with high accumulated influence); *recency* (publications from 2022 onward reflecting current hardware realities); *quantum paradigm diversity* (quantum annealing, gate-based QAOA, quantum machine learning, post-quantum cryptography, and quantum-enhanced communication); *ITS application breadth* (traffic optimization, vehicle routing, demand forecasting, vehicular security, and maritime and aviation scheduling); and *methodological variety* (theoretical frameworks, circuit simulations, NISQ hardware implementations, and automotive-grade hardware demonstrations). The resulting sample comprised 27 publications, approximately 25% of the full corpus, sized to capture the intellectual range of the field rather than to provide exhaustive or statistically representative coverage, consistent with purposive sampling conventions in mixed-methods bibliometric and systematic review designs [62, 63].

Each sampled publication was subjected to detailed critical reading using a standardized extraction protocol addressing (i) research objectives and motivating problem; (ii) the quantum technique employed and its theoretical justification; (iii) transportation problem formulation and modeling assumptions; (iv) computational complexity analysis and classical baseline comparisons; (v) experimental methodology, including dataset characteristics, problem scale, and evaluation metrics; (vi) key results and any claimed quantum advantage; (vii) explicitly acknowledged limitations; and (viii) proposed future directions. Findings were synthesized within each thematic cluster to identify common methodological patterns, convergent results, sites of empirical controversy, and structural gaps in the evidence base.

Critical evaluation was applied as a unifying lens, assessing each contribution against four dimensions of methodological quality. *Baseline fairness* examined whether classical comparisons represented the state of the art, specifically commercial integer-programming solvers such as CPLEX and Gurobi and specialized transportation metaheuristics such as the Lin–Kernighan–Helsgaun heuristic [34] and hybrid genetic search [35], or were selected to be easily outperformed. *Problem scale realism* assessed whether evaluated instances reflected operational transportation complexity or were confined to laboratory scales orders of magnitude below real deployment contexts. *Reproducibility* examined whether sufficient implementation detail, dataset description, and parameter specification were provided for independent replication. *Deployment practicality* assessed whether authors addressed the engineering, regulatory, and economic challenges of translating laboratory

demonstrations into operational systems. This framework was designed to test, rather than to validate, quantum advantage claims, a stance of calibrated skepticism appropriate given the documented tendency of emerging quantum literature to overstate near-term performance through selective baseline selection and small-scale evaluation [18, 55].

3.6 Methodological Rigor, Transparency, and Limitations

Several considerations bound the scope and generalizability of the findings reported in Sects. 4 and 5, stated explicitly to enable readers to interpret and contest the conclusions.

Database scope. The Scopus restriction supports metadata consistency and citation-network validity, but it also has consequences for specific findings that warrant explicit acknowledgment. Publications indexed exclusively in Web of Science, standalone IEEE Xplore proceedings, or regional repositories such as CNKI are not captured, and these exclusions have asymmetric effects across the four thematic clusters. For post-quantum cryptography, a substantial body of security-engineering work appears in IEEE Symposium on Security and Privacy and ACM CCS proceedings that may be incompletely indexed in Scopus, potentially understating the deployment maturity of this cluster. For quantum optimization, contributions appearing primarily in standalone IEEE Xplore proceedings may be undercounted, which could attenuate regional collaboration signals in the co-authorship analysis. Chinese-language contributions are also likely underrepresented because the search was restricted to English-language Scopus records, so the full scale of Chinese quantum-ITS activity may be understated. These exclusions are judged acceptable for the study's aims: the gains in metadata consistency, citation-network validity, and analytical reproducibility outweigh the marginal coverage benefit that multi-database merging would provide at the cost of substantial preprocessing uncertainty [68, 69]. Readers should nonetheless interpret the geographic concentration findings and the relative size of the PQC cluster with this limitation in mind.

Temporal boundary. The January 2014 start date was justified in Sect. 3.1. The January 28, 2026, search date represents a temporal snapshot; publications appearing after this date are not captured, and rapid hardware or algorithmic developments could alter specific conclusions.

Qualitative sample. The 27-paper sample was stratified across citation impact, recency, quantum paradigm diversity, ITS application breadth, and methodological variety to reflect the intellectual range of the full corpus rather than concentrating in the highest-citation or most recent subset. The central finding, the consistent absence of rigorous quantum advantage demonstration across the reviewed application domains and hardware modalities, is substantially robust to individual study exclusion precisely because it holds uniformly rather than resting on a single cluster's evidence. Where reviewer judgments diverged during coding, disagreements were resolved through discussion in which each reviewer documented the textual evidence supporting their position; cases of genuine interpretive ambiguity, principally regarding the sufficiency of classical baselines in five studies, were resolved conservatively, with the ambiguity acknowledged in the synthesized evaluation.

Evaluative stance. The critical evaluation framework reflects the authors' disciplinary expertise in transportation engineering and quantum computing. Other domain experts may weight the dimensions differently, particularly regarding the significance of simulator-based evaluations or the appropriate evidentiary threshold for claiming quantum advantage. The evaluation rationale is documented throughout Sect. 3.5 to support independent critical assessment.

Technological contingency. Maturity assessments and deployment-timeline estimates are grounded in the hardware, benchmark, and classical-computing evidence available within the review window. Rapid advances in either quantum or classical computing could alter specific conclusions, and the findings should be interpreted accordingly.

Reproducibility is supported across all phases. Table 2 provides the complete Scopus query, execution date, and field restrictions, enabling replication of data collection. Table 3 documents the eligibility criteria, and inter-rater agreement is reported throughout ($\kappa = 0.87$ for title and abstract screening; $\kappa = 0.91$ for full-text assessment; $\kappa = 0.84$ for content classification). The stratified sampling criteria and critical evaluation framework in Sect. 3.5 provide a transparent account of the interpretive phase. Together, these elements establish a reproducible methodological architecture appropriate for an evidence-grounded critical review in computational engineering.

4 Results

This section presents the bibliometric and qualitative findings from the 108-publication corpus on quantum computing applications in intelligent transportation systems. The analysis is organized into three layers: performance analysis, science mapping, and qualitative synthesis. Section 4.1.1 reports publication growth, citation impact, source fragmentation, author productivity, and geographic concentration. Section 4.2 examines intellectual structure through coauthorship, bibliographic coupling, keyword co-occurrence, and thematic evolution. Section 4.3 synthesizes 27 representative studies using the four-dimension evaluation framework. Together, these results show how the field has

expanded and whether quantum-enabled progress claims are supported by methodologically credible evidence.

4.1 Performance Analysis

4.1.1 Temporal Evolution and Publication Trends

Publication output increased from 2 papers in 2019 to 52 in 2025, corresponding to 2,500% growth over six years. The fitted exponential model shows strong agreement with observed annual counts ($R^2 = 0.982$), indicating sustained field acceleration rather than short-term fluctuation, as shown in Fig. 4. This growth rate is notable when compared with adjacent ITS subdomains assessed through recent bibliometric studies [68, 69]. However, rapid growth in a nascent interdisciplinary area should be interpreted cautiously. In this context, publication expansion may reflect increased access to quantum platforms, broadening keyword use, and speculative methodological transfer as much as the accumulation of validated transportation-specific evidence.

Three growth phases are identifiable. The *emergence phase* (2019–2021, 2–5 papers annually) reflects early conceptual exploration following the 2019 quantum supremacy demonstration [27] and the initial availability of cloud-accessible quantum hardware. The *acceleration phase* (2022–2023) saw output increase from 7 to 12 publications, coinciding with broader availability of cloud-based quantum computing platforms such as IBM Quantum, Amazon Braket, and Microsoft Azure Quantum. This wider access broadened participation in experimental quantum computing research, although access alone did not ensure transportation-scale validation. The *consolidation phase* (2024–2025) sustained the exponential trajectory, with 20 publications in 2024 and 52 in 2025. This expansion reflects the combined influence of hardware accessibility, national quantum investment programs in the United States, the European Union, and China, and growing computational demands in ITS research [77–79]. At the same time, the sharp rise in publication volume underscores the need to distinguish field visibility from demonstrated technological maturity.

The final corpus includes 108 publications written by 445 authors affiliated with 278 institutions across 39 countries. Journal articles account for 82.4% of the corpus, consistent with the archival publication culture of transportation engineering and applied computing. The corpus cites 3,892 unique sources, reflecting an interdisciplinary knowledge base that connects quantum computing, optimization theory, machine learning, cybersecurity, and transportation systems. Figure 5 presents the metadata-completeness profile of the Scopus-exported dataset, including missing counts and missing-value percentages for key bibliographic fields. This metadata profile supports the reliability of the bibliometric analysis while also showing that author-affiliation and funding-related fields contain more incomplete records than core bibliographic fields such as title, abstract, DOI, and publication year.

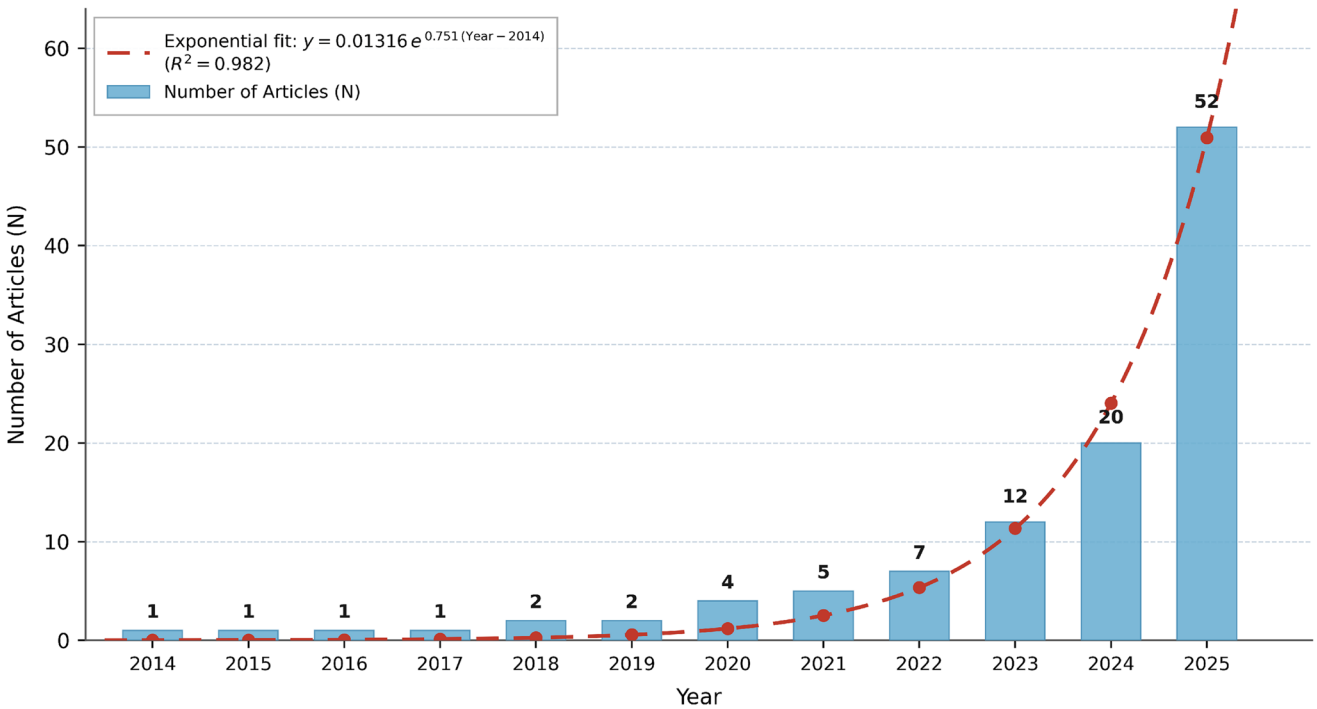


Fig. 4 Annual publication counts with exponential growth model ($R^2 = 0.982$), showing accelerating research output from 2014 to 2025

Metadata	Description	Missing Counts	Missing %	Status
AB	Abstract	0	0.00	Excellent
C1	Affiliation	0	0.00	Excellent
AU	Author	0	0.00	Excellent
CR	Cited References	0	0.00	Excellent
DT	Document Type	0	0.00	Excellent
LA	Language	0	0.00	Excellent
PY	Publication Year	0	0.00	Excellent
TI	Title	0	0.00	Excellent
TC	Total Citation	0	0.00	Excellent
SO	Journal	1	0.93	Good
DI	DOI	3	2.78	Good
RP	Corresponding Author	18	16.67	Acceptable
ID	Keywords Plus	18	16.67	Acceptable
DE	Keywords	24	22.22	Poor
WC	Science Categories	108	100.00	Completely missing

Fig. 5 Metadata completeness profile of the Scopus-exported quantum-ITS corpus, showing missing counts and missing-value percentages across key bibliographic fields

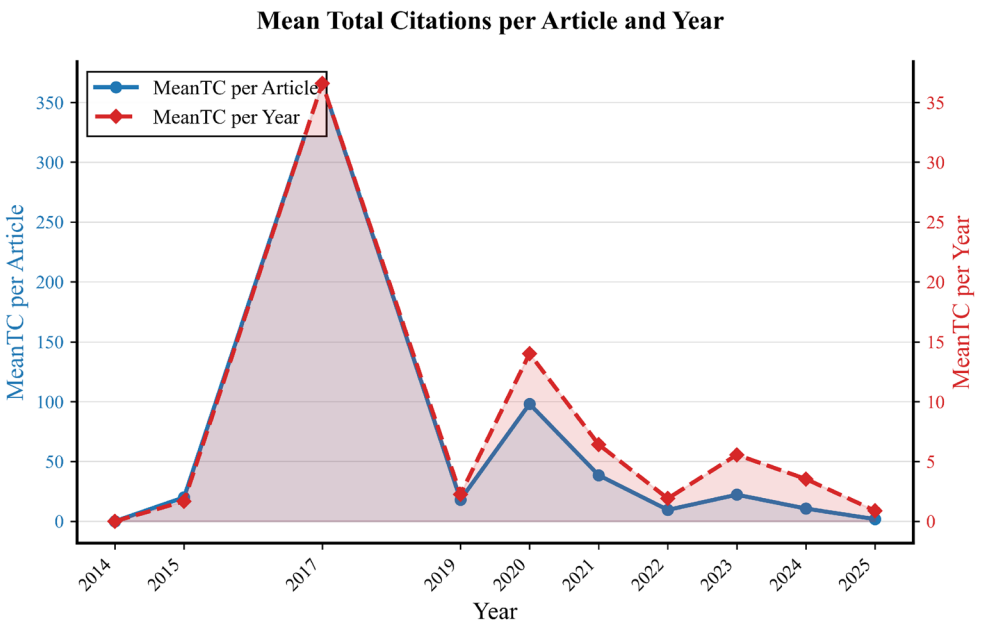


Fig. 6 Mean total citations per article (MeanTC) and citations per year (MeanTcpy) from 2014 to 2025

4.1.2 Citation Impact

Mean total citations per article (MeanTC) and mean citations per year (MeanTCpY) are reported in Fig. 6. Citation impact is uneven across publication years, with early publications showing higher citation intensity because they had longer accumulation windows and, in some years, very small publication counts. The prominent early-year peak in Figure 6 should therefore be interpreted cautiously: it reflects the influence of a small number of highly cited early

contributions rather than broad maturity across the field. Age-normalized MeanTCpY partially reduces this temporal bias and shows which publication years achieved stronger annual citation visibility. More recent papers from 2023 to 2025 have had less time to accumulate citations, so their lower MeanTC values should not be read as evidence of lower long-term influence.

The citation distribution is highly skewed. The top 10 publications, representing 9.3% of the corpus, account for approximately 47% of total citations. Such concentration is consistent with bibliometric power-law distributions [64, 80], but it also indicates that the field's intellectual direction is shaped by a small number of highly visible studies. These influential works mainly concern quantum optimization for traffic management, quantum machine learning for prediction, and quantum-resistant security for vehicular communication. Their dominance confirms the main thematic structure of the field, while also raising a methodological concern: citation visibility may amplify early conceptual claims before their empirical validity has been established through rigorous benchmarking.

4.1.3 Publication Sources and Venues

The source analysis shows a fragmented publication landscape, as illustrated in Fig. 7. *Scientific Reports* leads with 13 publications, followed by *IEEE Transactions on Intelligent Transportation Systems* with 11, *IEEE Internet of Things Journal* with 7, and *Human-Centric Computing and Information Sciences* with 3. This distribution reflects the dual audience of quantum-ITS research. Multidisciplinary journals provide broad visibility for emerging cross-domain work, while ITS, IoT, and embedded-systems journals provide discipline-specific validation. Bradford's law analysis identifies a productive core of approximately 8–10 journals, surrounded by a large periphery of single-contribution outlets. This confirms that the field has not yet developed a stable publication venue structure or a clearly defined disciplinary home.

The most globally cited documents appeared in *Frontiers in ICT*, *IEEE Transactions on Intelligent Transportation Systems*, *Human-Centric Computing and Information Sciences*, and *Scientific Reports*. These studies introduced early QUBO formulations for traffic optimization, quantum machine-learning frameworks for ITS intelligence, and

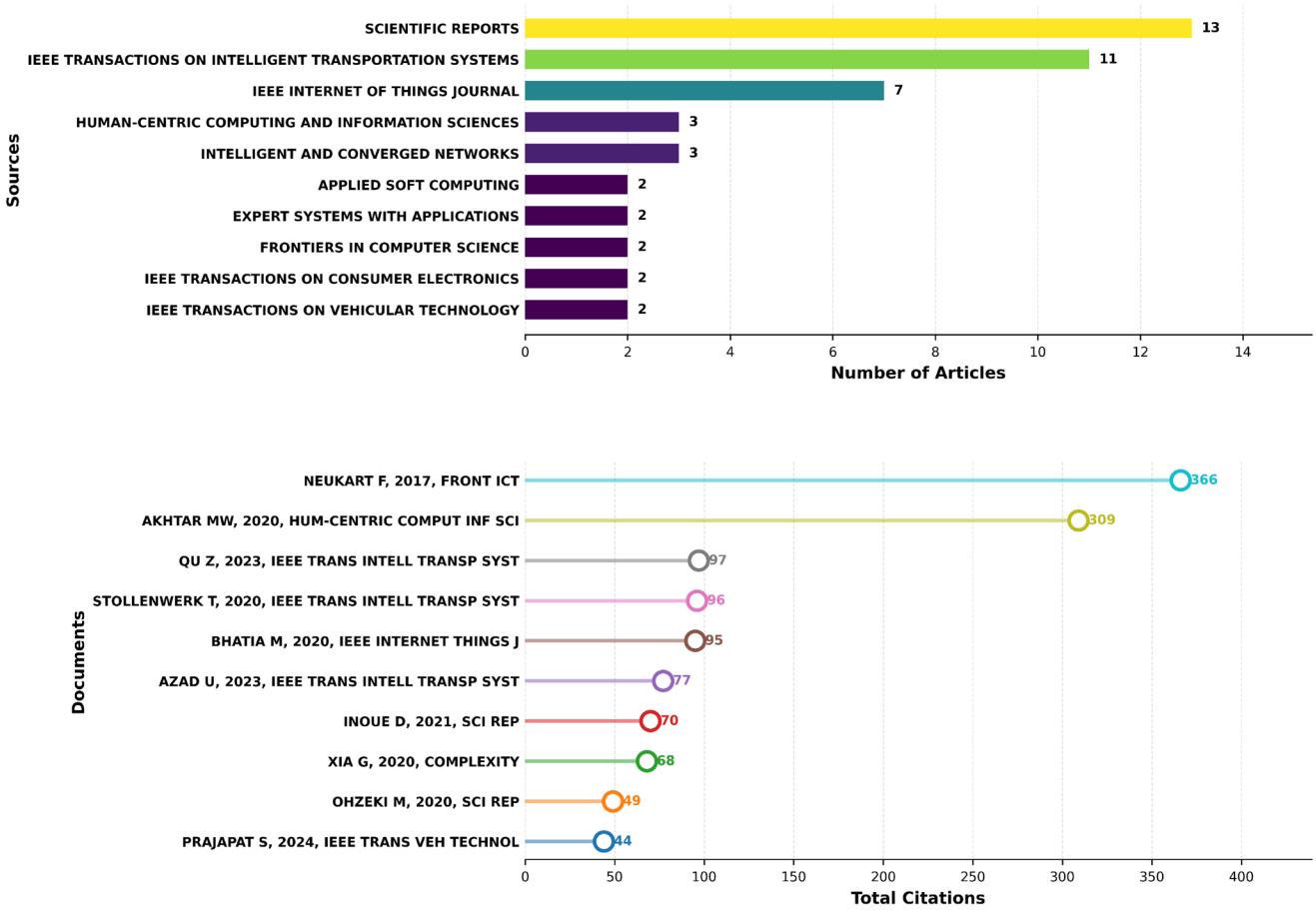


Fig. 7 Most relevant publication sources by article count and most globally cited documents by total citations

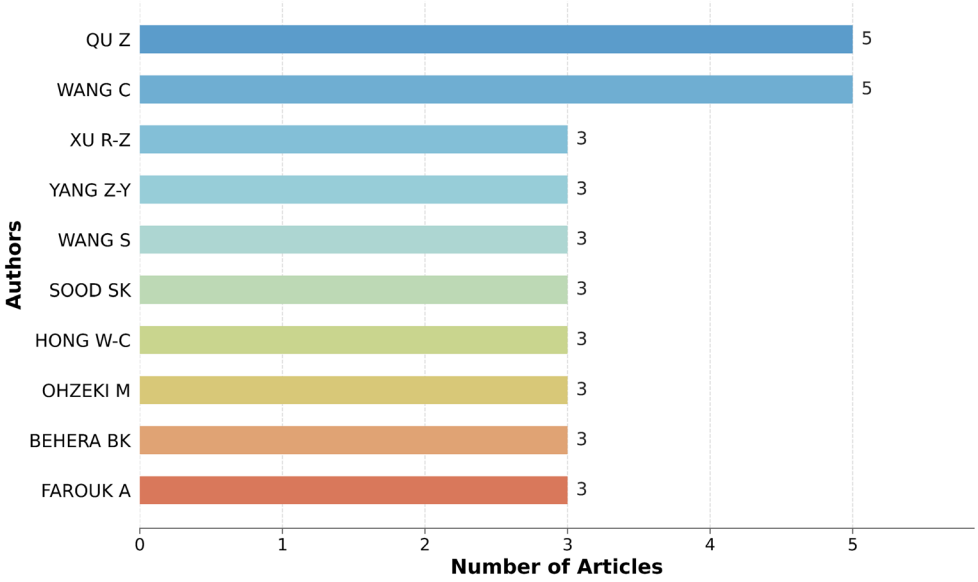


Fig. 8 Top contributing authors ranked by publication count

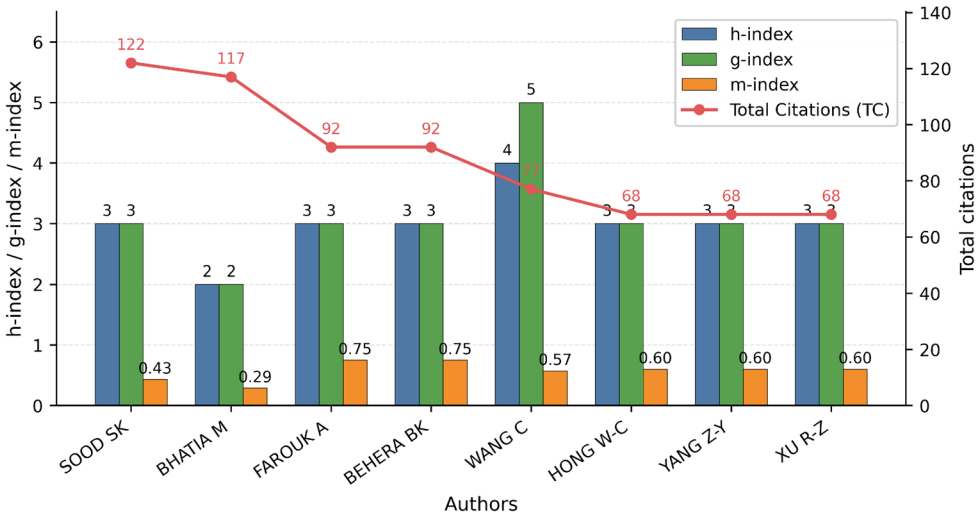


Fig. 9 Comparative author impact across h-index, g-index, m -index, and total citations

foundational discussions of quantum paradigms for vehicular networks. Their influence is substantial, but the venue dispersion suggests that quantum-ITS research remains in a formation stage, where conceptual exploration has advanced faster than methodological standardization.

4.1.4 Author Productivity and Impact

Leading authors contributed 3–5 papers each, indicating the presence of focused research programs rather than purely isolated contributions, as shown in Fig. 8. Among the displayed top contributors, author h-indices range from 2 to 4, with variation in g-index, m-index, and total citation counts indicating that productivity and influence do not align uniformly. This pattern is consistent with a young research area in which citation accumulation remains limited by the short publication history, while a small number of highly visible studies can strongly shape author-level impact. The g-index and total-citation results in Figure 9 therefore provide a complementary view of influence beyond publication count alone.

The temporal productivity analysis in Figure 10 shows that most researchers entered the field after 2020 and published intensively for two to three years. The absence of contributors active across the full study period indicates that the field has not yet formed a stable core of long-term research leaders. The highest density of productive authors occurs in 2022–2023, aligning with the acceleration phase identified earlier. This pattern suggests a rapidly expanding but still fluid research community, where continuity, cumulative benchmarking, and long-term agenda setting remain underdeveloped.

Local citation analysis, shown in Fig. 11, identifies authors with strong intra-corpus influence and recurring

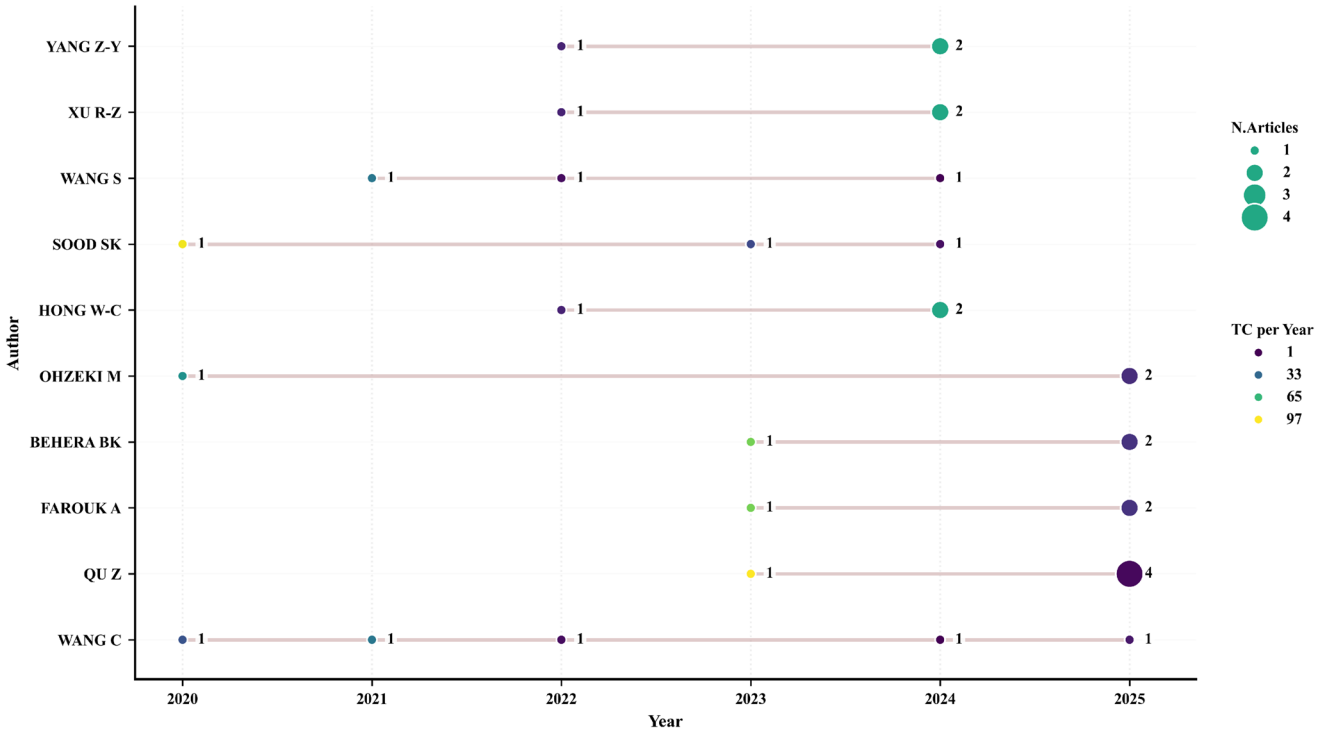


Fig. 10 Author productivity trends from 2020 to 2025. Bubble size represents annual publication count, and color intensity indicates citations per year

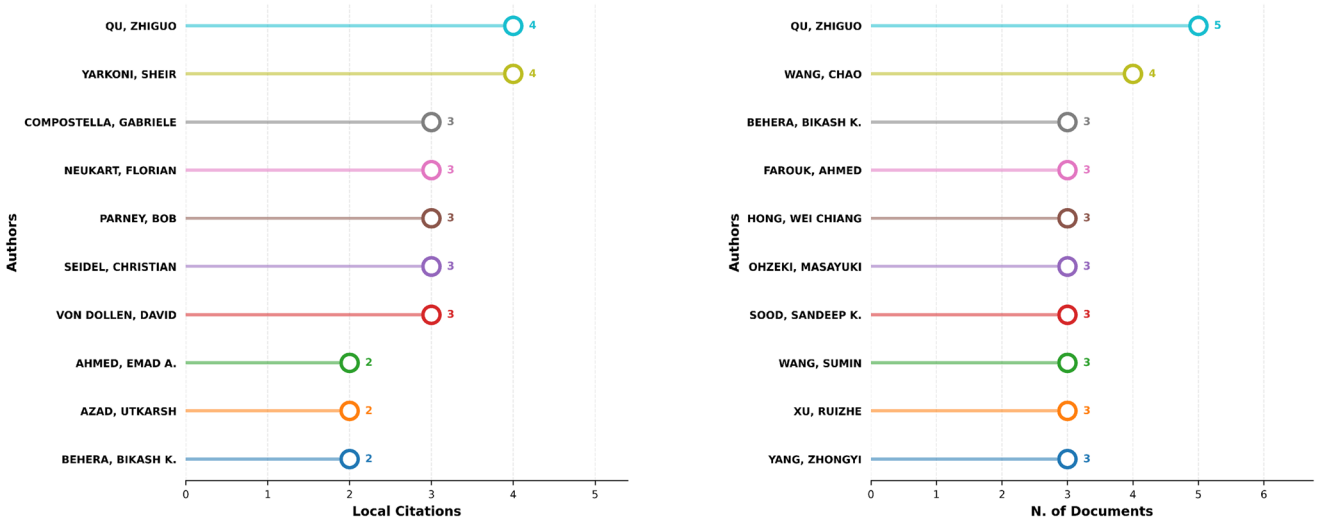


Fig. 11 Top authors by local citations within the study corpus and publication productivity

relevance within the quantum-ITS literature. These contributors typically published early or frequently cited studies that introduced shared problem formulations, terminology, or methodological templates. Their influence is therefore not only bibliometric but also conceptual, since they helped shape how transportation problems were translated into quantum computational terms.

4.1.5 Geographic Distribution and Collaboration Patterns

Research output is geographically concentrated, but the interpretation depends on the bibliometric counting convention. At the unique-publication level, the leading contributors are the United States, China, India, and the United Kingdom, consistent with the geographic concentration reported across the corpus. Figure 12, by contrast, visualizes country-level scientific production based on author-affiliation

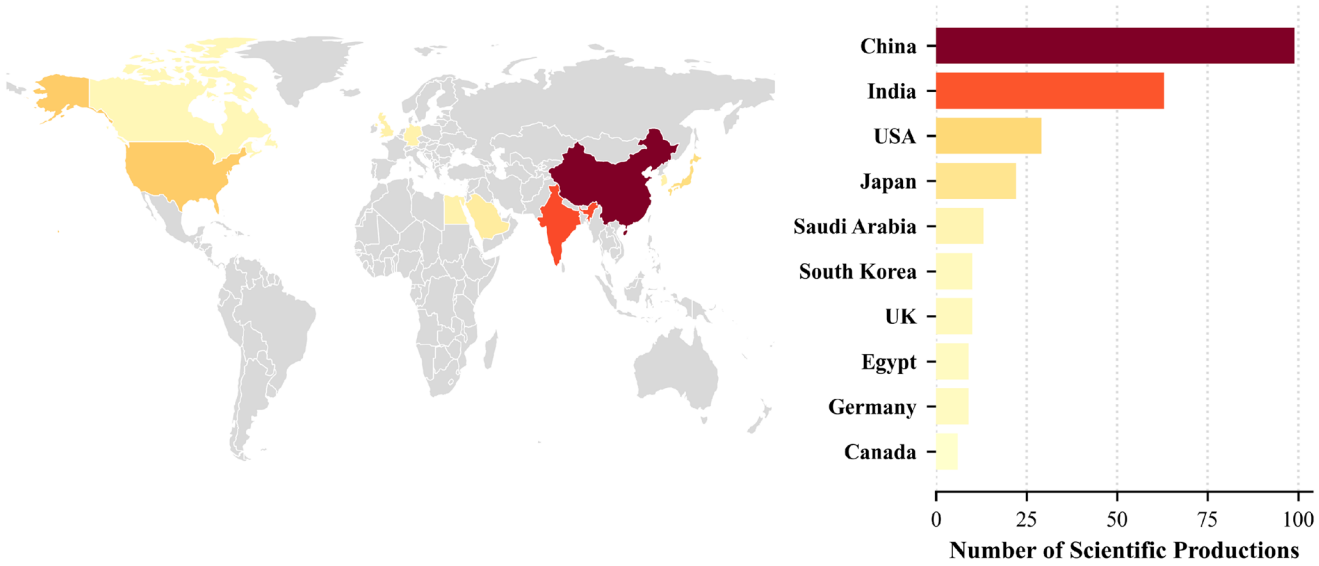


Fig. 12 Global distribution of quantum-ITS publications by country

occurrences, so countries with larger multi-author teams or repeated institutional participation can show larger production footprints than their unique-publication counts alone would suggest. Read in this way, the country map confirms a concentrated global structure rather than a fully distributed research landscape. This concentration broadly mirrors global quantum investment patterns, particularly the U.S. National Quantum Initiative [77], the European Quantum Flagship [78], and China's Quantum Information Science programs. These initiatives have created the infrastructure, funding base, and technical capacity needed to sustain quantum-ITS research.

A more critical pattern emerges when authorship is examined from an implementation perspective. Transportation agencies and infrastructure operators appear as direct contributors in fewer than 2% of publications. This is a consequential gap because the operational problems motivating quantum-ITS research are largely defined outside the communities producing the algorithms. The absence of agency participation limits the extent to which studies incorporate procurement constraints, certification requirements, safety responsibilities, liability considerations, and operational cost structures. As a result, many studies evaluate quantum methods on simplified or simulated settings that are not aligned with the decision environments in which transportation technologies must ultimately operate.

This disconnect has direct implications for translational readiness. A quantum optimization method evaluated on a small synthetic network cannot be credibly positioned for deployment without vendor-independent validation, operational data, regulatory compatibility, and transparent comparison with mature classical systems. The limited involvement of transportation agencies also means that benchmark design, problem-scale selection, and performance metrics are rarely shaped by real operational requirements, which helps explain why quantum annealing and quantum machine learning remain at low Technology Readiness Levels despite growing publication output. The structural mechanisms needed to close this gap are developed in Sects. 5 and 6; within the present analysis, the salient point is that the field may continue to expand bibliometrically while remaining weakly connected to deployment pathways.

International collaboration patterns show substantial heterogeneity, as presented in Fig. 13. The figure separates single-country publications (SCP) from multiple-country publications (MCP), showing that collaboration intensity varies across countries rather than simply tracking total output. Several smaller-output countries display comparatively higher MCP shares, suggesting reliance on international partnerships to access complementary quantum computing, transportation, and computational expertise. In contrast, the largest producers tend to show stronger domestic publication components, indicating greater capacity for self-contained research programs. Although this structure supports national research autonomy, it may also reinforce fragmented benchmark practices and reduce cross-system validation unless international standardization efforts become more prominent.

4.2 Science Mapping

4.2.1 Coauthorship Networks and Collaboration Structure

The author coauthorship network shows pronounced fragmentation, with many small disconnected components of 2–5 researchers and only a limited number of larger clusters

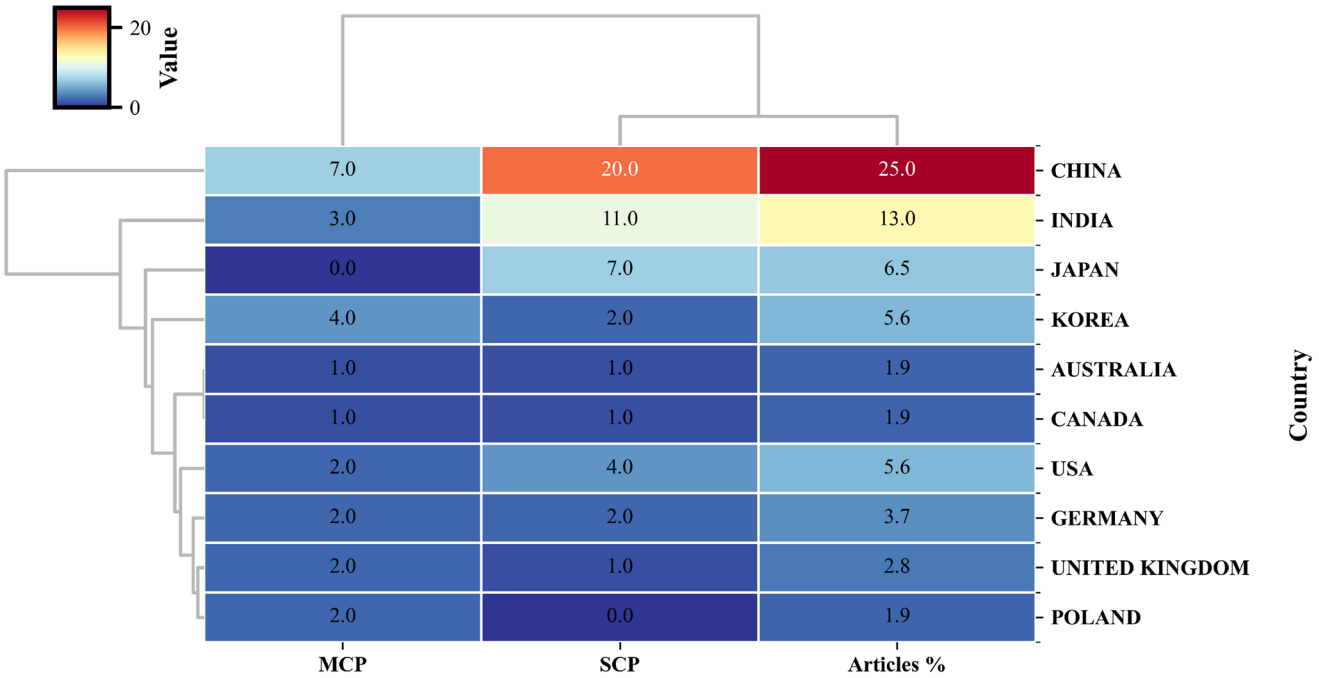


Fig. 13 Country-level scientific output and collaboration intensity, distinguishing single-country publications (SCP) from multiple-country publications (MCP)

of 8–12 authors. This pattern is consistent with emerging research fields in which stable epistemic communities have not yet formed and exploratory work proceeds through relatively isolated groups [81]. Degree centrality identifies directly active collaborators, betweenness centrality identifies authors who broker knowledge between otherwise separated communities, closeness centrality indicates how efficiently a node can access information across the network, and PageRank captures recursive influence gained through links to already well-connected collaborators. In this corpus, the strongest bridge positions are held by authors combining quantum-computing and transportation expertise, or by institutions active in both domains. The sparse network structure is therefore a methodological vulnerability, not merely a social pattern: datasets, benchmark designs, and evaluation practices are not circulating efficiently across the field. This fragmentation limits cumulative learning and reinforces the need for dedicated workshops, shared repositories, and community-accepted benchmark initiatives.

The country-level bibliographic coupling network in Fig. 14 reveals regional knowledge communities with distinct reference-overlap patterns. The United States and China occupy central positions and show the broadest connection profiles. Three regional clusters are visible: a North American cluster with dense shared bibliographic foundations, a European cluster with comparatively tight coupling that is consistent with coordinated quantum research programs, and an Asian cluster with moderate coupling that reflects an expanding but less consolidated regional knowledge base. This structure suggests that quantum-ITS research is globally visible, but its intellectual development remains regionally segmented.

4.2.2 Keyword Co-Occurrence and Thematic Structure

The keyword co-occurrence network for author-provided terms, using a minimum occurrence threshold of 5, yields 53 keywords from 1,086 unique terms and identifies four thematic clusters, as shown in Fig. 15. The *quantum optimization cluster* centers on quantum optimization, quantum annealing, QAOA [19, 20], and vehicle routing. It represents the dominant stream applying quantum algorithms to combinatorial transportation problems. The *quantum machine learning cluster* includes quantum machine learning, quantum neural networks, traffic prediction, and deep learning. This cluster captures studies that explore quantum-enhanced learning for transportation intelligence, although the methods remain heterogeneous and only weakly standardized. The *post-quantum and vehicular security cluster* groups post-quantum cryptography, lattice-based authentication, V2X communication, and cybersecurity. It forms a comparatively self-contained research community focused on connected-vehicle security. The *quantum communication and ITS application cluster* connects quantum communication, intelligent transportation systems, connected vehicles, autonomous vehicles, and smart cities. It functions as the

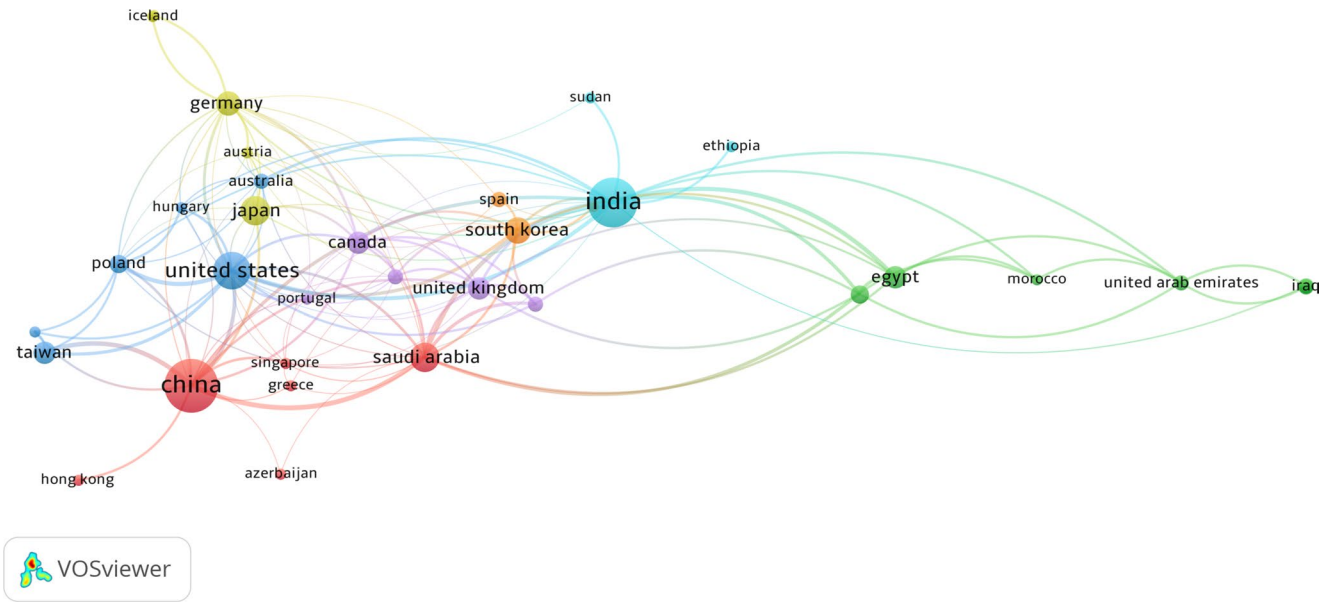


Fig. 14 Bibliographic coupling network among 39 countries

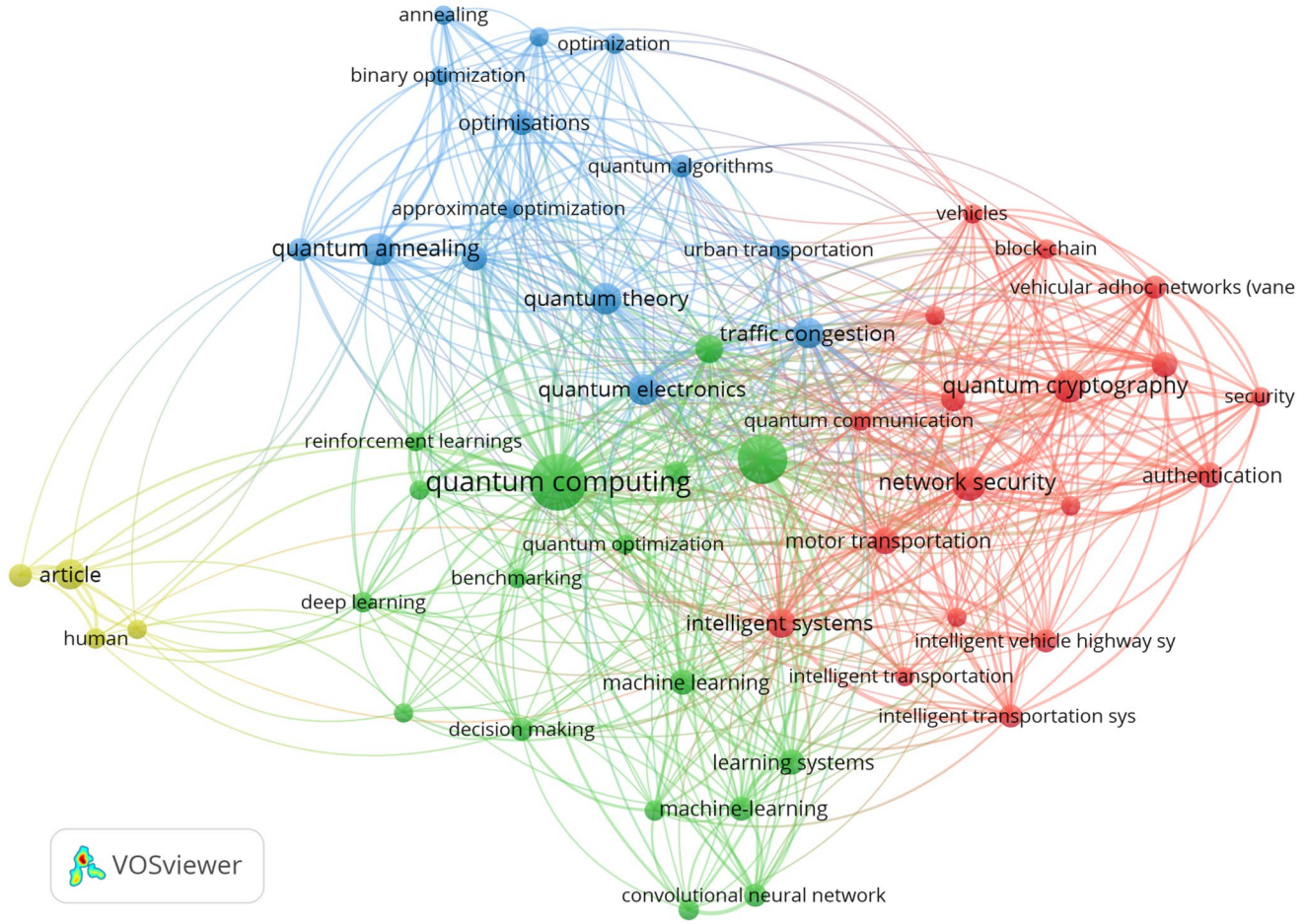


Fig. 15 Keyword co-occurrence network for all keywords with a minimum of 5 occurrences

domain bridge through which optimization, learning, and security themes are anchored to transportation problems.

The Scopus-indexed keyword network, presented in Fig. 16, confirms the four-cluster structure through standardized terminology while adding important temporal nuance. Terms such as NISQ [18], variational algorithms, and hybrid quantum-classical approaches appear more frequently from 2022 onward. This shift suggests that the field has begun to frame its claims more explicitly around current hardware constraints rather than idealized fault-tolerant quantum computing. Temporal keyword patterns show a clear transition. Publications from 2019 to 2020 were dominated by quantum annealing and routing optimization, reflecting early interest in D-Wave hardware [22]. From 2022 onward, machine-learning and neural-network terminology became more visible. More recent publications increasingly emphasize NISQ, variational, and hybrid formulations. This evolution indicates movement from hardware-specific demonstrations toward broader algorithmic experimentation, although the field still lacks stable evaluation standards.

4.2.3 Strategic Thematic Map

The strategic thematic map in Fig. 17 positions research themes according to centrality, which captures integration with other themes, and density, which captures internal coherence. The map classifies themes into four strategic quadrants following established bibliometric methodology [82].

Motor themes in the upper-right quadrant include quantum optimization algorithms and intelligent transportation systems as application contexts. These themes show both high internal development and strong cross-theme connectivity. They therefore represent the most structurally central area of the field. However, centrality should not be interpreted as deployment maturity. The prominence of this cluster reflects the stabilization of problem formulations, the growth of hardware demonstrations, and high citation visibility, but it does not by itself establish operational readiness or quantum advantage.

Basic themes in the lower-right quadrant include machine learning, traffic prediction, and autonomous vehicles. These themes have broad cross-theme relevance but weaker internal coherence. Their position reflects an active but unsettled

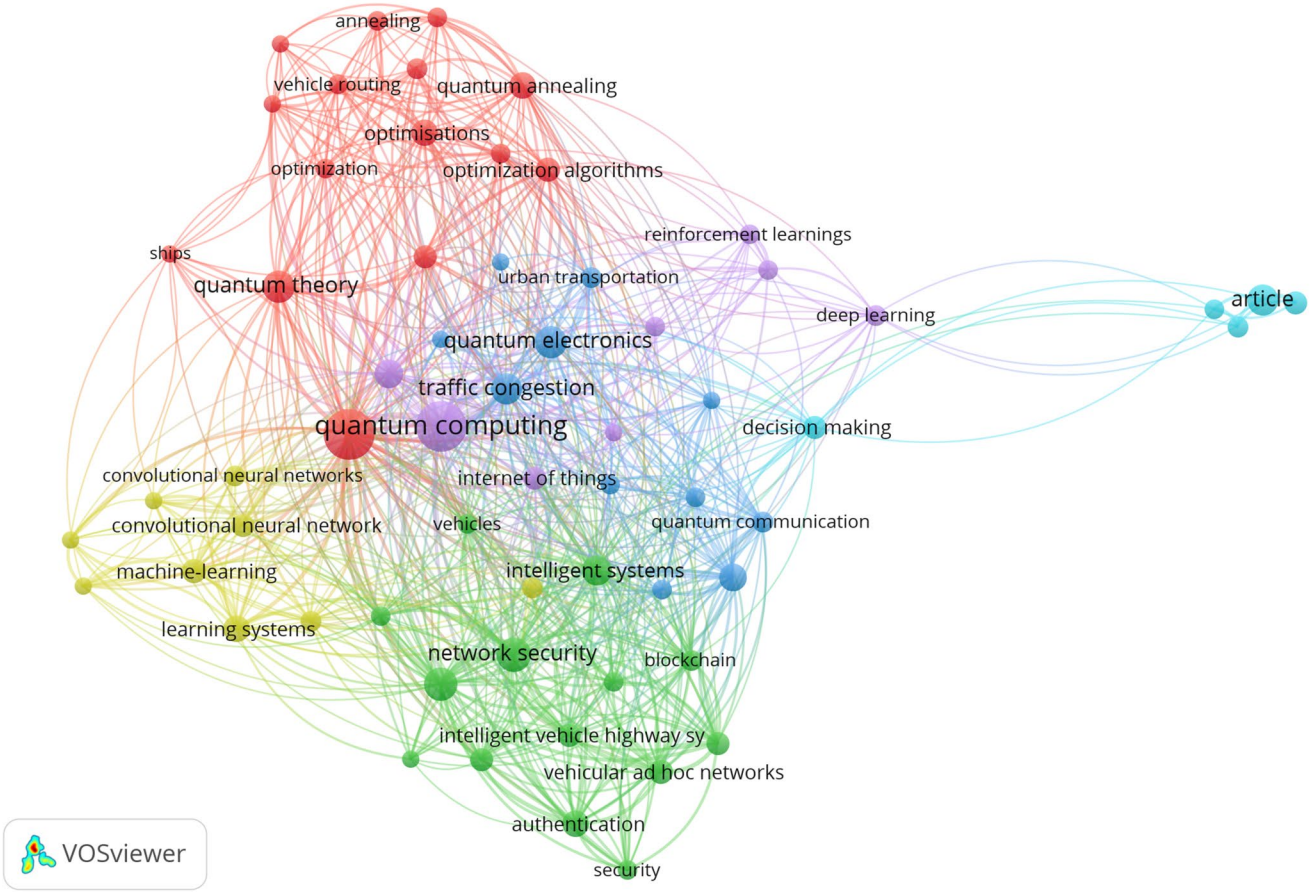


Fig. 16 Keyword co-occurrence network for Scopus-indexed keywords with minimum 4 occurrences

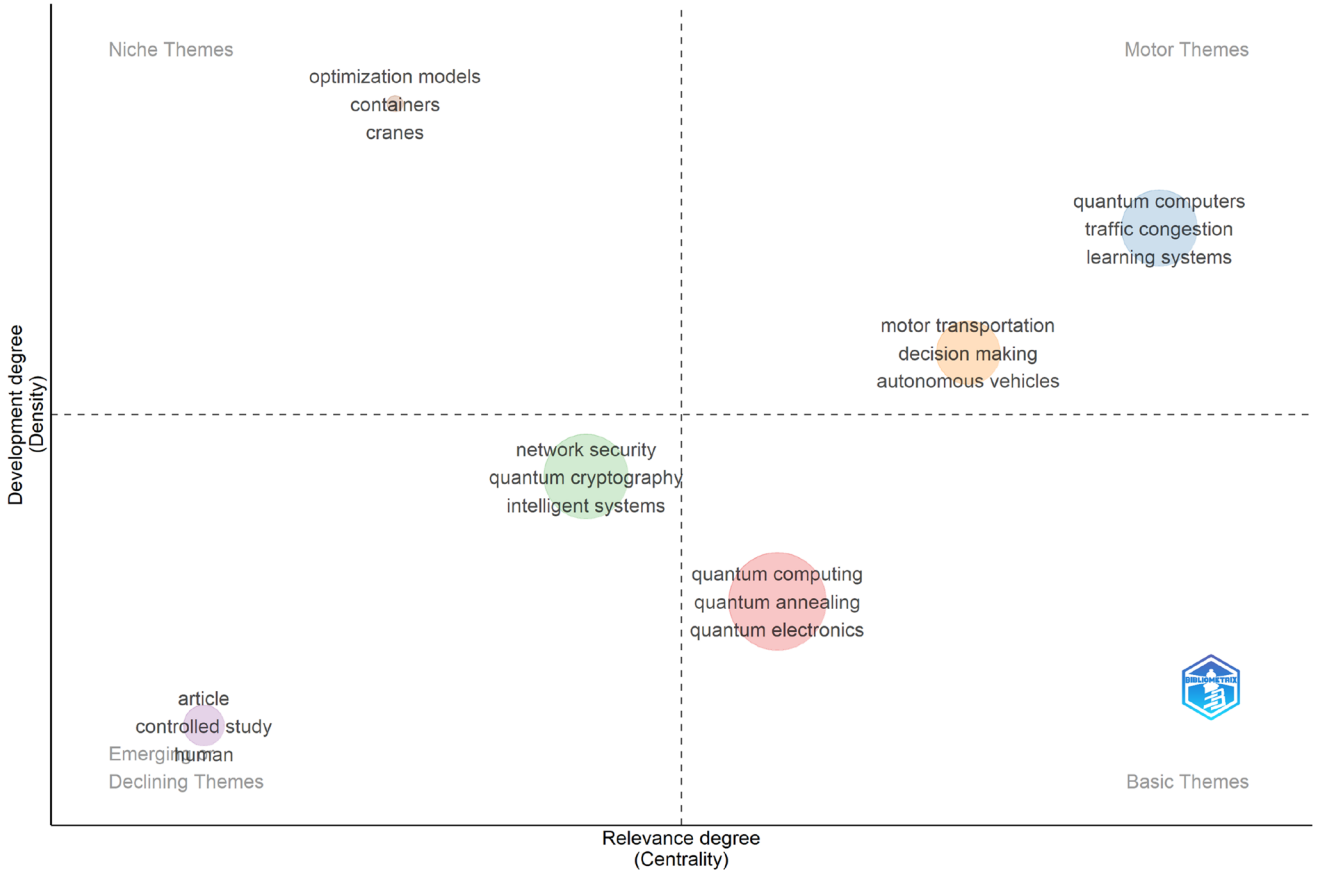


Fig. 17 Strategic thematic map classifying research themes by centrality and density

research stream in which architectures, datasets, evaluation metrics, and baselines vary substantially across studies. The absence of standardized benchmarks remains the main barrier preventing this cluster from reaching motor-theme status.

Emerging themes in the lower-left quadrant include quantum simulation, quantum sensing, and selected post-quantum security terms. Their low centrality and density indicate limited integration with the wider quantum-ITS literature. Post-quantum cryptography is an important exception. Its position reflects thematic isolation rather than technological immaturity, since its translational readiness is established more clearly in Sect. 4.3 through embedded and vehicular security studies [24, 25]. This distinction is important because bibliometric position and deployment readiness do not always move together.

Niche themes in the upper-left quadrant include quantum key distribution and V2X security. These topics exhibit high internal coherence but limited integration with optimization and machine-learning themes. This isolation suggests that the security community has developed a more specialized knowledge base, but it also limits methodological exchange with other quantum-ITS streams. As a result, the field remains thematically plural but weakly integrated.

Viewed developmentally, the optimization cluster is the most mature bibliometric cluster because it contains the earliest hardware demonstrations, the strongest citation density, and relatively stable QUBO/Ising terminology. The machine-learning and communication-security clusters are more emerging: both show recent keyword growth, but the former lacks stable benchmarks and the latter remains thematically isolated from the wider ITS optimization literature. This distinction between thematic maturity and technological readiness is essential; PQC is bibliometrically peripheral but practically more deployable than central optimization themes.

The Louvain community detection algorithm was applied under the standard modularity-maximization framework [75]. Sensitivity checks across lower and higher resolution settings supported the interpretability of the four-cluster structure without justifying a stronger parameter-exact stability claim. At lower resolution, optimization and machine-learning terminology tended to merge; at higher resolution, the optimization stream separated into hardware-executed annealing and quantum-inspired subcommunities. The

retained four-cluster interpretation therefore provides the most balanced summary of thematic structure while preserving the paradigm distinction used throughout the review.

4.2.4 Complementary Visualizations and Research Flow Analysis

The hierarchical treemap in Figure 18 confirms that research activity is concentrated around traffic signal control and vehicle routing within quantum optimization, traffic prediction and congestion forecasting within quantum machine learning, and quantum-resistant or quantum-security methods within vehicular communication security. In contrast, parking optimization, intersection management, autonomous-vehicle perception, and incident detection occupy relatively small areas despite their practical relevance. These patterns indicate that the field has prioritized mathematically convenient problem classes more strongly than the full range of operational ITS challenges.

The word cloud in Fig. 19 shows the coexistence of quantum computing vocabulary, including qubit, annealing, QAOA, and superposition, with transportation domain vocabulary, including intersection, congestion, route, and mobility. This pattern suggests substantive interdisciplinary engagement. However, term frequency alone cannot establish methodological integration. The stronger evidence of integration comes from the network and flow analyses, which show how quantum paradigms are being mapped onto specific transportation problems.

The Sankey diagram in Fig. 20 traces research flows from quantum computing paradigms through algorithmic techniques to ITS application domains. The strongest pathway links gate-based quantum computing through QAOA and variational algorithms to traffic optimization. A second major pathway connects quantum annealing through QUBO formulations to vehicle routing and traffic signal control, reflecting the influence of commercially accessible D-Wave systems. Quantum machine learning flows through quantum neural networks and kernel methods toward traffic prediction, but this pathway remains less mature and more methodologically diffuse. Quantum communication and post-quantum security flow narrowly toward V2X and

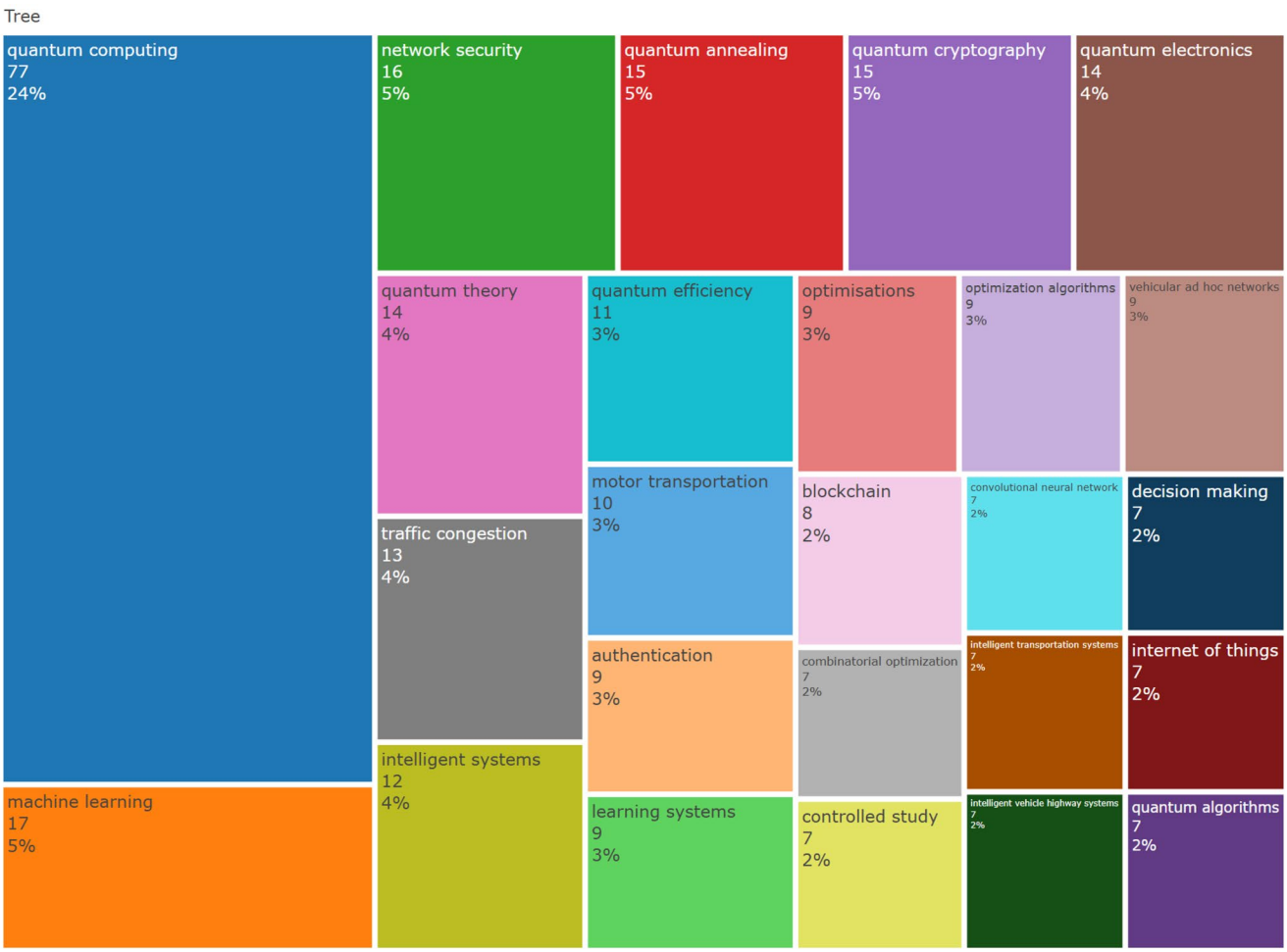
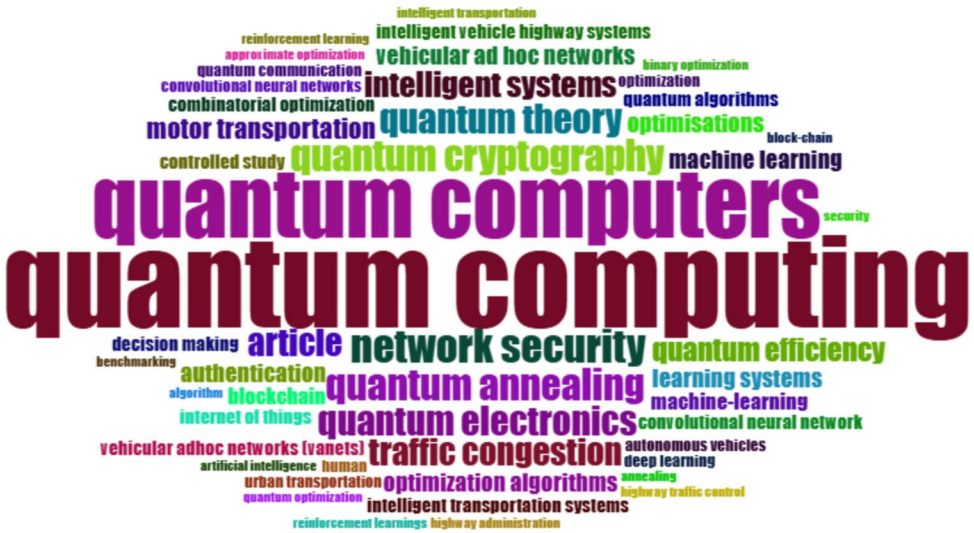


Fig. 18 Hierarchical treemap showing topic prevalence within the quantum-ITS corpus



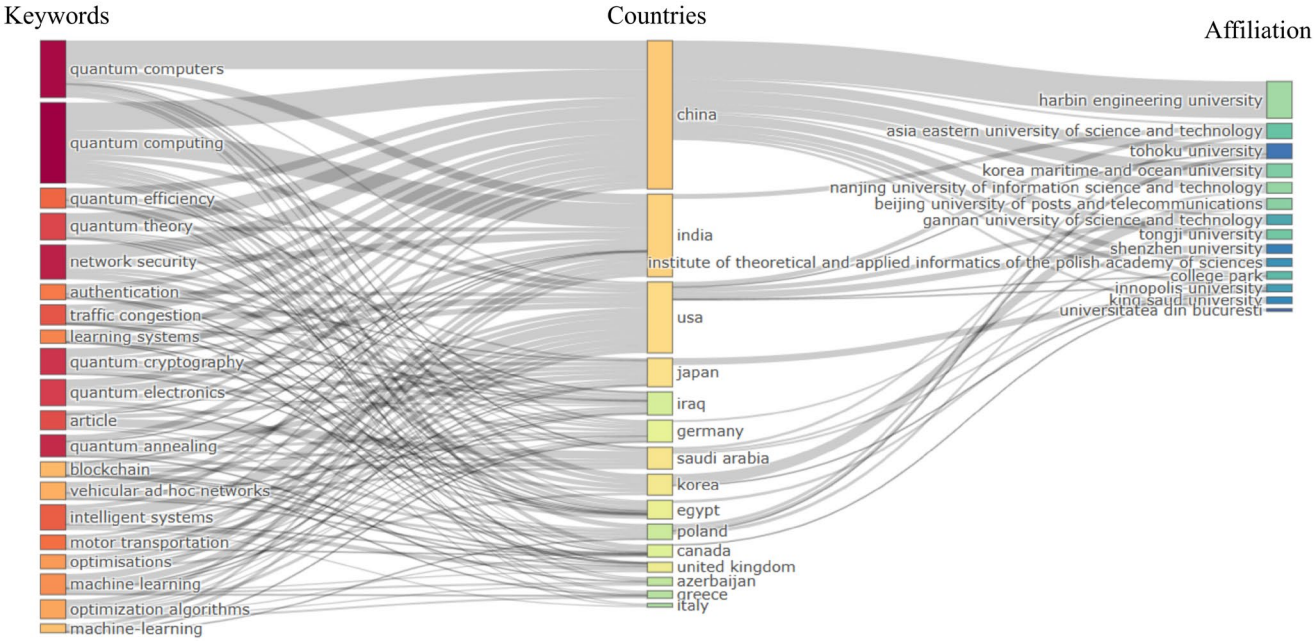


Fig. 19 Word cloud of frequent terms in titles, abstracts, and keywords

Fig. 20 Sankey diagram tracing flows from quantum paradigms to algorithms and ITS application domains

vehicular security, reinforcing the structural separation of the security stream from optimization and machine learning.

The Sankey structure also exposes important gaps. Within optimization, parking, demand management, and multimodal network design receive far less attention than traffic signal control and vehicle routing. Within machine learning, autonomous-vehicle perception and real-time incident detection remain underexplored despite their computational complexity and operational relevance. These thin or absent flows indicate that the current literature is shaped more by algorithmic tractability and platform availability than by a systematic mapping of quantum methods to high-priority transportation needs.

4.3 Qualitative Review and Critical Analysis

The critical synthesis presented in this subsection applies the four-dimension evaluation framework uniformly across all 27 reviewed publications, assessing each contribution against baseline fairness, problem scale realism, reproducibility, and deployment practicality. These dimensions were selected because they define the evidentiary conditions under which a quantum advantage claim can be considered credible rather than merely asserted. A favorable result against a weak classical baseline, evaluated on an unrealistically small problem instance, reported without sufficient implementation detail for replication, and disconnected

from the engineering constraints of operational deployment, does not constitute evidence of quantum advantage regardless of the performance metric reported. The analysis that follows applies this standard across six thematic domains, with Table 6 providing the structured cross-study reference for all 27 contributions.

4.3.1 Quantum Optimization for Traffic and Transportation Management

The ten quantum optimization studies in this review address the NP-hard combinatorial structure of transportation planning through three distinct approaches: quantum annealing on physical D-Wave hardware, gate-based QAOA on simulators, and quantum-inspired metaheuristics executing on classical processors. Across all three approaches, the central evaluative finding is uniform: no study demonstrates quantum computational advantage over state-of-the-art classical solvers under conditions reflecting operational transportation complexity. The reasons differ by paradigm, however, and understanding them separately is necessary for calibrating near-term investment.

Traffic flow optimization via quantum annealing. Four studies apply quantum annealing to urban traffic flow, with progressively elaborated formulations but a shared evidentiary ceiling: results are generated at problem scales and under baseline comparisons that prevent any confident conclusion about quantum advantage.

Neukart et al. [60] establish the foundational QUBO formulation for vehicle rerouting on D-Wave 2X, evaluated on 418 vehicles from the T-Drive Beijing dataset with 1,254 binary decision variables. Two limitations prevent this study from establishing quantum advantage. First, the authors report a minimum observed qbsolv runtime of 22 seconds, while noting that runtime varied because the D-Wave QPU was accessed as a shared cloud resource; consequently, the result demonstrates workflow feasibility rather than operational real-time readiness. Second, performance is benchmarked against random route assignment and historically observed routes rather than against commercial integer-programming solvers or specialized routing heuristics [34, 35]. The study is therefore best characterized as a proof-of-concept QUBO formulation and hybrid quantum–classical workflow, not as evidence of quantum computational advantage.

Ohzeki [83] addresses the constraint-handling problem inherent to D-Wave architectures through a Hubbard–Stratonovich transformation, applied to 350 vehicles across synthetic networks representing Sendai and Kyoto. The technique is technically sound, but iterative Lagrange-multiplier updates require numerous successive annealing cycles, substantially eroding the speed argument for quantum annealing. The quantum-annealing solution does improve on a naive shortest-path policy (for Sendai, roughly 830,000 versus 1,050,000 in cost), but its margin over classical Gibbs sampling within the same multiplier framework is only about two percent, too small to attribute to quantum computation without stronger statistical and state-of-the-art classical-solver comparisons.

Inoue et al. [84] scale the approach to 2,500 intersections on a 50×50 lattice, partitioned into 42 subproblems for D-Wave 2000Q. The scale is the largest hardware demonstration in this cluster and represents a genuine engineering contribution. Its external validity is nevertheless constrained by two structural choices: the periodic lattice topology with homogeneous turning probabilities has no counterpart in real urban networks, and the one-step-ahead objective function produces acknowledged oscillatory signal behavior. No comparison is provided against max-pressure controllers or deep reinforcement learning agents, both of which are relevant classical standards for this problem class.

Marchesin et al. [85] add density-weighted turning statistics to the QUBO formulation, reporting approximately 40% lower average Ising objective value on a 64-intersection network over 100 optimization cycles. The Ising objective is an internal optimization measure, not a traffic performance indicator such as average delay or throughput; the relationship between reduced Ising energy and operational traffic improvement is not validated against microsimulation. Pegasus topology embedding overhead further constrains feasible network size to well below metropolitan deployment scale.

Vehicle routing, network design, and combinatorial scheduling. Dixit and Niu [87] formulate the transport network design problem as a bilevel QUBO with user-equilibrium assignment and compare D-Wave quantum annealing with Tabu Search across benchmark networks. Pure quantum annealing is evaluated only on the smallest Sioux Falls network (76 links), where it produces solution quality comparable to Tabu Search and shorter reported computation times. For larger networks, the study uses D-Wave's hybrid solver rather than direct quantum annealing; this hybrid workflow combines classical heuristic search with quantum-module queries and therefore does not isolate the quantum contribution. The reported results are nevertheless important: on larger benchmark networks the hybrid solver reduces runtime relative to Tabu Search, with the largest reported speedup occurring on Hessen-Asym, and in several large-network cases the objective value is slightly better, by up to about 2%. The study therefore provides credible evidence that hybrid quantum-classical decomposition can be computationally useful for large TNDP instances, but it should not be interpreted as a standalone demonstration

Table 6 Literature matrix of the 27 studies included in the qualitative synthesis

Study	QC Paradigm	ITS Application	Method	Algorithm / Technique	Dataset / Scale	Key Metrics & Baselines	Maturity
Quantum Optimization for Traffic Management							
Neukart et al.[60]	QA D-Wave 2X	Traffic flow rerouting	Hybrid QA+classical	QUBO+*qbsolv* decomposition	T-Drive Beijing; 418 vehicles; 1,254 vars	Congestion ↓ vs random/original; 22 s min runtime	HW+qbsolv
Ohzeki [83]	QA D-Wave 2000Q	Traffic flow optimization	Constraint-handling transform	Hubbard–Stratonovich+Lagrange multipliers	Sendai & Kyoto; 350 vehicles; 3 routes/vehicle	Cost 830K vs 1050K shortest-path baseline	HW+iterative
Inoue et al.[84]	QA D-Wave 2000Q	Traffic signal control	Ising Hamiltonian global opt.	Ising flow-imbalance formulation+Metis partitioning	50×50 lattice; 2,500 signals; 42 partitions	$\bar{H}$ lower than local/SA; magnetization analysis	HW+partitioning
Marchesin et al.[85]	QA D-Wave Advantage	Traffic signal optimization	QUBO+parameter tuning	Density-based QUBO+Pegasus embedding	64 intersections; 196 lights; 100 cycles	∼40% Ising objective ↓ vs reference	HW+config study
Wang et al.[86]	QA-inspired simulation	Taxi-stand clustering	Quantum spectral+brain-inspired	QABICA MAX-CUT+selective attention	Xi'an & Chengdu; 45,331 pickups; 10×500 samples	42–83% bias ↓ vs k-means across cities	Quantum-inspired
Dixit & Niu [87]	QA D-Wave+hybrid	Transport network design	Bilevel QUBO+UE assignment	TNDP capacity-investment QUBO	Transportation Networks repo; 76–6,674 links	Pure QA faster on Sioux Falls; hybrid up to ∼ 11× faster than Tabu Search; ∼2% best objective gain on selected Hessen-Asym cases	HW+hybrid BQM
Azad et al.[88]	Gate QAOA Qiskit	Vehicle routing	Variational quantum algorithm	QAOA+COBYLA/SPSA optimizers	VRP(4,2)–(5,3); 12–20 qubits; 150 instances	No speedup vs CPLEX; p↑ improves quality	Qiskit simulation
Wang & Wang [89]	QI-QEA classical	Freight procurement bundling	Two-round CA bundling	QEA rotation-gate bundling optimization	Freight lane auction examples; random CA tests	QEA better than GA on small/mid cases; GA stronger on large cases	Quantum-inspired
Xia et al.[90]	QI-MQPSO	USV collision avoidance	Path planning+COLREGs	MQPSO Q-bit encoding+NOT-gate mutation	3 single-obstacle encounter cases; benchmarks	Lower cost than PSO/QBPSO; 149 vs 228 iters	MQPSO simulation
Haba et al.[91]	QA D-Wave Adv. 1.1	UAM routing & scheduling	MWIS QUBO+UTM simulator	MWIS interference graph+Pegasus embedding	Singapore airspace; 629 instances; ≤50 vars	Competitive with Gurobi only at small scales; no overall advantage	HW+UTM sim
Quantum Machine Learning for ITS Intelligence							
Harikrishnakumar & Nannapaneni [92]	Gate QBN Qiskit	Bike-sharing demand forecast	Hybrid ensemble+QBN	QBN+LSTM+GPR ensemble	NYC Citi Bike; 9.5 M records; 1,054 stations (ID 72)	RMSPE <2% QBN vs Netica BN; LSTM R^2=0.88–0.92	IBM Qiskit sim
Qu et al.[93]	Gate QGCNN TFQ	Traffic congestion prediction	Temporal-spatial QNN	TS-QGCNN Schrödinger+QGCNN layers	SiouxFalls dataset; 160–800 samples	Test MSE= 0.42; Acc= 0.88	TF Quantum

Table 6 (continued)

Study	QC Paradigm	ITS Application	Method	Algorithm / Technique	Dataset / Scale	Key Metrics & Baselines	Maturity
Bhatia & Sood [94]	QI-QCiO QDK	IoT sensor-space optimization	QCiO real-time acquisition	Q-bit parameters + q-gates OSS/SoI	90 WiSense highway nodes; 30,270 datasets	DA≈95%; DTE≈16.85 s vs 23.69 s DE	QDK sim
Bhatia et al.[95]	QI-QCIO ProjectQ	IoT temporal-space minimization	Energy-aware data acquisition	DoA/SoI + QCIO q-bit search	70 WiSense nodes; 1 km highway; 28,586 segments	Accuracy≈95.29%; delay≈15.69 s vs 26.69 s ACO	ProjectQ sim
Maritime & Aviation Applications							
Yang et al.[96]	QI-CQASBO	Port berth-tugboat-crane allocation	Multi-objective metaheuristic	CQASBO chaotic init + quantum revolving gates	Simulated 4 berths; 10 cranes; 6 tugs; 5–120 vessels	F=3224 avg; 20.32% vessel time ↓ vs E-B&QC	Quantum-inspired
Li et al.[97]	QI-CQASOA	Port berth-crane-truck + uncertainty	Green optimization metaheuristic	CQASOA tide windows + carbon + uncertainty	South China ports; large/small scale; robustness tests	5.53–14.19% F↓ vs SASS/COLSHADE/HHO	Quantum-inspired
Stollenwerk et al. [73]	QA D-Wave 2X/2000Q	Air-traffic conflict resolution	Strategic conflict-resolution QUBO	Departure-delay QUBO + decomposition	North Atlantic wind-optimal trajectories; 984 flights; embeddable subproblems	∼99% optimality probability within ∼1 s annealing time for hardest programmable subproblems; classical solver benchmarks	HW + decomposition
Lu et al.[98]	QRNG-assisted heuristic	Aircraft landing scheduling	Metaheuristic QRN + tunneling	QEWOA QRN init + tunneling jump + ABC	OR-Library ALP benchmarks; Friedman test	Best rank 2.38; mixed penalty ratios by instance	QRNG-assisted
Post-Quantum Security for Connected Vehicles							
Khalid et al.[99]	Lattice PQC	ITS end-to-end auth. + encrypt.	Protocol + SVO/AVISPA verification	RAVEN lattice discrete Gaussian	OMNeT++ 50-node scenarios; microbenchmarks	Delay≈0.22 ms; throughput↑50%; energy↓36%	OMNeT++ sim
Dharminder & Mishra [100]	Lattice SIS/ISIS	VANET auth. V2V/V2I	LCPPA + batch verification	Lattice signature with anonymous ID and traceability	Analytical protocol; security and cost comparison	ID privacy, traceability, and quantum resistance; no OBU/RSU test	Crypto protocol analysis
Prajapat et al.[101]	Lattice R-LWE/NTRU	VANET aggregate signature	IBAGS + QROM proof	Lattice identity-based aggregate signature	VANET model; desktop microbenchmarks	Lower signing/aggregation/verification cost vs baselines; desktop only	Lattice desktop
Wen et al.[102]	Lattice MLWE/MSIS	VANET revocable ring	Crypto scheme + ROM proof	LaRRS revocable-tag lattice PKE	Dilithium ref. + Python; ring sizes n=1–8	Sigs 10–62 KB; signing/verification scales with ring size	Lattice crypto
Zhu et al.[103]	PQC NTRUEncrypt	In-vehicle Ethernet key negotiation	Embedded benchmarking TLS-like	NTRUEncrypt vs RSA/ECDH	AURIX TC397 MCU; 1 Gb/s Ethernet; 1,000 runs	∼5 ms vs 330 ms ECDH; 7,655 ms RSA	Automotive MCU
Quantum-Enhanced Communication & Resource Allocation							
El Azzaoui et al.[104]	Gate QAOA + blockchain	Smart logistics routing + security	QAOA MaxCut + Hyperledger	QAOA + Hyperledger Fabric Curve25519	IBM Q Lab Qiskit + Hyperledger sim	91% route success; 10% cost↓; 6% carbon↓	Qiskit + blockchain

Table 6 (continued)

Study	QC Paradigm	ITS Application	Method	Algorithm / Technique	Dataset / Scale	Key Metrics & Baselines	Maturity
Chaudhary et al.[105]	Gate QFRL Torchquantum	V2X STAR-RIS resource allocation	Quantum federated RL	QNN + federated aggregation MDP	10 V2V pairs; 5 EVU pairs; convergence plots	20.5–46.7% improvement vs QFL/FRL/FL	Torchquantum sim
Survey & Vision Contributions							
Akhtar et al.[106]	Survey: QC/QML for 6G	6G vision V2X/AVs	Literature synthesis + taxonomy	Quantum comm./QML as 6G enablers	Conceptual; KPI targets (μs latency, 9-nines)	Requirements baseline (not empirical)	Conceptual survey
Wang et al.[28]	Survey: universal QC + QA	ITS path planning/operations	Technology categorization review	QAOA, D-Wave, quantum-inspired	Taxonomy + D-Wave toolchain	Application categories (not empirical)	Technology review

of quantum computational advantage over state-of-the-art classical optimization.

Azad et al. [88] provide the only direct QAOA-versus-CPLEX comparison in this cluster, and the result is unambiguous: CPLEX solves a small vehicle-routing instance in under one second, while QAOA on the Qiskit simulator requires substantially longer runtimes. All evaluations use simulated gate-based execution, meaning performance on physical NISQ hardware, where gate errors and decoherence would further degrade solution quality, is not characterized. Increasing circuit depth p improves solution quality but also extends runtime and increases exposure to decoherence, creating a trade-off that the simulation environment cannot resolve.

Three further studies in this cluster employ quantum-inspired algorithms that execute entirely on classical processors. Wang et al. [86] apply QABICA clustering to taxi-stand site selection; Wang and Wang [89] apply a quantum evolutionary algorithm to freight bundling in combinatorial auctions; and Xia et al. [90] develop a quantum-inspired particle swarm optimizer for unmanned surface vessel collision avoidance. Performance improvements in all three cases are attributable to metaheuristic design, not to evidence from physical quantum hardware; none involves quantum hardware, entanglement, or interference. Their classification as quantum-computing research is therefore a terminological artifact, and their results carry no implications for the deployment readiness of quantum hardware in transportation.

Haba et al. [91] formulate urban air mobility routing as a maximum weighted independent set problem, solved on D-Wave Advantage across synthetic urban traffic management scenarios. This is one of the more rigorous hardware-linked studies in the cluster because physical quantum hardware is used and Gurobi serves as the classical baseline. The study nevertheless does not demonstrate a general runtime or solution-quality advantage over Gurobi at operationally meaningful scales, and the evaluated instance sizes remain below the scale expected in deployed urban air mobility systems.

4.3.2 Quantum Machine Learning for ITS Intelligence

Four studies explore quantum-enhanced learning architectures for transportation intelligence. Despite theoretical predictions of advantage from exponential feature-space encoding and quantum kernel methods [23], three structural barriers collectively prevent empirically validated performance gains in this cluster: evaluation is dominated by simulators, problem scales are far smaller than real ITS deployments, and quantum components produce accuracy comparable to or below strong classical baselines.

Figure 21 synthesizes the reviewed literature into four linked layers: multimodal ITS data environments, classical preprocessing and feature preparation, hybrid quantum or quantum-inspired learning modules, and downstream ITS intelligence outputs. The visual structure makes explicit the central evaluative finding: quantum components are inserted into otherwise classical ITS pipelines, not replacing them. The field's contribution in this cluster lies in exploratory architectural formulation, not operationally validated performance gains.

Harikrishnakumar and Nannapaneni [92] apply quantum Bayesian networks to bike-sharing demand forecasting on 9.5 million Citi Bike trip records. The detailed evaluation focuses on station ID 72 within a 1,054-station dataset, all computations use Qiskit simulation rather than noisy quantum hardware, and no computational-time comparison between quantum and classical Bayesian-network inference is provided. The omission of runtime comparison is critical: it is the metric that would distinguish quantum advantage from accuracy parity with a more complex model architecture.

Qu et al. [93] propose a temporal-spatial quantum graph convolutional neural network for congestion prediction on the 24-node SiouxFalls benchmark. The authors explicitly acknowledge that their model does not outperform an optimally implemented classical graph-convolutional model, making this one of the few studies in the corpus that reports a negative comparative finding transparently. SiouxFalls is a standard pedagogical network; its 24-node structure is substantially smaller than real urban road networks, and generalization to realistic graph scales under NISQ constraints remains uncharacterized.

Bhatia and Sood [94] and Bhatia et al. [95] present two related quantum-inspired IoT sensing and data-acquisition frameworks on WiSense highway testbeds. The first optimizes an Optimal Sensor Space for real-time traffic data quality, whereas the second minimizes temporal IoT sensor space through Degree-of-Aptness and Sensors-of-Interest selection. Both report higher acquisition accuracy and lower acquisition delay than DE-, ACO-, PSO-, or GBiO-based baselines, depending on the study. However, both algorithms execute on classical processors or quantum simulators without physical quantum hardware; modeled efficiency gains are not equivalent to measured quantum energy savings; and the quantum-inspired encodings contribute no quantum mechanical effect. The reported improvements represent legitimate sensor-selection and data-acquisition advances, but they carry no implications for the viability of quantum machine learning in ITS.

4.3.3 Maritime and Aviation Quantum Applications

Four studies address quantum-algorithmic approaches to port terminal operations and air traffic management. As in the transportation optimization cluster, hardware-linked and quantum-inspired contributions require strict analytical separation: only studies using physical quantum hardware permit inference about quantum computational properties; quantum-inspired studies evidence only metaheuristic performance.

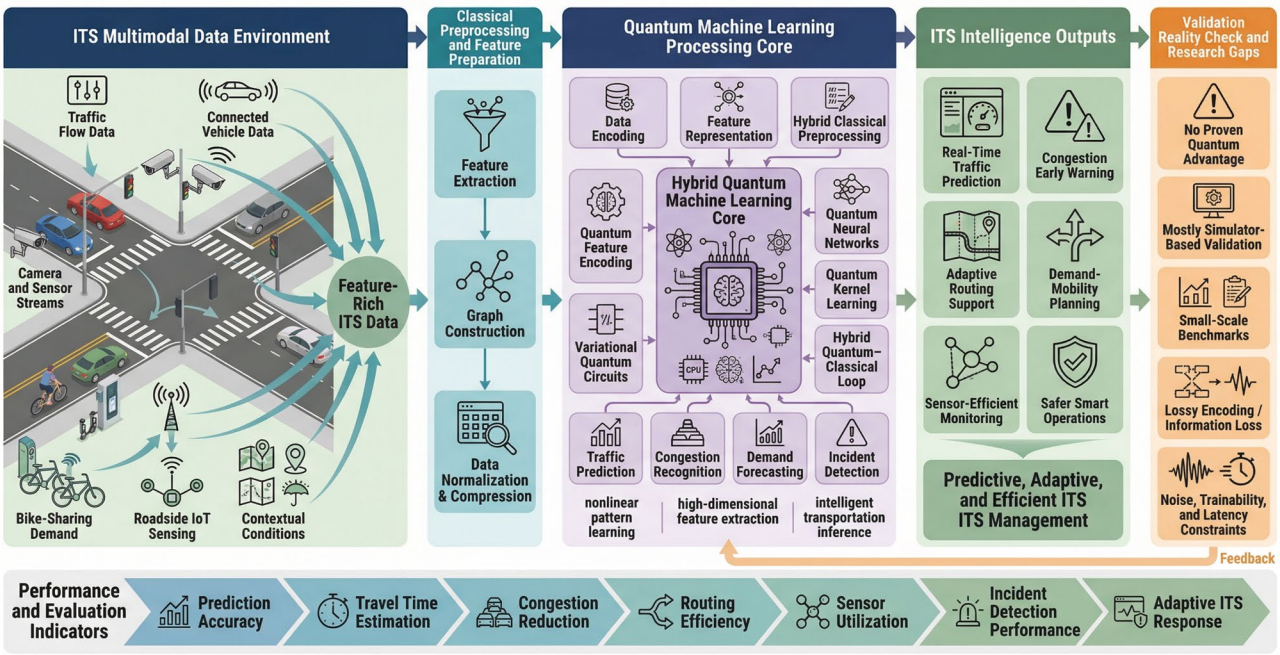


Fig. 21 Quantum machine-learning workflow for ITS intelligence and the main evidence constraints identified in the reviewed studies

Figure 22 maps the principal application pathways and distinguishes between hardware-linked optimization workflows and classical metaheuristics carrying quantum labels.

Yang et al. [96] and Li et al. [97] both apply quantum-inspired metaheuristics to port terminal scheduling, reporting improved objective values and vessel-time reductions relative to evolutionary or metaheuristic baselines. Neither study involves quantum hardware, entanglement, or interference, and neither provides computational-time comparisons against operations research methods such as branch-and-bound or commercial scheduling solvers. The reported improvements are plausible metaheuristic contributions, but characterizing them as quantum computing results is methodologically unjustified.

Stollenwerk et al. [73] apply D-Wave annealing to strategic air-traffic conflict resolution using wind-optimal North Atlantic trajectory data comprising 984 actual flight records. The conflict graph is decomposed into smaller embeddable subproblems and benchmarked using classical solvers and D-Wave 2X/2000Q quantum annealers. For reasonable modeling choices, the authors report that the most challenging programmable subproblems are solved to optimality with approximately 99% probability within about one second of annealing time. This makes the study one of the stronger hardware-linked demonstrations in the corpus, but the formulation remains a simplified departure-delay-based conflict-resolution model and does not establish end-to-end operational quantum advantage for full-scale air-traffic management. Lu et al. [98] label their aircraft landing scheduling algorithm quantum-enhanced, but the quantum contribution is limited to quantum random number initialization and a tunneling-inspired heuristic; the optimization logic is otherwise classical and therefore does not constitute demonstrated quantum optimization.

4.3.4 Post-Quantum Cryptography for Secure ITS Communications

Post-quantum cryptography is the only domain in this review where the distinction between theoretical promise and near-term deployment relevance substantially narrows. Five studies examine lattice-based cryptographic schemes for vehicular authentication, and the cluster is characterized by provable security reductions, simulation or hardware-grounded performance evaluation, and direct alignment with the wider post-quantum migration agenda [24, 25]. The critical challenges that remain are primarily engineering problems, not fundamental cryptographic obstacles.

Figure 23 illustrates the migration pathway from quantum-vulnerable legacy infrastructure through hybrid transition to PQC-ready deployment across V2X communication, trust management, and in-vehicle security components.

Khalid et al. [99] demonstrate RAVEN, a lattice-based V2X authentication protocol achieving 0.22 ms authentication latency in OMNeT++ simulations of 50-node vehicular network scenarios, with reported throughput and energy improvements relative to baseline schemes. The results are operationally promising, but validation remains confined to

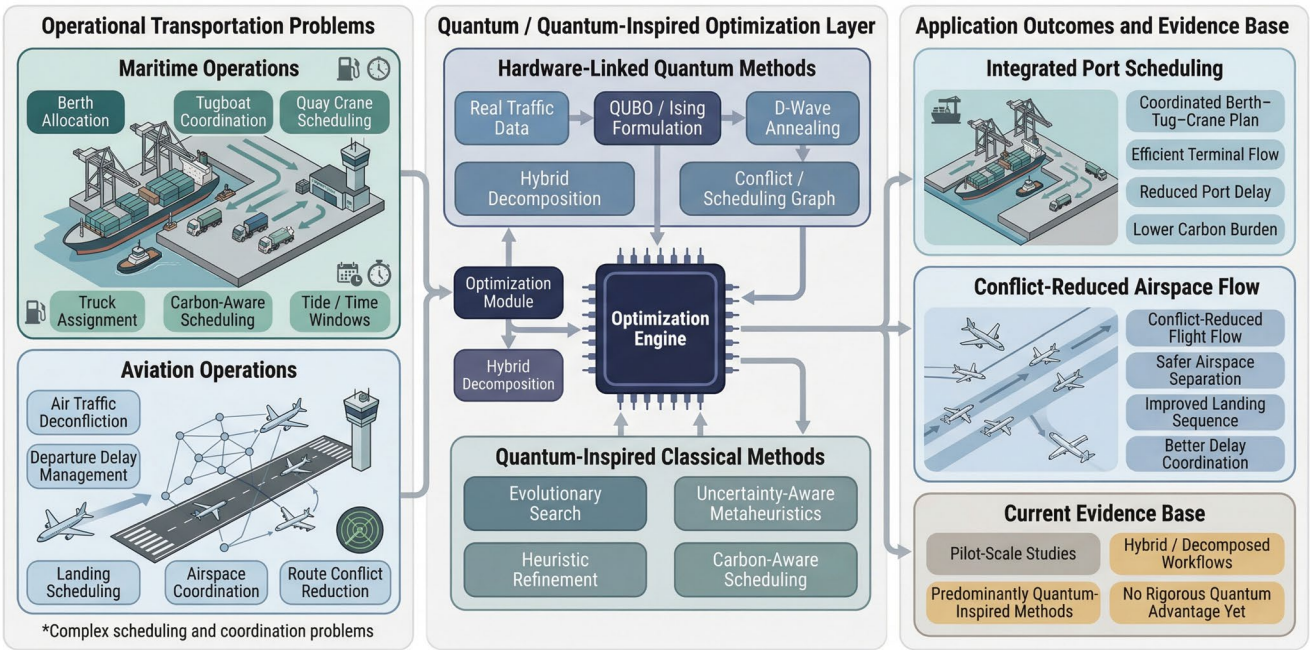


Fig. 22 Maritime and aviation quantum-application pathways, distinguishing hardware-linked workflows from quantum-inspired classical metaheuristics

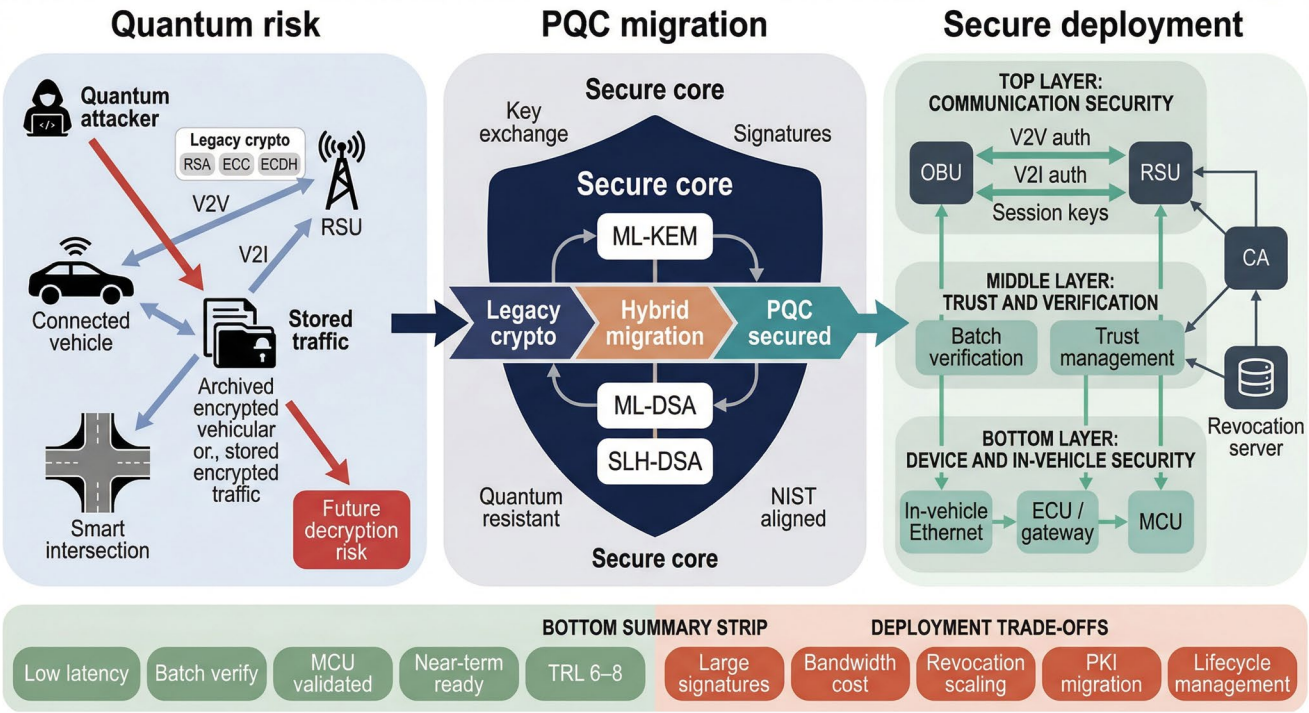


Fig. 23 Post-quantum cryptography migration pathway for secure ITS communications

network simulation without implementation on automotive-grade embedded processors, and the revocation mechanism essential for scalable vehicular PKI remains an open engineering concern.

Dharminder and Mishra [100] ground their LCPPA V2V/V2I authentication protocol in SIS and ISIS lattice hardness, combining anonymous identities, traceability through a trusted authority, and batch verification equations for high-volume vehicular messages. Its practical limitation is evidential rather than conceptual: the evaluation is based on formal security arguments and analytic communication, storage, and computation cost comparisons, not on implementation over representative on-board units, roadside units, or mobile vehicular network traces.

Prajapat et al. [101] and Wen et al. [102] extend this line toward lattice-based aggregate and revocable authentication. The first emphasizes identity-based aggregate signatures and batch verification for dense VANET message streams, whereas the second introduces a revocable ring signature that enables trusted-authority-controlled identity opening. In both cases, the security functionality is stronger than in many pre-quantum schemes, but communication size and verification cost remain sensitive to aggregation or ring-size choices, and neither study reports an end-to-end field trial on deployed vehicular communication infrastructure.

Zhu et al. [103] provides the strongest embedded-system deployment evidence in the reviewed corpus: NTRUEncrypt on the AURIX TC397 automotive microcontroller achieves key negotiation in approximately 5 ms, a 66-fold improvement over ECDH and an approximately 1,500-fold improvement over RSA at the evaluated security level, validated across 1,000 repeated trials. This result supports the computational feasibility of lattice-based post-quantum key establishment on automotive-grade embedded hardware. The evaluation addresses session key negotiation only, however, and does not characterize certificate management, trust anchors, revocation infrastructure, side-channel hardening, or full-stack automotive PKI replacement. These remaining gaps define the engineering roadmap.

4.3.5 Quantum-Enhanced Communication and Resource Optimization

El Azzaoui et al. [104] integrate QAOA-based route optimization with a Hyperledger Fabric blockchain for secure logistics coordination, reporting 91% route success, 10% cost reduction, and 6% carbon-footprint reduction in a simulation environment. The core quantum optimization problem is a very small Max-Cut instance, which is trivial for conventional solvers at the reported scale; the reported system-level performance improvements are therefore more plausibly attributable to the routing and blockchain architecture than to quantum computation. The QAOA component is implemented through Qiskit simulation, and the blockchain integration, while novel, is analytically independent of the quantum component.

Chaudhary et al. [105] propose quantum federated reinforcement learning for STAR-RIS-enabled V2X resource allocation, reporting 20.5–46.7% improvements over baseline learning methods. The improvement range is substantial, but all evaluations use a Torchquantum simulation environment on a scenario of 10 V2V pairs and 5 EVU pairs, and no ablation distinguishes whether gains arise from the quantum neural-network architecture, the federated aggregation strategy, or the reinforcement-learning design. Without this decomposition, attributing the performance gain to the quantum component is analytically unsupported.

4.3.6 Cross-Cutting Synthesis: Patterns, Gaps, and Research Priorities

Examined as a corpus, the 27 reviewed studies reveal systematic patterns that characterize the current state of quantum-ITS research. The first group of findings concerns the field's structure; the second identifies methodological deficiencies that limit the evidentiary value of reported results.

Absence of rigorous quantum advantage demonstration. This is the central finding of the qualitative synthesis. No study in the reviewed corpus provides a statistically robust speedup or empirically confirmed performance superiority over state-of-the-art classical algorithms under controlled experimental conditions. Reported improvements reflect one or more of four identifiable confounds: evaluation against weak baselines rather than commercial solvers such as CPLEX, Gurobi, or specialized routing heuristics [34, 35]; omission of runtime comparisons necessary to distinguish solution quality from computational efficiency claims; evaluation on instances solvable trivially by classical methods; and hybrid classical–quantum workflows where improvements may derive from the classical component. Post-quantum cryptography constitutes the sole near-term exception, for reasons analyzed separately above.

Implementation maturity bifurcation. A pronounced structural split exists between hardware-executed quantum studies and quantum-inspired classical algorithms. Within the 27-study qualitative sample, quantum annealing dominates hardware-based demonstrations (42%), gate-based quantum computing accounts for 23%, and quantum-inspired classical algorithms represent 35% of publications labeled as quantum-ITS research. This split matters because the categories have different implications for technology readiness: improvements from quantum-inspired algorithms say nothing about the operational potential of quantum hardware. Methodological rigor is also uneven: 63% of studies benchmark against generic metaheuristics rather than state-of-the-art solvers, only 31% report end-to-end solving times, and fewer than 15% employ formal hypothesis testing to validate performance claims.

Scale realism deficiency. The evaluated problem scales diverge systematically from operational ITS requirements. Traffic optimization studies address hundreds of vehicles or idealized lattice networks; metropolitan systems involve millions of vehicles and tens of thousands of intersections. Vehicle routing studies address small customer sets; operational logistics commonly involves dozens to hundreds of stops. Maritime and aviation studies examine tens of simultaneous agents; real terminals and airspaces involve hundreds to thousands. This scale gap is not merely a limitation of current hardware: it defines the evidentiary threshold that quantum methods must clear before advantage claims are meaningful.

Evaluation methodology heterogeneity. Disparate datasets, metrics, and baseline selections prevent cross-study comparison and foreclose meta-analysis. Standardized benchmark suites with public instances, reference classical solver performance profiles, and agreed metric definitions are a prerequisite for determining which quantum approaches, if any, outperform classical alternatives for specific ITS problem classes.

Noise and decoherence impacts undercharacterized. Most quantum optimization and machine-learning studies evaluate algorithms on simulators or omit systematic noise characterization and error-mitigation analysis. NISQ gate-based devices carry error and coherence constraints that limit achievable circuit depth [56, 57]. Simulator results cannot be assumed to transfer to physical hardware without noise analysis; studies that omit this analysis contribute exploratory formulations, not performance evidence.

System-level integration absent. All reviewed studies address isolated ITS subproblems without examining the functional coupling characteristic of real transportation systems, in which signal timing decisions alter route choice behavior, route choices redistribute link flows, demand forecasts drive predictive control, and incidents trigger coordinated responses. A quantum algorithm that optimizes a single isolated subproblem without accounting for these feedbacks cannot be evaluated for operational value.

Economic and sustainability analyses absent. No reviewed study provides a cost-benefit analysis comparing quantum infrastructure investment against performance improvements relative to classical alternatives. Quantum cloud processing on D-Wave or IBM Quantum hardware incurs access costs and scheduling overhead that are not characterized in any reviewed study, nor are they compared against classical HPC costs for equivalent transportation problem scales. Cryogenic cooling requirements for superconducting quantum processors impose a continuous energy burden that hybrid quantum–classical workflows inherit; this net energy cost is similarly uncharacterized. These omissions are not peripheral: without economic and

Table 7 Structured comparison of quantum advantage claims across quantum-ITS paradigms

Paradigm	Claimed advantage	Critical limitations	Verdict
Quantum annealing (D-Wave)	Faster combinatorial optimization for traffic signal control, vehicle routing, and network design	Weak classical baselines; problem sizes orders of magnitude below metropolitan scale; cloud-queuing latency; hybrid workflows embed classical logic whose quantum contribution cannot be isolated	No demonstrated advantage; TRL 3–4
Gate-based QAOA	Improved solution quality for vehicle routing and logistics optimization	QAOA remains outperformed by CPLEX on tested VRP instances; simulator evaluations do not transfer directly to physical NISQ hardware; circuit-depth and trainability constraints persist	No demonstrated advantage; TRL 2–3
Quantum machine learning	Enhanced prediction accuracy for traffic demand, congestion prediction, transport sensing, and resource-allocation tasks	Accuracy is comparable to or below strong classical baselines; data encoding, measurement overhead, and barren-plateau risks remain unresolved	No demonstrated advantage; TRL 2–3
Post-quantum cryptography	Quantum-resistant authentication and key exchange for V2X, VANET, and in-vehicle communication	Larger signatures and revocation scalability remain engineering challenges; full-stack ITS PKI migration is not yet demonstrated	Comparatively deployment-ready; TRL 6–8

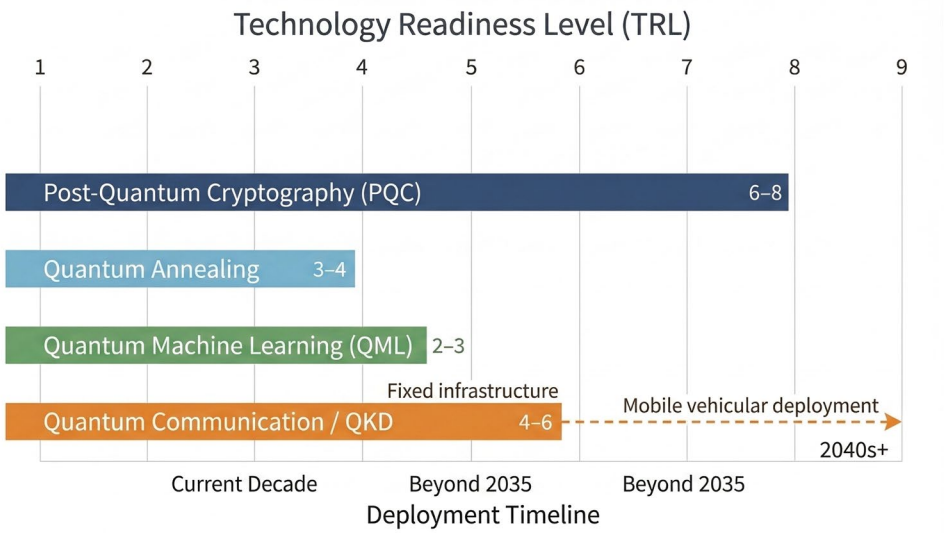


Fig. 24 Technology readiness level assessment across four quantum-ITS paradigms

sustainability analysis, a quantum-ITS deployment recommendation cannot be made responsibly.

Reproducibility and open-science limitations. Source code is publicly available in fewer than 30% of reviewed studies. Proprietary datasets, incomplete hyperparameter specifications, unreported random seeds, and underspecified hardware configurations collectively impede independent verification. A field that cannot reproduce its own results cannot accumulate evidence systematically.

Post-quantum cryptography as the near-term deployment priority. In contrast to every other cluster, PQC provides provable security reductions to well-studied lattice hardness problems, authentication latencies competitive with current elliptic-curve standards in selected vehicular studies, and automotive hardware validation in the in-vehicle Ethernet case [24, 25, 103, 107]. The harvest-now-decrypt-later threat to currently deployed V2X infrastructure is operationally relevant [13]. Remaining challenges, including signature size and revocation scalability, are engineering problems with plausible solution pathways, not fundamental cryptographic barriers. PQC is therefore the clearest near-term deployment priority in the quantum-ITS landscape.

Structured comparison of quantum advantage claims. Table 7 consolidates the quantum advantage assessment across four major paradigms, making explicit the evidentiary basis, specific methodological limitations, and maturity verdict for each.

Technology readiness and deployment timeline assessment. Quantum-ITS applications span a wide spectrum of technology readiness, summarized in Fig. 24. Post-quantum cryptography occupies TRL 6–8, with deployment feasible and urgent within the current decade. Quantum annealing for combinatorial optimization occupies TRL 3–4, with progress contingent on hardware scaling, stronger classical benchmarking, and credible advantage evidence at operational problem scales. Quantum machine learning occupies TRL 2–3, with deployment timelines extending beyond 2035 and conditional on resolving barren-plateau and data-encoding challenges [56, 57]. Quantum communication occupies TRL 4–6 for fixed or well-controlled infrastructure

links, but vehicular mobile deployment remains substantially less mature than PQC and extends further into the long-term research horizon.

Taken together, these readiness levels imply a clear ordering of evidential maturity within the corpus: post-quantum cryptography is the only domain at a deployment-ready level, quantum annealing sits at a pilot-research level contingent on commercial-solver benchmarking, quantum machine learning remains exploratory, and the quantum-inspired classical methods in the corpus carry no implications for quantum-hardware readiness. The corresponding stakeholder and investment recommendations are developed in Sects. 5 and 6.

5 Discussion

This section interprets the bibliometric and qualitative findings rather than repeating the Results section. It synthesizes the four research questions, positions the evidence against technology readiness and deployment timelines, identifies opportunity domains and persistent barriers, relates the findings to the broader quantum computing landscape, assesses Sustainable Development Goal (SDG) alignment, states review limitations, and provides stakeholder recommendations.

5.1 Addressing the Research Questions

5.1.1 RQ1: Publication Trends, Geographic Distribution, and Research Networks

The 108-publication corpus from 2014 to 2025 indicates rapid field formation, but publication growth should not be equated with scientific maturity. The exponential fit reported in Sect. 4 ($R^2 = 0.982$) confirms that output growth is systematic rather than episodic; it does not, by itself, show that quantum-ITS methods have become operationally reliable. The more important interpretation is that the field is expanding faster in visibility than in evidential quality. This interpretation is supported by three internal signals from the qualitative synthesis: 63% of studies use generic metaheuristics rather than state-of-the-art classical solvers as baselines; 35% of publications labeled as quantum-ITS research are quantum-inspired classical metaheuristics with no quantum hardware involvement; and fewer than 15% apply formal hypothesis testing. Thus, the observed growth reflects a combination of genuine interdisciplinary progress, wider access to cloud quantum platforms, and broadened use of quantum terminology.

The geographic pattern also has a dual meaning. The United States, China, India, and the United Kingdom account for most global output, with the European Union contributing through several distributed national clusters. These patterns are consistent with broader national and supranational quantum investment programs [77, 78]. However, concentration of output does not imply coordinated deployment capacity. The co-authorship structure remains fragmented, with many small research groups and limited cross-cluster knowledge flow. More critically, transportation agencies and infrastructure operators appear as direct contributors in fewer than 2% of publications. This is not merely a participation statistic; it explains why many studies use simplified networks, synthetic instances, and evaluation metrics that do not reflect procurement, certification, latency, liability, maintenance, and interoperability constraints faced by real transportation systems. The translational gap is therefore institutional as much as technical.

5.1.2 RQ2: Thematic Clusters, Application Domains, and Intellectual Structure

The field is organized around four recurring themes: quantum optimization, quantum machine learning, post-quantum cryptography, and quantum communication. The optimization cluster is bibliometrically central because traffic signal control, vehicle routing, network design, and scheduling map naturally to QUBO, Ising, and variational formulations. Yet bibliometric centrality should not be read as deployment readiness. Optimization studies have generated the largest body of proof-of-concept work, but they remain constrained by weak baselines, small problem scales, and uncertain quantum contribution in hybrid workflows [60, 84, 87].

Quantum machine learning is more exploratory. Studies on bike-sharing demand, traffic congestion prediction, and quantum graph neural architectures show methodological creativity, but they do not establish consistent superiority over classical deep learning or graph neural-network baselines [92, 93]. Post-quantum cryptography has the opposite profile: it is less central in the science maps but substantially more mature in deployment terms because it runs on classical hardware and responds to a present cryptographic migration problem [24–26]. Quantum communication and QKD form a coherent but comparatively isolated security theme; their strongest near-term case lies in fixed infrastructure links rather than mobile vehicular deployment [61]. This mismatch between bibliometric prominence and technology readiness is one of the main conceptual findings of the review.

5.1.3 RQ3: Quantum Techniques, Performance Claims, and Advantage Validation

The qualitative synthesis does not support a claim of demonstrated quantum advantage for ITS. In this review, a credible advantage claim requires more than successful quantum formulation: it requires comparison with strong classical baselines, operationally meaningful problem scales, reproducible implementation details, and statistically defensible performance evidence. Across the 27 representative studies, this threshold is not met.

For quantum annealing, D-Wave-based studies demonstrate that selected transportation problems can be encoded and solved on quantum annealing hardware, but they do not show robust superiority over optimized classical methods. Early traffic-flow and signal-control studies established the feasibility of QUBO or Ising formulations, while later network-design and air-traffic applications expanded the problem range [60, 73, 84, 87]. The evidentiary problem is that solution quality and runtime are often compared against weak or incomplete baselines, and hybrid workflows make it difficult to isolate whether any observed gain comes from the quantum processor or the classical decomposition strategy.

Gate-based QAOA remains at an even earlier deployment stage. Vehicle-routing studies demonstrate formal mapping and simulator-level feasibility, but tested instances remain far below operational logistics scale, and reported runtimes do not outperform CPLEX on the evaluated cases [88]. Quantum machine-learning studies face a different barrier: even when reported accuracy is competitive, current evidence does not separate quantum representational benefit from classical model design, feature engineering, or dataset-specific effects. Barren plateaus, noise sensitivity, measurement overhead, and data-encoding costs remain unresolved constraints for transportation-scale learning [37, 56, 57].

Post-quantum cryptography is the principal exception, but it should be interpreted correctly. PQC is not evidence that quantum computers improve ITS performance; it is a classical security response to quantum-enabled attacks. Its importance lies in deployment readiness and threat urgency. Shor's algorithm establishes the long-term vulnerability of RSA and elliptic-curve cryptography [12], and the harvest-now-decrypt-later model makes migration planning a current issue for safety-critical V2X and in-vehicle communication [13]. In the reviewed corpus, vehicular PQC protocols and in-vehicle Ethernet evaluations provide the strongest embedded-system deployment evidence for near-term quantum-related ITS security [99, 103]; however, full-stack certificate management, revocation, interoperability, and side-channel hardening remain unresolved.

5.1.4 RQ4: Research Gaps and Future Directions

Four gaps define the research agenda. First, no current study validates quantum methods at operational metropolitan scale. Future work must therefore focus on decomposition strategies that preserve solution quality while scaling beyond toy routing, lattice, and small-network instances. Second, the field lacks standardized benchmark suites that combine public transportation instances, reference classical solver profiles, end-to-end runtime reporting, and statistical validation protocols [58]. Without such benchmarks, individual studies cannot accumulate into a reliable evidence base. Third, noise characterization, error mitigation, and hardware-transfer analysis are still underreported, especially in simulator-based optimization and machine-learning work. Fourth, cost-benefit and lifecycle analysis are almost absent, even though public transportation agencies cannot justify quantum investment without transparent comparison against classical high-performance computing, commercial optimization software, and existing ITS platforms.

5.2 Maturity Assessment and Deployment Timelines

Technology readiness across quantum-ITS is highly uneven. The TRL spectrum summarized in Figure 24 should be read as an evidence-based maturity assessment, not as formal industrial certification.

Post-quantum cryptography occupies TRL 6–8. It is implementable on existing classical and embedded hardware, directly addresses the quantum threat to public-key cryptography, and is supported by a growing migration literature [24–26]. For transportation, the realistic timeline is the current decade: agencies, OEMs, and infrastructure operators should begin PQC transition planning now, with phased V2X and in-vehicle pilots before fleet-wide migration.

Quantum annealing for optimization occupies TRL 3–4. Laboratory-scale and cloud-hardware demonstrations exist, but quantum advantage over state-of-the-art classical solvers remains unproven. Its most plausible pathway is conditional medium- to long-term use in offline planning problems where latency is less critical, including strategic network design, robust timetable planning, and multi-objective logistics optimization. Operational deployment before the 2030–2040 horizon would require a clear demonstration of advantage on realistic instances and a defensible total-cost comparison.

Gate-based QAOA is also a low-maturity optimization pathway, at TRL 2–3: current vehicle-routing evidence remains simulator-heavy, small-scale, and outperformed by strong classical benchmarks on the tested instances. QAOA should therefore be considered a research direction for

transportation optimization, not a deployment-ready planning tool.

Quantum machine learning occupies TRL 2–3. Current studies remain exploratory, and the field lacks evidence that QML can outperform well-tuned classical models on transportation-scale datasets. Given unresolved trainability, encoding, noise, and measurement challenges, QML deployment in ITS should be treated as a beyond-2035 possibility with high uncertainty rather than a near-or medium-term investment priority.

Quantum communication and QKD occupy TRL 4–6 for fixed or well-controlled infrastructure links, but vehicular mobile QKD is much less mature. Range, alignment, miniaturization, cost, and network-integration constraints make mobile vehicular deployment a long-term prospect beyond 2040 in the absence of major technological breakthroughs [61]. For the current decade, PQC is therefore the more practical security pathway for V2X and in-vehicle communication.

5.3 Strategic Opportunities and Persistent Barriers

5.3.1 Opportunity Domains

The strongest near-term opportunity is PQC migration for connected and automated transportation infrastructure. This opportunity is operationally important because V2X authentication, in-vehicle Ethernet, fleet communication, and roadside infrastructure security rely on cryptographic foundations that must remain trustworthy throughout long vehicle and infrastructure lifecycles. Concrete priorities include embedded implementation studies, aggregate signature design for bandwidth-constrained V2X channels, certificate and revocation scalability, hybrid transition protocols, and interoperability testing across OEMs, suppliers, and infrastructure operators.

A second opportunity lies in quantum optimization for offline and planning-oriented transportation problems. Quantum annealing and QAOA are not credible substitutes for real-time signal control or collision-avoidance decision loops under current hardware constraints, but they may become useful for planning tasks that tolerate longer solving times. Candidate domains include network design, depot and charging infrastructure placement, long-horizon timetabling, freight consolidation, and multi-objective logistics. The correct research posture is controlled experimentation with real data and strong classical baselines, not premature deployment framing.

Quantum sensing is a longer-term adjacent opportunity rather than a central finding of the present corpus. Atomic-clock synchronization, quantum inertial navigation, and single-photon lidar could eventually support positioning, timing, and perception functions in ITS, but these technologies face cost, ruggedization, and automotive-grade integration barriers [108]. They should be tracked in the research roadmap without being conflated with near-term quantum computing deployment.

5.3.2 Persistent Barriers

The scale-realism gap is the dominant barrier. Current demonstrations remain far below metropolitan traffic networks, full-scale logistics fleets, and multi-operator infrastructure environments. This gap cannot be solved by adding larger toy examples alone; it requires problem decompositions that preserve solution quality, benchmark instances reflecting operational constraints, and comparison against commercial and domain-specialized classical solvers.

The second barrier is methodological fragmentation. Studies use different datasets, baselines, metrics, runtime definitions, hardware configurations, and reporting standards. This prevents meta-analysis and weakens cumulative learning. A field-wide benchmark suite, with public instances, reference CPLEX/Gurobi and specialized heuristic profiles, statistical testing requirements, and mandatory end-to-end runtime reporting, is a prerequisite for credible progress.

The third barrier is terminological inflation. Quantum-inspired classical metaheuristics may be useful transportation algorithms, but they are not quantum computing. They should be reported as classical algorithms and evaluated under conventional metaheuristic standards. Similarly, PQC should be described as quantum-resistant cybersecurity, not as quantum hardware deployment. Clear terminology is necessary because transportation agencies and funding bodies may otherwise mistake metaphorical or security-oriented quantum terminology for operational quantum computing readiness.

The fourth barrier is institutional. The near absence of transportation agency authorship indicates that many research problems are still being defined from the algorithm side rather than from the operational side. Bridging this gap requires agency co-design of benchmark problems, controlled pilots with real operational datasets, joint testbeds linking traffic simulation with quantum cloud resources, and workforce programs that build enough quantum literacy for transportation professionals to evaluate vendor claims independently.

5.4 Positioning Within the Broader Quantum Computing Landscape

Quantum-ITS follows a pattern visible across applied quantum computing: optimization attracts the most near-term

experimentation, quantum machine learning attracts speculative interest ahead of robust validation, and PQC offers the clearest practical deployment path [79]. The transportation context, however, raises the evidentiary threshold. Safety-critical systems require certification, auditability, fail-safe behavior, and interoperability. Real-time traffic control, cooperative perception, and collision-avoidance applications impose latency requirements incompatible with current cloud-access quantum workflows. Public infrastructure procurement also requires transparent cost-benefit evidence, not only technical novelty.

For these reasons, quantum-ITS deployment is likely to lag domains where each computation has high economic value, latency requirements are less stringent, and certification burdens are lower. This does not make quantum-ITS research unimportant; it means that the field should learn from earlier-adopting domains while maintaining transportation-specific validation criteria. In the near term, the rational strategy is selective investment: prioritize PQC transition, conduct limited quantum optimization pilots under strict benchmarking rules, monitor QML without treating it as deployment-ready, and build institutional capacity through shared testbeds and interdisciplinary training.

5.5 Alignment with Sustainable Development Goals

Quantum-ITS research can be connected to selected SDGs, but these connections must be stated cautiously. The reviewed corpus supports technology-specific links to SDG 3, SDG 9, SDG 11, SDG 13, and SDG 16; it does not support a broad claim that quantum computing currently delivers validated sustainability gains for transportation. Table 8 summarizes only those connections grounded in application areas actually covered by the corpus.

The strongest SDG contribution is in cybersecurity resilience. PQC supports SDG 3 and SDG 16 because transportation safety and institutional reliability depend on trustworthy communication. The SDG 9 and SDG 11 connections are credible but conditional on future validation of optimization methods at operational scale. The SDG 13 connection is the weakest: reported carbon or fuel reductions in the corpus are simulation-based or quantum-inspired, so they should not be generalized to real-world climate benefits.

5.6 Limitations of the Review

Four limitations bound the interpretation of this Discussion. First, the corpus is restricted to Scopus-indexed English-language publications. This improves metadata consistency and science-mapping reliability, but it may underrepresent regional publications, non-English studies, selected conference proceedings, and recent preprints. The geographic findings and the relative size of the PQC cluster should therefore be interpreted as Scopus-grounded patterns rather than complete global counts.

Second, the qualitative synthesis covers 27 representative studies rather than all 108 publications. The sampling design intentionally covers citation impact, recency, paradigm diversity, ITS domain breadth, and methodological variety. Nevertheless, unexamined studies may refine local judgments for particular subdomains. The central conclusion is considered robust because the absence of rigorous quantum advantage appears across all major paradigms rather than depending on a single study or cluster.

Third, the maturity assessment is interpretive. The TRL ranges synthesize evidence on baseline fairness, scale realism, reproducibility, and deployment practicality; they are not formal certifications issued by industry or regulatory

Table 8 Evidence-grounded alignment between quantum-ITS research and selected UN Sustainable development goals

SDG	Goal Title	Specific Quantum-ITS Connection	Evidence Base	TRL
3	Good Health and Well-Being	PQC for V2X and in-vehicle communication can help preserve the integrity of safety-critical messages under future quantum-enabled attacks. The connection is indirect but defensible: secure authentication supports, rather than guarantees, road-safety functions.	[99, 103]	6–8 (PQC)
9	Industry, Innovation and Infrastructure	Quantum annealing and QAOA studies contribute to the knowledge base for post-classical infrastructure planning, including transport network design, routing, and fleet scheduling. Current evidence supports research value, not operational infrastructure readiness.	[87, 88, 91]	3–4 (QA); 2–3 (QAOA)
11	Sustainable Cities and Communities	Quantum optimization studies target congestion reduction, traffic signal coordination, and more efficient use of existing road infrastructure. These remain proof-of-concept contributions until validated at metropolitan scale against strong classical traffic-control methods.	[60, 84, 85]	3–4 (QA)
13	Climate Action	Freight routing, fleet dispatch, and port scheduling studies report simulated fuel, cost, or CO_2 reductions, but most evidence comes from quantum-inspired classical algorithms or small simulator instances. These results should be treated as mechanistic hypotheses, not validated real-world emission reductions attributable to quantum computing.	[96, 97, 104]	N/A (QI classical or simulator)
16	Peace, Justice and Strong Institutions	PQC schemes for VANET, V2X, and fleet communication strengthen the cybersecurity foundation of civil transportation institutions by protecting authentication and message integrity against future quantum-capable adversaries.	[99, 101, 102]	6–8 (PQC)

bodies. Fourth, the timeline estimates are technologically contingent. Faster progress in error correction, hardware connectivity, or fault-tolerant architectures could improve readiness, while continued advances in classical optimization and machine learning could raise the comparison threshold for quantum approaches.

5.7 Stakeholder Recommendations

5.7.1 For Quantum Computing Researchers

Future quantum-ITS studies should treat strong classical benchmarking as a minimum requirement. Optimization papers should compare against commercial solvers and specialized transportation heuristics, including established Lin-Kernighan and hybrid genetic search families where relevant [34, 35]. Reported runtimes should include data encoding, preprocessing, queueing or access delay, quantum execution, postprocessing, and classical coordination overhead. Authors should publish code, benchmark instances, random seeds, hyperparameters, annealing or circuit settings, transpilation details, and noise models. Most importantly, problem formulations should be co-designed with transportation experts so that benchmarks reflect real operational constraints rather than mathematically convenient toy instances.

5.7.2 For Transportation Practitioners

Transportation agencies, road operators, OEMs, and fleet managers should treat PQC migration as the immediate action item. The first steps are inventorying current V2X and in-vehicle cryptographic dependencies, testing PQC candidates on representative embedded hardware, evaluating certificate and revocation overheads, and designing phased hybrid transition plans. Quantum optimization should be explored only through controlled offline pilots with agency-owned data, independent benchmarking, and transparent total-cost comparison against existing classical tools. Vendor claims should not be accepted unless they report end-to-end performance and isolate the quantum contribution from the classical workflow.

5.7.3 For Funding Agencies

Funding should follow maturity. PQC migration, benchmark development, and transportation-agency testbeds deserve near-term support because they address clear operational needs and benefit the whole community. Quantum optimization should be funded as controlled research infrastructure and pilot evaluation, not as near-term deployment. QML for ITS should remain primarily a foundational research topic until it demonstrates advantages over classical models on standardized transportation datasets. Calls for proposals should require terminological precision, strong baselines, reproducibility artifacts, and transportation stakeholder participation.

5.7.4 For Quantum Technology Industry

Quantum hardware and software vendors should prioritize credible, reproducible transportation benchmarks over promotional demonstrations. Useful industry outputs would include QUBO formulation templates for common ITS problems, embedding strategies tailored to transportation network topology, APIs that connect hybrid solvers to traffic simulation and planning software, and public benchmark reports comparing quantum, hybrid, and classical alternatives on the same instances. In the near term, the most credible commercial opportunity is not real-time quantum traffic control but PQC implementation support and controlled optimization pilots for offline planning.

5.8 Summary of Discussion

The evidence supports a calibrated rather than promotional interpretation of quantum-ITS. Publication growth is strong, but evidential maturity remains limited. PQC is the only near-term deployment-ready quantum-related domain because it is a classical security migration pathway responding to a present quantum threat. Quantum annealing and QAOA remain promising but unproven for transportation optimization, with deployment contingent on strong classical benchmarking, scale realism, and cost justification. QML remains exploratory and should not be presented as deployment-ready for ITS. The field will mature only if it moves from toy-scale demonstrations toward operational benchmarks, from quantum terminology toward terminological discipline, and from isolated algorithm studies toward sustained collaboration with the transportation agencies and operators that would ultimately deploy these technologies.

6 Open Challenges and Future Research Directions

This section delineates the critical open challenges impeding the adoption of quantum computing in intelligent transportation systems and articulates prioritized research directions at the technical, methodological, infrastructural, and socio-institutional levels. The analysis is grounded in the systematic review findings and structured to guide coordinated investment across research communities, funding agencies, and operational stakeholders. Two synthesizing

tables accompany this section: Table 9 taxonomizes technical challenges across quantum paradigms and ITS application domains with associated research priorities and success metrics, and Table 10 maps strategic research directions to stakeholder responsibilities, implementationtimelines, expected outcomes, and measurable indicators.

6.1 Technical and Algorithmic Challenges

6.1.1 Quantum Hardware Limitations

Qubit Count, Connectivity, and Embedding Overhead. Current NISQ processors remain constrained by qubit count, hardware connectivity, noise, limited circuit depth, and access latency; available devices span tens to hundreds of gate-based qubits and thousands of annealing qubits, but effective logical problem sizes remain far below operational ITS-scale QUBO and circuit-model requirements [37, 44, 45, 58]. ITS optimization problems formulated naturally require dense variable interactions: traffic signal coordination requires strong coupling among neighboring signal decisions, and vehicle routing often requires pairwise distance or precedence terms across stops. These interaction structures do not map directly to sparse hardware topologies, and embedding logical QUBO problems onto physical hardware graphs can introduce substantial physical-qubit overhead and chain-breaking errors that degrade solution quality [22, 48–50].

Research in this area should prioritize ITS-specific embedding strategies that exploit the spatial locality and temporal decomposability inherent in transportation networks, the design of quantum algorithms whose operator structures respect hardware connectivity constraints, and compressed QUBO representations that trade controlled optimality loss for feasibility at larger problem scales.

Decoherence, Gate Errors, and Fault Tolerance Requirements. Quantum states in superconducting processors are subject to coherence and gate-error constraints that limit practical circuit depth on NISQ hardware [44, 45]. Gate errors compound across circuit layers, and ITS applications requiring expressive quantum models, including iterative QAOA for signal control and quantum neural networks for demand forecasting, may require circuit depths well beyond current hardware tolerances. Fault-tolerant quantum computing would address these limitations through quantum error correction, but surface-code approaches may require on the order of hundreds to thousands of physical qubits per logical qubit depending on physical error rates, target logical error rates, and circuit requirements [59]. This overhead keeps effective problem sizes far below operational ITS requirements.

Research priorities include error mitigation strategies adapted to transportation problem structure [109, 112], hybrid algorithmic designs that minimize quantum circuit depth through classical warm-starting and shallow quantum refinement, and formal verification methods that can provide mathematical correctness evidence for quantum circuits and QUBO implementations used in safety-critical ITS settings [113].

6.1.2 Algorithmic Maturity and Advantage Validation

Absence of Demonstrated Quantum Advantage. As established in Sect. 5.1.3, no study in the reviewed corpus provides rigorous evidence of quantum computational advantage over state-of-the-art classical algorithms under operationally realistic conditions. Performance comparisons frequently rely on weak or incomplete baselines rather than commercial optimization solvers or established transportation heuristics, including Lin-Kernighan-type methods and hybrid genetic search for routing problems [34, 35]. Studies that do compare against stronger classical baselines generally report comparable performance, scale limitations, or quantum disadvantage. Consistent with the structured advantage assessment summarized in Table 7, the evidence base remains methodologically thin: fewer than 15% of publications employ formal hypothesis testing, 69% omit end-to-end computational time reporting, and most studies evaluate single problem instances, rendering robust statistical inference infeasible.

Addressing this challenge requires community benchmark protocols mandating repeated experiments over multiple instances, confidence interval reporting, and full computational accounting across preprocessing, quantum execution, and postprocessing time [58]. Rigorous controlled comparative studies should identify the instance characteristics and problem structures under which quantum approaches exhibit advantage, parity, or disadvantage. Theoretical complexity analysis identifying ITS problem classes for which quantum algorithms can provide provable asymptotic speedups would provide principled guidance for experimental prioritization.

Problem Scale and Decomposition Quality. Demonstrated instances across the corpus remain systematically undersized relative to operational ITS requirements. Traffic-signal evaluations address hundreds of vehicles or idealized lattice networks; vehicle-routing studies often test three-to-four-customer instances; and metropolitan operational systems involve substantially larger vehicle populations, intersection networks, and logistics tasks. Bridging this scale gap requires decomposition strategies, including spatial partitioning, temporal rolling horizons, and hierarchical problem structures. Yet the approximation error introduced

Table 9 Technical and algorithmic challenges in quantum computing for intelligent transportation systems: taxonomy across quantum paradigms and ITS application domains with research priorities and success metrics

Quantum Paradigm	ITS Application Domain	Challenge Description	Research Priorities	Success Metrics
Quantum Annealing (D-Wave, Fujitsu DA)	Traffic Signal Control & Network Optimization	*Embedding Overhead*: Dense signal-coordination QUBO formulations do not map directly to sparse hardware graphs; embedding can require substantial physical-qubit overhead and may induce chain-breaking errors [48–50].	ITS-specific embedding exploiting spatial locality; hierarchical decomposition with boundary coordination between local signal clusters; compressed QUBO representations.	Target metric: reduced embedding overhead and chain-break frequency relative to generic embedding methods, reported together with solution quality and end-to-end runtime.
	Traffic Signal Control & Network Optimization	*Real-Time Latency*: Cloud-mediated quantum execution may involve queueing, compilation, execution, and communication delays incompatible with second-scale traffic-control windows.	Latency-aware hybrid workflows; priority quantum job scheduling; classical fallback controllers; empirical latency characterization across hardware platforms.	Target metric: complete latency distribution reported for preprocessing, quantum execution, postprocessing, and fallback control under traffic-control time budgets.
	Vehicle Routing & Logistics	*Scale Gap*: Demonstrated quantum-routing instances remain far below operational logistics scale; decomposition approximation error is rarely quantified.	Hybrid algorithms combining classical route construction with quantum local refinement; provable approximation bounds.	Target metric: benchmarked performance on progressively larger VRP instances, with optimality gap, runtime, and decomposition error reported against strong classical solvers.
	Vehicle Routing & Logistics	*Baseline Comparison Weakness*: Quantum VRP studies are often benchmarked against generic GA and PSO rather than commercial MIP solvers, OR-based heuristics, and established routing heuristics [34, 35].	Rigorous comparative studies with commercial MIP solvers; standardized benchmark instances with reference classical solver performance profiles.	Target metric: demonstrated improvement, parity, or disadvantage relative to strong classical baselines, with full statistical uncertainty and computational accounting.
Gate-Based (IBM, IonQ, Rigetti)	Traffic Optimization (QAOA)	*Circuit Depth Constraints*: NISQ noise limits practical QAOA circuit depth and expressiveness for complex traffic problems [44, 45].	Error mitigation exploiting transportation problem symmetries [109]; shallow hybrid QAOA with classical warm-starting.	Target metric: error-mitigated performance compared with noiseless simulation and strong classical baselines under the same problem instances.
	Traffic Optimization (QAOA)	*Parameter Optimization Overhead*: QAOA angle tuning can require many classical optimizer evaluations, dominating total time-to-solution.	Meta-learning QAOA parameters; parameter transfer across related traffic instances; low-evaluation classical optimizers.	Target metric: number of objective evaluations, total wall-clock time, and final solution quality reported jointly.
	Vehicle Routing (QAOA)	*Quantum Advantage Absent*: Existing VRP-QAOA evidence does not show advantage over CPLEX on small benchmark instances [88].	Identification of VRP structural properties enabling QAOA advantage; complexity-theoretic characterization of advantage conditions.	Target metric: reproducible quantum-classical comparison on predefined VRP classes, including positive, negative, and parity results.
	Vehicle Routing (QAOA)	*Barren Plateaus*: Variational circuits may exhibit exponentially vanishing gradients with increasing qubit count and circuit depth, limiting trainability at operational scales [56, 57].	Problem-aware ansatz design; gradient-free optimization alternatives for QAOA angle optimization.	Target metric: gradient statistics, convergence behavior, and solution quality reported as instance size and circuit depth increase.
Quantum Machine Learning	Demand Forecasting & Traffic Prediction	*No Demonstrated Advantage*: QBN and TS-QGCNN studies do not establish consistent superiority over strong classical forecasting baselines [92, 93].	Theoretical analysis identifying transportation forecasting tasks with provable quantum learning advantage; empirical QML studies with rigorous classical baselines including LSTM, Transformer, and GNN architectures.	Target metric: accuracy, calibration, robustness, and time-to-solution reported against state-of-the-art classical models on standardized transportation datasets.
	Demand Forecasting & Traffic Prediction	*Barren Plateaus*: Deep QNN parameterized circuits may exhibit exponential gradient vanishing, preventing stable training convergence [56, 57].	Shallow quantum models with classical preprocessing; quantum kernel methods [36]; reservoir quantum computing [110].	Target metric: convergence stability and gradient statistics reported across increasing dataset size and circuit depth.

Table 9 (continued)

Quantum Paradigm	ITS Application Domain	Challenge Description	Research Priorities	Success Metrics
	Perception & Autonomous Vehicle ML	*Data Encoding Overhead*: Encoding lidar point clouds and camera imagery into quantum states may require state-preparation circuits whose cost outweighs downstream quantum benefit.	Efficient quantum encoding exploiting spatial correlations; hybrid quantum-classical perception.	Target metric: encoding cost, circuit depth, inference time, and predictive performance reported together on transportation sensor data.
	Perception & Autonomous Vehicle ML	*Real-Time Inference Latency*: Circuit-shot requirements make real-time inference difficult under current quantum hardware constraints.	Low-shot quantum ML achieving target accuracy with fewer circuit executions; classical fallback for safety-critical perception.	Target metric: complete inference latency, shot count, uncertainty estimates, and fallback behavior reported for time-critical ITS tasks.
Post-Quantum Cryptography	V2X Security & Authentication	*Signature Size and Bandwidth*: PQC primitives may increase message size and verification cost in bandwidth-constrained vehicular networks [24, 26, 99].	Batch verification amortizing per-message cost; aggregate signature protocols; RSU-side caching and geocasting.	Target metric: end-to-end latency, channel load, verification throughput, and packet-delivery impact evaluated under high-density V2X scenarios.
	V2X Security & Authentication	*Revocation Infrastructure Scalability*: Pseudonym management and revocation are already complex in V2X PKI, and PQC migration may increase credential-management overhead [111].	Delta-CRL incremental updates; Bloom filter revocation; certificate-status stapling; privacy-preserving revocation.	Target metric: revocation-check latency, communication overhead, privacy leakage, and false-positive behavior reported at fleet scale.
	Fleet Management & Logistics Security	*PQC Computational Overhead*: PQC migration must be validated on automotive-grade processors rather than desktop-only testbeds.	Hardware-aware PQC implementation on representative automotive-grade embedded platforms, including AURIX-class controllers [103].	Target metric: signing, verification, and key-establishment latency measured on representative embedded platforms under realistic load.
	Fleet Management & Logistics Security	*Hybrid Transition Complexity*: Backward compatibility with legacy cryptographic vehicles will be necessary during gradual fleet turnover.	Hybrid authentication supporting both classical and PQC cipher suites; graceful degradation for legacy vehicles.	Target metric: multi-vendor interoperability validated in testbed environments with both classical and PQC-capable devices.
Quantum Communication & QKD	V2I/V2V Communications	*Mobile QKD Limitations*: QKD is most credible for fixed or well-controlled links; mobile vehicular deployment faces channel, alignment, loss, integration, and cost constraints [61].	Prioritize PQC for vehicular security; reserve QKD for fixed infrastructure links such as RSU-to-RSU fiber connections.	Target metric: fixed-link QKD evaluated against PQC alternatives using security, cost, latency, and operational-maintenance criteria.
	V2I/V2V Communications	*Economic Barriers*: QKD link infrastructure costs present uncertain ROI for transportation agencies.	Cost-benefit analyses comparing QKD, PQC, and hybrid approaches; deployment decision frameworks adopted by transportation agencies.	Target metric: peer-reviewed economic models comparing total cost of ownership and security benefit across QKD, PQC, and hybrid options for ITScontexts.

QA = quantum annealing; QML = quantum machine learning; PQC = post-quantum cryptography; QKD = quantum key distribution

by decomposition is rarely quantified, and decomposition overhead may offset the computational gains that originally motivated the quantum approach.

Research should prioritize hybrid quantum–classical decomposition frameworks with solution-quality bounds relating decomposed solutions to global objectives, architectures that isolate hard combinatorial core subproblems most likely to benefit from quantum processing, and systematic scalability studies that characterize performance degradation and classical crossover points as problem size increases.

6.1.3 Quantum Machine Learning Barriers

Barren Plateaus and Trainability Collapse. Parameterized quantum circuits face the barren plateau phenomenon, in which cost function gradients may vanish exponentially with increasing qubit count and circuit depth, preventing convergence of variational optimization [56, 57]. Transportation applications requiring expressive quantum models to capture spatiotemporal traffic dynamics and high-dimensional sensor inputs may require precisely the deeper circuits where trainability becomes most difficult. The reviewed QML contributions in this corpus do not demonstrate consistent performance advantage over strong

Table 10 Strategic research priorities for quantum-ITS: Roadmap mapping research directions to stakeholder responsibilities, timelines, expected outcomes, and success indicators

Tier	Research Direction	Stakeholder Responsibilities	Timeline	Expected Outcomes	Success Indicators
Tier 1: Critical Near-Term Initiated (2025–2027)					
T1	Post-Quantum Cryptography V2X Transition	Automotive OEMs; Tier-1 suppliers; chipset vendors; standards bodies; transportation and cybersecurity regulators	2025–2030	PQC algorithms from the NIST standardization process, especially ML-KEM and ML-DSA, evaluated for automotive-grade embedded platforms; compact-signature alternatives such as Falcon/FN-DSA considered where bandwidth constraints dominate; hybrid classical–PQC protocol stacks assessed for mixed fleets [24–26, 103]	PQC V2X testbeds reporting latency, bandwidth, and interoperability under realistic vehicular communication loads; published migration guidance for mixed classical-PQC fleets
T1	Quantum-ITS Benchmark Repository	Academic transport engineering and quantum computing centers; national labs; professional societies; transportation agencies	2025–2027	Open-access benchmark suites for traffic signal control, VRP, multimodal demand forecasting, and aviation/maritime scheduling; standardized evaluation protocols; reference classical solver performance profiles and metric definitions	Repository publicly released with versioned datasets, solver baselines, and documentation; adoption by multiple independent research groups; recurring open quantum-ITS benchmark challenge established
T1	Rigorous Quantum Advantage Validation Studies	Academic researchers; quantum hardware vendors; national transportation agencies; ITS peer-reviewed journals	2025–2028	Controlled comparative experiments with CPLEX, Gurobi, and specialized heuristics; statistical testing across multiple instances; full computational accounting; ITS problem classes with advantage, parity, or disadvantage identified; negative results reported	Peer-reviewed studies with reproducible quantum-classical benchmarks; transparent negative and parity results published; research community consensus on ITS contexts warranting continued quantum investigation
Tier 2: High-Strategic Research Initiated (2026–2032)					
T2	Scalable Hybrid Quantum-Classical Algorithms	Academic institutions; quantum software companies; transportation planning software vendors; operations research groups	2026–2030	Hybrid optimization frameworks with solution-quality bounds; quantum solver APIs interoperable with traffic planning and logistics software; classical warm-starting and shallow quantum refinement workflows validated on larger benchmark instances	Hybrid workflows demonstrated on operationally motivated benchmark instances; solution quality, runtime, and decomposition error reported against strong classical baselines
T2	Quantum-ITS Field Trials	State and local transportation agencies; city traffic management centers; quantum cloud platforms; academic research centers; automotive OEMs	2026–2031	Quantum-assisted traffic signal control pilots; quantum logistics routing in commercial freight operations; PQC V2X multi-vendor testbeds; hardware-in-the-loop validation at connected vehicle proving grounds	Peer-reviewed field-trial reports comparing quantum-hybrid workflows with classical operational baselines; documented deployment lessons; cost and latency data reported transparently
T2	Quantum ML Architectures for ITS Intelligence	Academic QML and transportation AI research groups; quantum hardware vendors; autonomous vehicle technology companies	2027–2033	ITS-specific QML ansätze and kernel methods addressing barren plateaus; hybrid quantum-classical perception pipelines; theoretical advantage conditions for transportation forecasting and decision tasks characterized; benchmark comparisons with LSTM, Transformer, and GNN baselines	QML models evaluated against classical deep learning on standardized transportation datasets; ablation studies isolating the quantum component; advantage, parity, or disadvantage reported transparently
T2	Quantum Algorithm Safety Certification Frameworks	Academic formal methods researchers; regulatory bodies; automotive and aviation safety assessors; standards bodies	2027–2034	Formal verification tools for QUBO formulations and quantum circuits; safety case frameworks incorporating quantum failure modes into fault tree analysis; pilot guidance for quantum-enabled traffic control and AV perception; regulatory sandbox environments	Formal verification tools publicly released; safety-case templates documented; regulatory sandbox studies reported; quantum failure modes incorporated into safety analysis examples
Tier 3: Long-Term Exploratory Directions (2028–2040+)					
T3	Quantum Sensing for ITS Positioning and Perception	Academic quantum physics and automotive engineering groups; quantum sensor companies; automotive OEMs; public research agencies	2028–2038	Quantum sensing concepts assessed for GPS-denied positioning, synchronization, and perception support; miniaturization, robustness, and cost barriers quantified for automotive environments [108]	Prototype evaluations reporting positioning accuracy, environmental robustness, integration complexity, and comparative benefit over classical sensor systems

Table 10 (continued)

Tier	Research Direction	Stakeholder Responsibilities	Timeline	Expected Outcomes	Success Indicators
T3	Fault-Tolerant Quantum Computing for Metropolitan ITS	Academic institutions; national quantum computing programs; quantum hardware scaling companies; transportation optimization researchers	2032–2040+	Error-corrected quantum resources assessed for metropolitan-scale combinatorial optimization; quantum simulation of traffic flow dynamics; deep quantum ML reconsidered only when logical resources become sufficient	Peer-reviewed ITS benchmarks showing whether fault-tolerant quantum systems outperform mature classical methods at operationally meaningful scales
T3	Economic and Investment Modeling for Quantum-ITS	Transportation economics researchers; national and regional transportation agencies; quantum industry consortia; policy research organizations	2028–2033	Standardized cost-benefit analysis frameworks for quantum-ITS investment; total cost of ownership models encompassing hardware, cloud access, integration, workforce, and sustainability costs; break-even analyses identifying investment-justifying performance thresholds	Peer-reviewed case studies quantifying quantum-ITS return on investment for representative use cases; lifecycle sustainability analyses reported for quantum-hybrid workflows
Cross-Cutting Enablers (Continuous, 2025–2035)					
CE	Interdisciplinary Workforce Development	Universities; professional societies; government workforce programs; online education platforms; transportation agencies	2025–2035	Graduate and professional programs bridging quantum computing and transportation engineering; shared technical vocabulary established; industry-academic-government partnerships enabling joint research programs; quantum literacy built in the operational transportation community	Documented curricula, short courses, and professional training modules; participation from both quantum researchers and transportation practitioners; joint publications and pilots involving transportation agencies
CE	Quantum-ITS Standards Development	Standards bodies; industry consortia; regulatory bodies; academia; industry	2026–2032	Standardized QUBO templates for common ITS problem classes; quantum-ITS evaluation protocol guidance; quantum-safe vehicular security migration guidance; quantum solver API interfaces for transportation software platforms	Draft protocols, reference implementations, and multi-vendor test reports published; standards discussions informed by reproducible benchmark evidence
CE	Open-Source Quantum-ITS Software Ecosystem	Academic research groups; open-source foundations; quantum software companies; transportation agencies	2025–2030	QUBO formulation libraries with documented derivations for traffic, VRP, and scheduling problems; quantum algorithm reference implementations; integration tools for transportation planning software; curated benchmark datasets with version-controlled provenance	Actively maintained public repositories; reproducible examples covering major quantum-ITS application domains; documented use in research and teaching
CE	Economic and Sustainability Frameworks for Quantum-ITS	Transportation economics and sustainability science researchers; national transportation funding agencies; quantum technology assessment organizations; SDG-aligned policy research groups	2026–2035	Standardized economic impact assessment methodologies for quantum-ITS investment across hardware, cloud, integration, and workforce costs; lifecycle sustainability analyses of quantum ITS workflows including cryogenic cooling overhead; SDG-aligned reporting frameworks; investment guidance for funding bodies	Peer-reviewed economic and sustainability frameworks; case studies applying the frameworks to optimization, security, and sensing use cases; funding programs requiring transparent cost and sustainability reporting

classical machine-learning baselines [92, 93, 105], consistent with the present immaturity of QML for ITS-scale prediction and decision tasks.

Research directions include ITS-specific ansatz designs that exploit transportation network structure to maintain trainability, alternative quantum learning paradigms avoiding deep variational optimization, such as quantum kernel methods [36], quantum reservoir computing [110], and theoretical complexity analysis establishing the conditions under which quantum learning architectures could outperform classical counterparts on transportation data distributions.

Data Encoding and Measurement Overhead. Quantum machine learning requires encoding classical transportation data into quantum states through mechanisms that each introduce distinct limitations. Amplitude encoding can reduce state dimension but requires complex state-preparation circuits vulnerable to gate errors. Basis encoding scales linearly in qubit count, limiting compression benefits. Angle encoding is hardware-friendly but may discretize continuous values and introduce information loss. Measurement overhead compounds these encoding challenges: obtaining expectation-value estimates with precision ϵ generally requires $O(\epsilon^{-2})$ circuit repetitions, a measurement burden

that is difficult to reconcile with real-time ITS inference requirements.

Efficient quantum encoding strategies exploiting the sparse structure and temporal correlations characteristic of transportation data represent a high-priority research need, as do hybrid quantum-classical machine learning architectures that apply classical feature engineering to reduce data dimensionality before quantum processing, and online incremental quantum learning approaches supporting city-to-city transfer.

6.1.4 Post-Quantum Cryptography Deployment Challenges

Signature Size and V2X Bandwidth Constraints. Post-quantum cryptographic primitives can impose larger keys, ciphertexts, or signatures than conventional elliptic-curve schemes, creating bandwidth and processing concerns for resource-constrained IoT and vehicular environments [24, 26]. Safety-critical V2X applications operating at high message frequency on shared DSRC and C-V2X channels [4, 114] already face bandwidth pressure under current protocols. PQC overhead may therefore exacerbate channel congestion and verification latency at high vehicle densities, particularly when pseudonymous certificate rotation increases transmission and credential-management volume [99, 111].

Research priorities include batch verification schemes that amortize per-message cryptographic costs, aggregate signature protocols for vehicle platoon authentication, and application-layer optimization techniques including message suppression, geocasting, and roadside-unit caching.

Trust Infrastructure and Revocation at Scale. Vehicular public key infrastructure must manage certificate issuance, pseudonym rotation, and revocation for large fleets while preserving both accountability and privacy [111]. PQC migration substantially complicates this infrastructure because larger credentials may inflate certificate distribution and revocation overhead. Centralized certificate authorities also introduce single points of failure and privacy risks through traffic pattern inference.

Research should address scalable PQC revocation through incremental update mechanisms, probabilistic revocation filters with controlled false-positive rates, and efficient online certificate-status checking, and should develop decentralized quantum-resistant trust architectures based on threshold cryptography and permissioned blockchain PKI, along with lattice-based anonymous credential schemes providing privacy-preserving authentication with efficient revocation.

6.2 Methodological and Evaluation Challenges

6.2.1 Benchmark Standardization and Reproducibility

The reviewed literature employs heterogeneous datasets spanning T-Drive Beijing GPS traces, NYC Citi Bike trip records, OR-Library combinatorial benchmarks, the Transportation Networks for Research repository, proprietary agency data, and custom synthetic instances. This diversity, combined with inconsistent performance metrics, varied quantum software platforms, and the frequent absence of implementation details, prevents comparative evaluation across studies and forecloses cumulative scientific progress. Quantum-ITS currently lacks community benchmark infrastructure analogous to established benchmarking ecosystems in optimization and machine learning, such as OpenML for reproducible machine-learning experiments [115].

Establishing an open-access quantum-ITS benchmark repository spanning traffic signal control, vehicle routing, multimodal demand forecasting, and maritime and aviation scheduling represents a foundational investment for the field. This repository should adapt general quantum benchmarking principles toward transportation-specific problem families, with benchmark instances, reference classical solver performance, hardware and software configuration metadata, and time-to-solution accounting reported consistently [58]. Standardized evaluation protocols requiring statistical hypothesis testing, confidence intervals, full computational accounting distinguishing preprocessing, quantum execution, and postprocessing time, and explicit hardware and software configuration reporting should accompany benchmark adoption. An open-source software ecosystem providing reference QUBO formulation libraries, QAOA solver implementations, and quantum machine learning pipelines with documented APIs would lower barriers to reproducible research.

6.2.2 Real-World Validation and the Simulation-to-Reality Gap

Laboratory demonstrations operate on historical datasets and simplified models that abstract away operational realities including unpredictable demand fluctuations, sensor noise, legacy system constraints, regulatory requirements, and stakeholder acceptance dynamics. Within the reviewed corpus, no quantum optimization or quantum machine-learning study provides real-world field validation in an operational transportation environment. This absence matters because methods performing well in simulation may fail to transfer when behavioral adaptation, sensor uncertainty,

network disruptions, procurement constraints, and safety responsibilities are introduced.

Quantum-ITS field trials conducted in partnership between quantum vendors, academic institutions, and transportation agencies, encompassing quantum-assisted traffic signal control pilots, quantum logistics routing in commercial freight fleets, and PQC V2X multi-vendor testbeds, are essential for establishing whether laboratory results transfer to operational environments. Hardware-in-the-loop testing integrating quantum infrastructure with connected vehicle proving grounds and traffic management center simulators would provide an intermediate validation tier. Systematic quantum algorithm robustness analysis under input perturbations, constraint violations, and anomalous conditions would characterize failure modes and intervention requirements.

6.2.3 Economic Viability and Cost-Benefit Analysis

Quantum infrastructure costs are poorly characterized in the ITS literature. Cloud QPU access, dedicated annealer use, gate-based task execution, hybrid workflow orchestration, and specialized workforce requirements create cost structures that differ substantially from classical high-performance computing. Because pricing and availability change rapidly, exact cost multipliers should be treated as procurement-specific rather than general scientific claims. More importantly, no reviewed study conducts a transportation-specific cost-benefit analysis comparing quantum infrastructure investment against performance improvements relative to mature classical alternatives.

Standardized quantum-ITS cost-benefit analysis methodologies incorporating infrastructure investment, performance improvement valuation through transportation-specific metrics, and sensitivity analysis across quantum hardware cost trajectories are a high-priority research need. Break-even analyses identifying the problem scales, improvement magnitudes, and quantum hardware cost reductions required to justify investment relative to classical alternatives would provide transportation agencies with principled decision frameworks. Lifecycle total cost of ownership models encompassing deployment, maintenance, workforce upskilling, technology refresh cycles, and exit strategies are prerequisites for procurement decision-making.

6.3 Infrastructural and System Integration Challenges

6.3.1 Quantum-Classical Hybrid Workflow Architecture

ITS operational environments require quantum solver modules to integrate within complex classical ecosystems encompassing traffic management systems, logistics dispatch platforms, transit planning tools, and automotive V2X communication stacks. Quantum solvers require classical preprocessing for problem translation, QUBO formulation, and embedding initialization; quantum execution through cloud job submission and hardware resource allocation; and classical postprocessing for measurement interpretation, constraint enforcement, and output translation into actionable transportation decisions. Standardized application programming interfaces enabling transportation software platforms to invoke quantum solvers through high-level abstractions remain underdeveloped, and hybrid workflow orchestration tools supporting asynchronous job submission, fault tolerance, and result consistency are still immature.

Research priorities encompass the design of quantum solver API standards enabling interoperability between transportation planning software and heterogeneous quantum hardware platforms, hybrid workflow orchestration frameworks with fault tolerance and consistency guarantees under quantum hardware failures, and edge-cloud quantum architectures managing the latency and bandwidth trade-offs between local classical computation and remote quantum processing.

6.3.2 Real-Time Responsiveness and Latency Management

Safety-critical transportation applications impose stringent latency bounds that current quantum hardware and cloud-mediated quantum workflows cannot reliably satisfy. Autonomous vehicle perception and control require rapid sensor processing and decision cycles, while V2X services are designed around low-latency communication requirements [16, 114]. Adaptive traffic signal control typically operates on second-scale cycles, where stale optimization outputs can degrade performance or create unsafe control actions. Public quantum cloud execution may involve queueing, compilation, execution, and network delays that are not aligned with these time-critical operational windows.

Systematic characterization of end-to-end quantum latency distributions across problem sizes, hardware platforms, time-of-day demand variations, and network conditions would provide the empirical foundation for latency-aware system design. Priority-aware quantum job scheduling and circuit compilation optimizations targeting minimal execution latency should be developed specifically for transportation control applications. Latency-adaptive control architectures that exploit quantum computation opportunistically when results arrive within the control budget, while falling back to certified classical heuristics otherwise, represent a pragmatic near-term research and pilot

architecture for latency-aware hybrid quantum-classical traffic-control evaluation.

6.3.3 Cybersecurity and Adversarial Robustness of Quantum-ITS Systems

Quantum-enabled ITS introduces a distinct attack surface beyond the vulnerabilities of conventional ITS security. Quantum cloud dependency exposes transportation optimization to denial-of-service attacks targeting cloud infrastructure, interception of problem submissions containing sensitive network state data, insider threats from service-provider environments, and supply-chain risks from hardware or software components. Transportation operational technology security has historically emphasized controlled connectivity and strong isolation; quantum cloud integration therefore requires explicit security analysis before operational adoption.

Research should address encrypted problem submission and result delivery, privacy-preserving computation for sensitive transportation data, adversarial attack characterization for quantum optimization and machine learning components in transportation contexts, and resilient hybrid architectures implementing defense-in-depth with quantum output validation against transportation safety invariants, anomaly detection for implausible quantum solutions, and graceful degradation to certified classical fallbacks.

6.4 Socio-Institutional and Workforce Challenges

6.4.1 Interdisciplinary Collaboration Barriers

Quantum computing and transportation engineering communities exhibit pronounced cultural, terminological, and methodological divides that impede collaborative research. Quantum researchers optimize for algorithmic novelty and hardware demonstration milestones, while transportation researchers prioritize operational impact, practitioner adoption pathways, and policy relevance. Bibliometric analysis quantified this divide: transportation agencies appear in fewer than 2% of author affiliations across the corpus, and the co-authorship network is dominated by small, fragmented clusters with sparse broker nodes. This structural disconnection means that quantum algorithm designs are not validated against realistic operational requirements, and transportation practitioners lack the quantum computing literacy to evaluate vendor claims or design meaningful pilots.

Institutional structures co-locating quantum physicists, computer scientists, and transportation engineers, whether through interdisciplinary research centers, industry-academic consortia, or government-sponsored institutes, are necessary conditions for productive quantum-ITS collaboration. Cross-training curricula teaching QUBO formulation and quantum algorithm design to transportation engineers, and ITS domain knowledge to quantum researchers, would build the shared technical vocabulary currently absent. Community evaluation campaigns and joint challenge competitions would create structured incentives for interdisciplinary team formation.

6.4.2 Regulatory Frameworks and Safety Certification

Automotive and aviation transportation sectors operate under formal safety assurance and certification regimes that require systematic hazard analysis, software verification, validation, and traceable evidence before operational deployment. Quantum computing presents novel certification challenges: quantum algorithm correctness verification is substantially harder than classical software verification due to probabilistic measurement, quantum hardware nondeterminism makes reproducible testing difficult, and formal verification tools for quantum circuits remain nascent. Regulatory uncertainty compounds these challenges, as it remains unclear whether existing safety frameworks can be adapted to cover quantum components or whether quantum-specific regulatory guidance will be required.

Formal verification methods providing mathematical correctness proofs for QUBO formulations and quantum circuit implementations, runtime monitoring systems detecting anomalous quantum output distributions, and safety case frameworks incorporating quantum failure modes into fault tree analysis represent necessary research investments. Engagement with regulatory bodies and standards organizations should focus first on pilot guidance, sandbox testing, and evidence templates rather than premature certification claims.

6.4.3 Public Acceptance and Equity Considerations

Within the reviewed corpus, public acceptance and equity implications are largely absent from quantum-ITS evaluation. Algorithmic opacity concerns apply to quantum optimization and quantum machine learning because quantum-computed plans are less intuitively explainable than many classical decision rules, complicating public explanation of transportation interventions derived from quantum-assisted workflows. This concern is consistent with wider ITS evidence showing that explainability, safety, and trust are central requirements for autonomous and AI-supported transportation systems [116]. Privacy implications of quantum-enabled traffic monitoring and the harvest-now-decrypt-later threat to historical location data require proactive policy attention. Equity considerations should be investigated explicitly, especially where uneven sensor

coverage and data availability may cause advanced optimization systems to benefit well-instrumented urban corridors more than underserved areas.

Research addressing explainability methods for quantum optimization and machine learning, fairness constraints embedded within QUBO formulations, and participatory design processes engaging public stakeholders in quantum-ITS pilot decisions would establish a stronger sociotechnical foundation for legitimate quantum-ITS evaluation.

6.5 Cross-Cutting Research Priorities

Table 9 synthesizes the technical challenges identified above, organizing them by quantum paradigm and ITS application domain with associated research priorities and measurable success metrics. Table 10 extends this synthesis into a strategic research roadmap, mapping priority directions to stakeholder responsibilities, implementation timelines, expected outcomes, and success indicators for evidence-based resource allocation across the quantum-ITS research ecosystem.

The open challenges identified through systematic synthesis are organized into four structural pillars and linked to priority research pathways, as illustrated in Fig. 25. These pillars capture the technical and algorithmic, methodological and evaluation, infrastructural and system integration, and socio-institutional barriers that must be addressed before operational quantum-ITS deployment.

The challenges and research directions identified above organize into three actionable priority tiers that provide a principled basis for resource allocation across the quantum-ITS research landscape. Figure 26 presents the prioritized three-tier roadmap, structuring investment guidance from near-term actionable priorities through long-term exploratory directions, underpinned by cross-cutting enablers spanning workforce, standards, open-source infrastructure, and economic frameworks.

Tier 1 (2025–2027) encompasses actions warranted on the basis of immediate need, evidentiary necessity, and comparatively mature technology readiness. PQC V2X transition is the critical priority: the harvest-now-decrypt-later threat creates a present security-planning problem, and PQC can be deployed on conventional computing hardware rather than waiting for fault-tolerant quantum infrastructure. Quantum-ITS benchmark repository establishment and rigorous quantum advantage validation studies are foundational investments that the entire research community requires before further specialization is productive. Without shared evaluation infrastructure and honest comparative studies, the field cannot distinguish genuine quantum contributions from algorithmic noise.

Tier 2 (2026–2032) encompasses high-value directions requiring sustained interdisciplinary collaboration and infrastructure investment whose success depends on Tier 1 foundations. Scalable hybrid quantum-classical algorithms, field trials validating real-world operational performance, quantum machine learning architectures tailored to transportation data structures, and quantum algorithm safety certification frameworks each represent multi-year research programs that will determine whether quantum computing

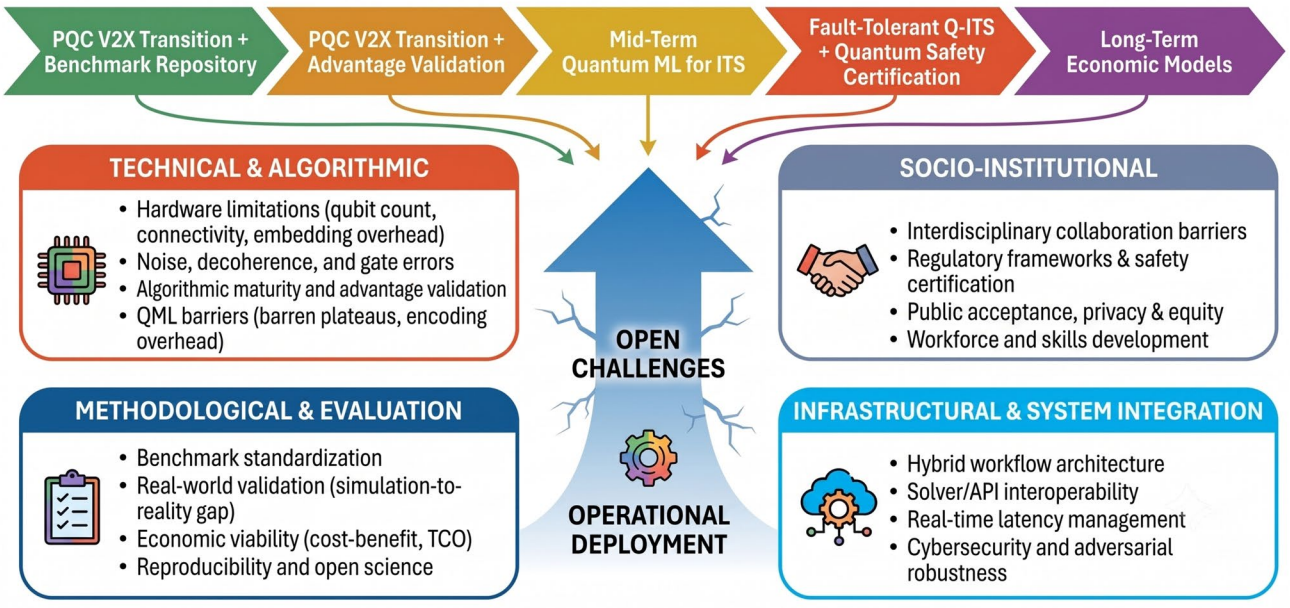


Fig. 25 Open challenges and priority pathways for quantum-enabled intelligent transportation systems (ITS), organized across technical and algorithmic, methodological and evaluation, infrastructural and system integration, and socio-institutional barriers

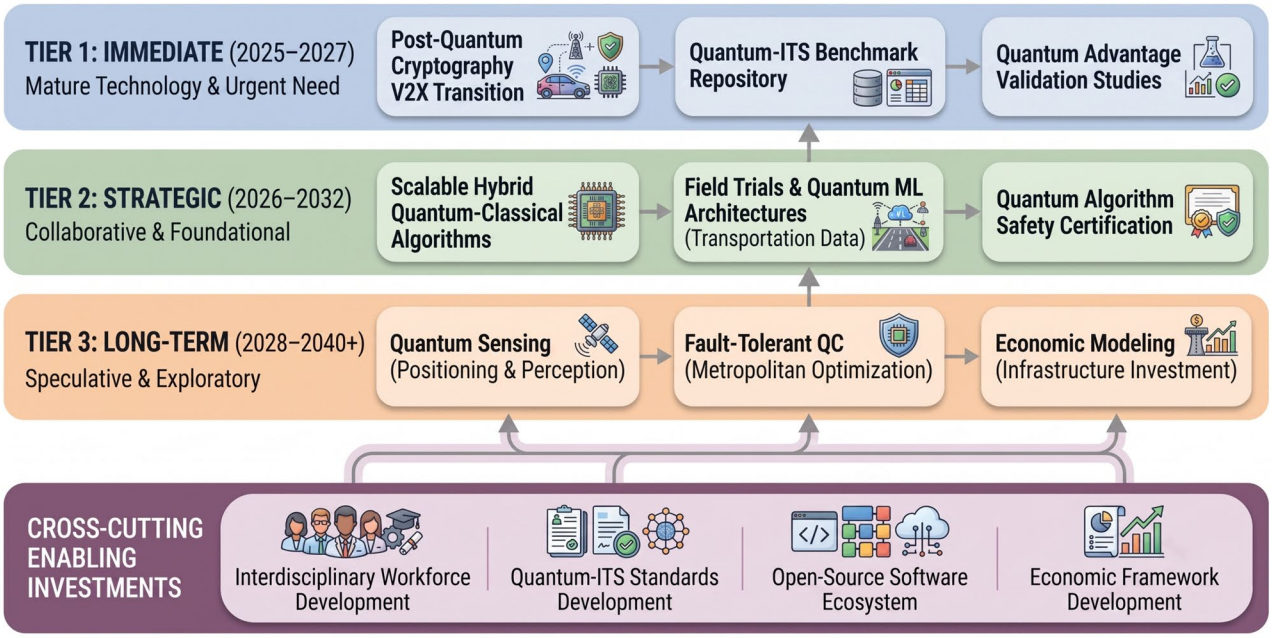


Fig. 26 Three-tier strategic research roadmap organizing quantum-ITS priorities across immediate (2025–2027), strategic (2026–2032), and long-term (2028–2040+) horizons, supported by four cross-cutting enabling investments

delivers measurable operational value to transportation systems within the next decade.

Tier 3 (2028–2040+) encompasses exploratory long-term directions warranting carefully bounded investment as technology options rather than operational commitments. Quantum sensing for positioning and perception, fault-tolerant quantum computing for metropolitan-scale optimization, and economic modeling for quantum infrastructure investment belong in this tier because their value depends on hardware, integration, and cost conditions that remain uncertain.

The four cross-cutting enabling investments interdisciplinary workforce development, quantum-ITS standards development, open-source software ecosystem construction, and economic framework development underpin all three tiers and warrant continuous funding independent of quantum hardware trajectory uncertainties. The value of quantum literacy, interdisciplinary collaboration infrastructure, benchmark discipline, and standards foundations accrues regardless of whether quantum-ITS applications ultimately deliver broad operational impact or remain a specialized research program. The roadmap presented in Tables 9 and 10 is intended to provide researchers selecting problem areas, funding agencies designing programs, practitioners evaluating pilot investments, and policymakers allocating public resources with an evidence-grounded structure for navigating the quantum-ITS landscape with calibrated ambition and appropriate epistemic humility.

7 Conclusion

This review examined the emerging intersection of quantum computing and intelligent transportation systems through an integrated bibliometric and qualitative synthesis. Drawing on 108 Scopus-indexed peer-reviewed publications from 2014 to 2025 and a structured qualitative assessment of 27 representative studies, the review addressed a central gap in the field: the absence of an evidence-grounded evaluation of scientific structure, methodological quality, and translational readiness. The four-dimension evaluation framework, covering baseline fairness, problem scale realism, reproducibility, and deployment practicality, provided a consistent basis for distinguishing credible progress from speculative or weakly benchmarked performance claims.

The bibliometric analysis shows a rapidly expanding but still fragmented field. Research output is concentrated geographically and institutionally, with limited participation from transportation agencies and infrastructure operators. Science mapping identifies four dominant clusters: quantum optimization, quantum machine learning, post-quantum cryptography, and quantum communication. These clusters confirm that quantum-ITS research has moved beyond isolated conceptual proposals, but they also reveal uneven maturity, weak benchmark convergence, and limited integration between algorithm developers and operational transportation stakeholders.

The qualitative synthesis provides the review's central analytical finding: current quantum-ITS literature does not yet demonstrate rigorous quantum computational advantage under fair classical benchmarking, reproducible experimental conditions, and operationally realistic problem scales. Quantum annealing and QAOA studies show that selected transportation problems can be formulated for quantum or hybrid quantum–classical workflows, but most evidence remains constrained by small instances, incomplete baselines, limited end-to-end runtime reporting, and difficulty isolating the quantum contribution. Quantum machine learning remains exploratory, with unresolved barriers in data encoding, measurement overhead, trainability, noise sensitivity, and comparison against strong classical models. Quantum-inspired classical metaheuristics therefore require careful separation from genuine quantum computing, because their inclusion can inflate perceived field maturity without demonstrating quantum-hardware readiness.

Post-quantum cryptography is the clearest near-term priority, but it must be interpreted correctly. PQC is a classical quantum-resistant security pathway, not evidence that quantum computers improve ITS optimization or prediction performance. Its relatively higher readiness, assessed at TRL 6–8, reflects its compatibility with existing embedded and communication infrastructure and the urgency of preparing V2X and in-vehicle networks for quantum-enabled cryptographic threats. By contrast, quantum annealing remains at TRL 3–4, while gate-based QAOA and quantum machine learning remain largely at TRL 2–3. Quantum communication and QKD occupy an intermediate position, around TRL 4–6 for fixed or controlled links, but mobile vehicular deployment remains substantially less mature.

The roadmap developed in this review translates these maturity distinctions into staged research priorities. The immediate horizon, 2025–2027, should prioritize PQC migration planning, benchmark standardization, reproducibility protocols, and agency co-designed test problems. The strategic horizon, 2026–2032, should focus on controlled hybrid quantum–classical pilots, scalability studies, error-aware evaluation, and integration with realistic traffic simulation and operational datasets. The long-term horizon, 2028–2040+, should remain reserved for fault-tolerant quantum optimization, advanced quantum communication infrastructure, quantum sensing, and broader economic evaluation, all contingent on hardware progress and demonstrated cost-effectiveness.

Several limitations frame these findings. The review is based on an English-language Scopus corpus and reflects the literature available at the January 2026 search date. The 27-study qualitative synthesis was designed to capture the intellectual range of the field rather than to provide exhaustive coverage of every publication. In addition, all maturity assessments remain technology-contingent: rapid advances in quantum hardware, quantum error correction, classical optimization, or transportation AI could alter specific readiness judgments.

Overall, quantum computing should be treated neither as an imminent replacement for mature ITS optimization and learning systems nor as an irrelevant speculative technology. Its near-term value lies most clearly in quantum-resistant security planning, while its longer-term transportation value depends on disciplined evidence building. Progress in quantum-ITS research will require stronger classical baselines, public benchmark suites, transparent computational accounting, reproducible implementations, operationally realistic datasets, and sustained collaboration with transportation agencies. With these conditions in place, the field can move from conceptual alignment toward credible assessment of where quantum technologies may eventually provide practical value for intelligent transportation systems.

Acknowledgements The authors would like to thank their respective institutions for academic and research support.

Author Contributions Pretom Sarkar: Data Curation, Visualization, Methodology, Formal Analysis, Writing-Original Draft, Writing-Review & Editing. Mahbub Hassan: Conceptualization, Methodology, Validation, Formal Analysis, Supervision, Writing-Original Draft, Writing-Review & Editing. Ali Hanafiah Rambe: Writing-Review & Editing, Validation, Funding Acquisition. Muzammil Jusoh: Validation, Writing-Review & Editing.

Funding This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data Availability The data presented in this study are available from the corresponding author upon reasonable request.

Declarations

Ethical approval Not Applicable. This study is based entirely on publicly indexed peer-reviewed publications retrieved from Scopus through institutional subscriptions in compliance with their respective terms of use. The study involved no collection of personal data, no human subjects, and no animal experiments. Therefore, formal ethical approval was not required.

Generative AI and AI-Assisted Technologies in the Writing Process During the preparation of this work, the author(s) utilized GPT 5.1, to assist with grammar correction, language refinement, and overall clarity of expression. Following the use of this AI tool, the author(s) thoroughly reviewed and edited the content to ensure accuracy, integrity, and alignment with the intended scholarly rigor. The author(s) take full responsibility for the final content of the published article. AI and AI-assisted technologies were not used to generate, analyze, or interpret research data, nor to produce or alter scientific conclusions.

Competing Interests The authors declare no competing interests.

References

1. Dimitrakopoulos G, Demestichas P (2010) Intelligent transportation systems. IEEE Veh Technol Mag 5(1):77–84. https://doi.org/10.1109/MVT.2009.935537
2. Kaffash S, Nguyen AT, Zhu J (2021) Big data algorithms and applications in intelligent transportation system: a review and bibliometric analysis. Int J Prod Econ 231:107868. https://doi.org/10.1016/j.ijpe.2020.107868
3. Yuan T, da Rocha Neto W, Rothenberg CE, Obraczka K, Barakat C, Turletti T (2022) Machine learning for next-generation intelligent transportation systems: a survey. Trans Emerging Telecommun Technol 33(4):e4427. https://doi.org/10.1002/ett.4427
4. Kenney JB (2011) Dedicated short-range communications (dsrc) standards in the United States. Proc IEEE 99(7):1162–1182. https://doi.org/10.1109/JPROC.2011.2132790
5. Papadimitratos P, Fortelle ADL, Evenssen K, Brignolo R, Cosenza S (2009) Vehicular communication systems: enabling technologies, applications, and future outlook on intelligent transportation. IEEE Commun Mag 47(11):84–95. https://doi.org/10.1109/MCOM.2009.5307471
6. Zhang X, Junxian L, Zhou J, Zhang S, Wang J, Yuan Y, Liu J, Jun L (2025) Vehicle-to-everything communication in intelligent connected vehicles: a survey and taxonomy. Automot Innov 8(1):13–45. https://doi.org/10.1007/s42154-024-00310-2
7. Papageorgiou M, Diakaki C, Dinopoulou V, Kotsialos A, Wang Y (2003) Review of road traffic control strategies. Proc IEEE 91(12):2043–2067. https://doi.org/10.1109/JPROC.2003.819610
8. Varaiya P (2013) Max pressure control of a network of signalized intersections. Transp Res Part C: Emerging Technol 36:177–195. https://doi.org/10.1016/j.trc.2013.08.014
9. Haydari A, Yılmaz Y (2022) Deep reinforcement learning for intelligent transportation systems: a survey. IEEE Trans Intell Transp Syst 23(1):11–32. https://doi.org/10.1109/TITS.2020.3008612
10. Braekers K, Ramaekers K, Van Nieuwenhuyse I (2016) The vehicle routing problem: state of the art classification and review. Comput Ind Eng 99:300–313. https://doi.org/10.1016/j.cie.2015.12.007
11. Costa L, Contardo C, Desaulniers G (2019) Exact branch-price-and-cut algorithms for vehicle routing. Transp Sci 53(4):946–985. https://doi.org/10.1287/trsc.2018.0878
12. Shor PW (1999) Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer. SIAM Rev 41(2):303–332. https://doi.org/10.1137/S0036144598347011
13. Mosca M (2018) Cybersecurity in an era with quantum computers: will we be ready? IEEE Secur Privacy 16(5):38–41. https://doi.org/10.1109/MSP.2018.3761723
14. Yisheng L, Duan Y, Kang W, Zhengxi L, Wang F-Y (2015) Traffic flow prediction with big data: a deep learning approach. IEEE Trans Intell Transp Syst 16(2):865–873. https://doi.org/10.1109/TITS.2014.2345663
15. Manibardo EL, Laña I, Manibardo EL, Del Ser J (2022) Deep learning for road traffic forecasting: does it make a difference? IEEE Trans Intell Transp Syst 23(7):6164–6188. https://doi.org/10.1109/TITS.2021.3083957
16. Grigorescu S, Trasnea B, Cocias T, Macesanu G (2020) A survey of deep learning techniques for autonomous driving. J Field Rob 37(3):362–386. https://doi.org/10.1002/rob.21918
17. Montanaro A (2016) Quantum algorithms: an overview. Npj Quantum Inf 2:15023. https://doi.org/10.1038/npjqi.2015.23
18. Preskill J (2018) Quantum computing in the nisq era and beyond. Quantum 2:79. https://doi.org/10.22331/q-2018-08-06-79
19. Blekos K, Brand D, Ceschini A, Chou C-H, Rui-Hao L, Pandya K, Summer A (2024) A review on quantum approximate optimization algorithm and its variants. Phys Rep 1068:1–66. https://doi.org/10.1016/j.physrep.2024.03.002
20. Zhou L, Wang S-T, Choi S, Pichler H, Lukin MD (2020) Quantum approximate optimization algorithm: performance, mechanism, and implementation on near-term devices. Phys Rev X 10(2):021067. https://doi.org/10.1103/PhysRevX.10.021067
21. Kadowaki T, Nishimori H (1998) Quantum annealing in the transverse ising model. Phys Rev E 58(5):5355. https://doi.org/10.1103/PhysRevE.58.5355
22. Johnson MW, Amin MHS, Gildert S, Lanting T, Hamze F, Dickson N, Harris R, Berkley AJ, Johansson J, Bunyk P, Chapple EM, Enderud C, Hilton JP, Karimi K, Ladizinsky E, Ladizinsky N, Oh T, Perminov I, Rich C, Thom MC, Tolkacheva E, Truncik CJS, Uchaikin S, Wang J, Wilson B, Rose G (2011) Quantum annealing with manufactured spins. Nature 473(7346):194–198. https://doi.org/10.1038/nature10012
23. Biamonte J, Wittek P, Pancotti N, Rebentrost P, Wiebe N, Lloyd S (2017) Quantum machine learning. Nature 549(7671):195–202. https://doi.org/10.1038/nature23474
24. Bernstein DJ, Lange T (2017) Post-quantum cryptography. Nature 549(7671):188–194. https://doi.org/10.1038/nature23461
25. Joseph D, Misoczki R, Manzano M, Tricot J, Pinuaga FD, Lacombe O, Leichenauer S, Hidary J, Venables P, Hansen R (2022) Transitioning organizations to post-quantum cryptography. Nature 605(7909):237–243. https://doi.org/10.1038/s41586-022-04623-2
26. Fernández-Caramés TM (2020) From pre-quantum to post-quantum iot security: a survey on quantum-resistant cryptosystems for the internet of things. IEEE Internet Things J 7(7):6457–6480. https://doi.org/10.1109/JIOT.2019.2958788
27. Arute F, Arya K, Babbush R, Bacon D, Bardin JC, Barends R, Biswas R, Boixo S, Brandao FG, Buell DA et al. (2019) Quantum supremacy using a programmable superconducting processor. Nature 574(7779):505–510. https://doi.org/10.1038/s41586-019-1666-5
28. Wang S, Pei Z, Wang C, Jie W (2021) Shaping the future of the application of quantum computing in intelligent transportation system. Intell Converged Networks 2(4):259–276. https://doi.org/10.23919/ICN.2021.0019
29. Yadav M, Rawat KS (2026) Knowledge mapping of quantum computing for climate change research and sustainable innovation. Qual Quantity 60(2):3511–3544. https://doi.org/10.1007/s11135-025-02413-2
30. Aria M, Cuccurullo C (2017) Bibliometrix: an r-tool for comprehensive science mapping analysis. J Informetrics 11(4):959–975. https://doi.org/10.1016/j.joi.2017.08.007
31. Van Eck N, Waltman L (2010) Software survey: Vosviewer, a computer program for bibliometric mapping. Scientometrics 84(2):523–538. https://doi.org/10.1007/s11192-009-0146-3
32. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, Shamseer L, Tetzlaff JM, Akl EA, Brennan SE et al. (2021) The prisma 2020 statement: an updated guideline for reporting systematic reviews. BMJ 372. https://doi.org/10.1136/bmj.n71
33. Laporte G (2009) Fifty years of vehicle routing. Transp Sci 43(4):408–416. https://doi.org/10.1287/trsc.1090.0301
34. Helsgaun K (2000) An effective implementation of the lin–kernighan traveling salesman heuristic. Eur J Oper Res 126(1):106–130. https://doi.org/10.1016/S0377-2217(99)00284-2
35. Vidal T (2022) Hybrid genetic search for the cvrp: open-source implementation and swap* neighborhood. Comput Operations Res 140:105643. https://doi.org/10.1016/j.cor.2021.105643
36. Schuld M, Killoran N (2019) Quantum machine learning in feature hilbert spaces. Phys Rev Lett 122(4):040504. https://doi.org/10.1103/PhysRevLett.122.040504

37. Bharti K, Cervera-Lierta A, Kyaw TH, Haug T, Alperin-Lea S, Anand A, Degroote M, Heimonen H, Kottmann JS, Menke T et al. (2022) Noisy intermediate-scale quantum algorithms. Rev Mod Phys 94(1):015004. https://doi.org/10.1103/RevModPhys.94.015004
38. Xu M, Liu H, Yang H (2024) Ensemble learning based approach for traffic incident detection and multi-category classification. Eng Appl Artif Intel 132:107933. https://doi.org/10.1016/j.engappai.2024.107933
39. Ladd TD, Jelezko F, Laflamme R, Nakamura Y, Monroe C, O'Brien JL (2010) Quantum computers. Nature 464(7285):45–53. https://doi.org/10.1038/nature08812
40. Horodecki R, Horodecki P, Horodecki M, Horodecki K (2009) Quantum entanglement. Rev Mod Phys 81(2):865–942. https://doi.org/10.1103/RevModPhys.81.865
41. Harrow AW, Hassidim A, Lloyd S (2009) Quantum algorithm for linear systems of equations. Phys Rev Lett 103(15):150502. https://doi.org/10.1103/PhysRevLett.103.150502
42. Lloyd S, Mohseni M, Rebentrost P (2014) Quantum principal component analysis. Nat Phys 10(9):631–633. https://doi.org/10.1038/nphys3029
43. Grover LK (1997) Quantum mechanics helps in searching for a needle in a haystack. Phys Rev Lett 79(2):325. https://doi.org/10.1103/PhysRevLett.79.325
44. Krantz P, Kjaergaard M, Yan F, Orlando TP, Gustavsson S, Oliver WD (2019) A quantum engineer's guide to superconducting qubits. Appl Phys Rev 6(2). https://doi.org/10.1063/1.5089550
45. Kjaergaard M, Schwartz ME, Braumüller J, Krantz P, Wang JI-J, Gustavsson S, Oliver WD (2020) Superconducting qubits: current state of play. Annu Rev Condens Matter Phys 11(1):369–395. https://doi.org/10.1146/annurev-conmatphys-031119-050605
46. Bruzewicz CD, Chiaverini J, McConnell R, Sage JM (2019) Trapped-ion quantum computing: progress and challenges. Appl Phys Rev 6(2). https://doi.org/10.1063/1.5088164
47. King AD, Raymond J, Lanting T, Harris R, Zucca A, Altomare F, Berkley AJ, Boothby K, Ejtemaee S, Enderud C et al. (2023) Quantum critical dynamics in a 5,000-qubit programmable spin glass. Nature 617(7959):61–66. https://doi.org/10.1038/s41586-023-05867-2
48. Boothby T, King AD, Roy A (2016) Fast clique minor generation in chimera qubit connectivity graphs. Quantum Inf Process 15(1):495–508. https://doi.org/10.1007/s11128-015-1150-6
49. Könz MS, Lechner W, Katzgraber HG, Troyer M (2021) Embedding overhead scaling of optimization problems in quantum annealing. PRX Quantum 2(4):040322. https://doi.org/10.1103/PRXQuantum.2.040322
50. Grant E, Humble TS (2022) Benchmarking embedded chain breaking in quantum annealing. Quantum Sci Technol 7(2):025029. https://doi.org/10.1088/2058-9565/ac26d2
51. Henriet L, Beguin L, Signoles A, Lahaye T, Browaeys A, Reymond G-O, Jurczak C (2020) Quantum computing with neutral atoms. Quantum 4:327. https://doi.org/10.22331/q-2020-09-21-327
52. Ebadi S, Wang TT, Levine H, Keesling A, Semeghini G, Omran A, Bluvstein D, Samajdar R, Pichler H, Wen Wei H et al. (2021) Quantum phases of matter on a 256-atom programmable quantum simulator. Nature 595(7866):227–232. https://doi.org/10.1038/s41586-021-03582-4
53. Bravyi S, Kliesch A, Koenig R, Tang E (2020) Obstacles to variational quantum optimization from symmetry protection. Phys Rev Lett 125(26):260505. https://doi.org/10.1103/PhysRevLett.125.260505
54. Tang E (2021) Quantum principal component analysis only achieves an exponential speedup because of its state preparation assumptions. Phys Rev Lett 127(6):060503. https://doi.org/10.1103/PhysRevLett.127.060503
55. Aaronson S (2015) Read the fine print. Nat Phys 11(4):291–293. https://doi.org/10.1038/nphys3272
56. McClean JR, Boixo S, Smelyanskiy VN, Babbush R, Neven H (2018) Barren plateaus in quantum neural network training landscapes. Nat Commun 9(1):4812. https://doi.org/10.1038/s41467-018-07090-4
57. Cerezo M, Sone A, Volkoff T, Cincio L, Coles PJ (2021) Cost function dependent barren plateaus in shallow parametrized quantum circuits. Nat Commun 12(1):1791. https://doi.org/10.1038/s41467-021-21728-w
58. Lubinski T, Johri S, Varosy P, Coleman J, Zhao L, Necaise J, Baldwin CH, Mayer K, Proctor T (2023) Application-oriented performance benchmarks for quantum computing. IEEE Trans. Quantum Eng 4:1–32. https://doi.org/10.1109/TQE.2023.3253761
59. Fowler AG, Mariantoni M, Martinis JM, Cleland AN (2012) Surface codes: towards practical large-scale quantum computation. Phys Rev A—At Mol Optical Phys 86(3):032324. https://doi.org/10.1103/PhysRevA.86.032324
60. Neukart F, Compostella G, Seidel C, Dollen DV, Yarkoni S, Parney B (2017) Traffic flow optimization using a quantum annealer. Front ICT 4:29. https://doi.org/10.3389/fict.2017.00029
61. Pirandola S, Andersen UL, Banchi L, Berta M, Bunandar D, Colbeck R, Englund D, Gehring T, Lupo C, Ottaviani C et al. (2020) Advances in quantum cryptography. Adv Opt Photonics 12(4):1012–1236. https://doi.org/10.1364/AOP.361502
62. Marzi G, Balzano M, Caputo A, Pellegrini MM (2025) Guidelines for bibliometric-systematic literature reviews: 10 steps to combine analysis, synthesis and theory development. Int J Manag Rev 27(1):81–103. https://doi.org/10.1111/ijmr.12381
63. Linnenluecke MK, Marrone M, Singh AK (2020) Conducting systematic literature reviews and bibliometric analyses. Australian J Manag 45(2):175–194. https://doi.org/10.1177/0312896219877678
64. Donthu N, Kumar S, Mukherjee D, Pandey N, Lim WM (2021) How to conduct a bibliometric analysis: an overview and guidelines. J Bus Res 133:285–296. https://doi.org/10.1016/j.jbusres.2021.04.070
65. Mongeon P, Paul-Hus A (2016) The journal coverage of web of science and scopus: a comparative analysis. Scientometrics 106(1):213–228. https://doi.org/10.1007/s11192-015-1765-5
66. Singh VK, Singh P, Karmakar M, Leta J, Mayr P (2021) The journal coverage of web of science, scopus and dimensions: a comparative analysis. Scientometrics 126(6):5113–5142. https://doi.org/10.1007/s11192-021-03948-5
67. Moral-Muñoz JA, Herrera-Viedma E, Santisteban-Espejo A, Cobo MJ (2020) Software tools for conducting bibliometric analysis in science: an up-to-date review. El Profesional de la Información 29(1):e290103. https://doi.org/10.3145/epi.2020.ene.03
68. Hassan M, Kabir ME, Jusoh M, Ki An H, Negnevitsky M, Chengjiang L (2025) Large language models in transportation: a comprehensive bibliometric analysis of emerging trends, challenges, and future research. IEEE Access 13:132547–132598. https://doi.org/10.1109/ACCESS.2025.3589319
69. Hassan M, Kabir ME, Islam MK, Alam E, Rambe AH, Jusoh M, Sameer M (2025) Mapping the machine learning landscape in autonomous vehicles: a scientometric review of research trends, applications, challenges, and future directions. IEEE Access 13:182036–182077. https://doi.org/10.1109/ACCESS.2025.3620637
70. Lamba YS, Sood SK, Rawat KS, Chopra M, Singh AK (2022) Knowledge mapping of 4D printing technologies in computer engineering. Comput Appl Eng Educ 30(6):1959–1978. https://doi.org/10.1002/cae.22561
71. Sharma G, Rawat KS, Sood SK (2025) Adoption of controls and saga of development in 3-D printed bioimplants: a business

perspective. IEEE Eng Manag Rev 53(4):246–270. https://doi.org/10.1109/EMR.2024.3457804
72. Rawat KS, Patel P, Sharma T (2025) Heart diseases prediction system using quantum technology. Netw Model Anal Health Inf Bioinform 14(1):123. https://doi.org/10.1007/s13721-025-00623-0
73. Stollenwerk T, O'Gorman B, Venturelli D, Mandrà S, Rodionova O, Hokkwan N, Sridhar B, Rieffel EG, Biswas R (2020) Quantum annealing applied to de-conflicting optimal trajectories for air traffic management. IEEE Trans Intell Transp Syst 21(1):285–297. https://doi.org/10.1109/TITS.2019.2891235
74. Cohen J (1960) A coefficient of agreement for nominal scales. Educ Psychological Meas 20(1):37–46. https://doi.org/10.1177/001316446002000104
75. Blondel VD, Guillaume J-L, Lambiotte R, Lefebvre E (2008) Fast unfolding of communities in large networks. J Stat Mech 2008(10):P10008. https://doi.org/10.1088/1742-5468/2008/10/P10008
76. Blei DM (2012) Probabilistic topic models. Commun ACM 55(4):77–84. https://doi.org/10.1145/2133806.2133826
77. Raymer MG, Monroe C (2019) The us national quantum initiative. Quantum Sci Technol 4(2):020504. https://doi.org/10.1088/2058-9565/ab0441
78. Acín A, Bloch I, Buhrman H, Calarco T, Eichler C, Eisert J, Esteve D, Gisin N, Glaser SJ, Jelezko F et al. (2018) The quantum technologies roadmap: a European community view. New J Phys 20(8):080201. https://doi.org/10.1088/1367-2630/aad1ea
79. Bova F, Goldfarb A, Melko RG (2021) Commercial applications of quantum computing. EPJ Quantum Technol 8(1):2. https://doi.org/10.1140/epjqt/s40507-021-00091-1
80. Newman MEJ (2005) Power laws, pareto distributions and Zipf's law. Contemp Phys 46(5):323–351. https://doi.org/10.1080/00107510500052444
81. Newman MEJ (2001) The structure of scientific collaboration networks. Proc Natl Acad Sci, India, Sect B Biol Sci 98(2):404–409. https://doi.org/10.1073/pnas.98.2.404
82. Cobo MJ, López-Herrera AG, Herrera-Viedma E, Herrera F (2011) An approach for detecting, quantifying, and visualizing the evolution of a research field: a practical application to the fuzzy sets theory field. J Informetrics 5(1):146–166. https://doi.org/10.1016/j.joi.2010.10.002
83. Ohzeki M (2020) Breaking limitation of quantum annealer in solving optimization problems under constraints. Sci Rep 10(1):3126. https://doi.org/10.1038/s41598-020-60022-5
84. Inoue D, Okada A, Matsumori T, Aihara K, Yoshida H (2021) Traffic signal optimization on a square lattice with quantum annealing. Sci Rep 11(1):3303. https://doi.org/10.1038/s41598-021-82740-0
85. Marchesin A, Montrucchio B, Graziano M, Boella A, Mondo G (2023) Improving urban traffic mobility via a versatile quantum annealing model. IEEE Trans Quantum Eng 4:1–13. https://doi.org/10.1109/TQE.2023.3312284
86. Wang N, Guo G, Wang B, Wang C (2020) Traffic clustering algorithm of urban data brain based on a hybrid-augmented architecture of quantum annealing and brain-inspired cognitive computing. Tsinghua Sci Technol 25(6):813–825. https://doi.org/10.26599/TST.2020.9010007
87. Dixit VV, Niu C (2023) Quantum computing for transport network design problems. Sci Rep 13(1):12267. https://doi.org/10.1038/s41598-023-38787-2
88. Azad U, Behera BK, Ahmed EA, Panigrahi PK, Farouk A (2023) Solving vehicle routing problem using quantum approximate optimization algorithm. IEEE Trans Intell Transp Syst 24(7):7564–7573. https://doi.org/10.1109/TITS.2022.3172241
89. Wang D, Wang N (2015) Quantum computation based bundling optimization for combinatorial auction in freight service procurements. Comput Ind Eng 89:186–193. https://doi.org/10.1016/j.cie.2014.11.014
90. Xia G, Han Z, Zhao B, Wang X (2020) Local path planning for unmanned surface vehicle collision avoidance based on modified quantum particle swarm optimization. Complexity 1:3095426. https://doi.org/10.1155/2020/3095426
91. Haba R, Mano T, Ueda R, Ebe G, Takeda K, Terabe M, Ohzeki M (2025) Routing and scheduling optimization for urban air mobility fleet management using quantum annealing. Sci Rep 15(1):4326. https://doi.org/10.1038/s41598-025-86843-w
92. Harikrishnakumar R, Nannapaneni S (2023) Forecasting bike sharing demand using quantum bayesian network. Expert Syst Appl 221:119749. https://doi.org/10.1016/j.eswa.2023.119749
93. Zhiguo Q, Liu X, Zheng M (2023) Temporal-spatial quantum graph convolutional neural network based on schrödinger approach for traffic congestion prediction. IEEE Trans Intell Transp Syst 24(8):8677–8686. https://doi.org/10.1109/TITS.2022.3203791
94. Bhatia M, Sood SK (2020) Quantum computing-inspired network optimization for iot applications. IEEE Internet Things J 7(6):5590–5598. https://doi.org/10.1109/JIOT.2020.2979887
95. Bhatia M, Sood S, Sood V (2023) A novel quantum-inspired solution for high-performance energy-efficient data acquisition from iot networks. J Ambient Intel Humanized Comput 14(5):5001–5020. https://doi.org/10.1007/s12652-020-02494-x
96. Yang Z-Y, Cao X, Rui-Zhe X, Hong W-C, Sun S-L (2024) Applications of chaotic quantum adaptive satin bower bird optimizer algorithm in berth-tugboat-quay crane allocation optimization. Expert Syst Appl 237:121471. https://doi.org/10.1016/j.eswa.2023.121471
97. Ming-Wei L, Rui-Zhe X, Yang Z-Y, Hong W-C, Xiao-Gang A, Yeh Y-H (2024) Optimization approach of berth-quay crane-truck allocation by the tide, environment and uncertainty factors based on chaos quantum adaptive seagull optimization algorithm. Appl Soft Comput 152:111197. https://doi.org/10.1016/j.asoc.2023.111197
98. Yong L, Chen S, Zhang X, Pan X, Gang Y, Wang C (2025) A quantum-enhanced heuristic algorithm for optimizing aircraft landing problems in low-altitude intelligent transportation systems. Sci Rep 15(1):21606. https://doi.org/10.1038/s41598-025-05261-0
99. Khalid H, Hashim SJ, Hashim F, Al-Jawher WAM, Chaudhary MA, Altarturi HHM (2024) Raven: robust anonymous vehicular end-to-end encryption and efficient mutual authentication for post-quantum intelligent transportation systems. IEEE Trans Intell Transp Syst 25(11):17574–17586. https://doi.org/10.1109/TITS.2024.3416060
100. Dharminder D, Mishra D (2020) Lcppa: lattice-based conditional privacy preserving authentication in vehicular communication. Trans Emerging Telecommun Technol 31(2):e3810. https://doi.org/10.1002/ett.3810
101. Prajapat S, Gautam D, Kumar P, Jangirala S, Das AK, Park Y, Lorenz P (2024) Secure lattice-based aggregate signature scheme for vehicular ad hoc networks. IEEE Trans Veh Technol 73(9):12370–12384. https://doi.org/10.1109/TVT.2024.3383967
102. Wen J, Bai L, Yang Z, Zhang H, Wang H, Debiao H (2024) Larrs: lattice-based revocable ring signature and its application for vanets. IEEE Trans Veh Technol 73(1):739–753. https://doi.org/10.1109/TVT.2023.3305037
103. Zhu Y, Liu Y, Mingzhi W, Jinzhao L, Liu S, Zhao J (2022) Research on secure communication on in-vehicle ethernet based on post-quantum algorithm ntruencrypt. Electronics 11(6):856. https://doi.org/10.3390/electronics11060856
104. Azzaoui AE, Kim TW, Pan Y, Park JH (2021) A quantum approximate optimization algorithm based on blockchain heuristic approach for scalable and secure smart logistics systems.

Hum-Centric Comput Inf Sci 11:46. https://doi.org/10.22967/HCIS.2021.11.046
105. Chaudhary S, Budhiraja I, Chaudhary R, Kumar N, Garg D, Almuhaideb AM (2024) Quantum federated reinforcement-learning-based joint mode selection and resource allocation for STAR-RIS-aided VRCS. IEEE Internet Things J 11(22):36242–36256. https://doi.org/10.1109/JIOT.2024.3464853
106. Akhtar MW, Hassan SA, Ghaffar R, Jung H, Garg S, Hossain MS (2020) The shift to 6g communications: vision and requirements. Hum-Centric Comput Inf Sci 10(1):53. https://doi.org/10.1186/s13673-020-00258-2
107. Regev O (2009) On lattices, learning with errors, random linear codes, and cryptography. J ACM (JACM) 56(6):1–40. https://doi.org/10.1145/1568318.1568324
108. Degen CL, Reinhard F, Cappellaro P (2017) Quantum sensing. Rev Mod Phys 89(3):035002. https://doi.org/10.1103/RevModPhys.89.035002
109. Temme K, Bravyi S, Gambetta JM (2017) Error mitigation for short-depth quantum circuits. Phys Rev Lett 119(18):180509. https://doi.org/10.1103/PhysRevLett.119.180509
110. Mujal P, Martínez-Peña R, Nokkala J, García-Beni J, Giorgi GL, Soriano MC, Zambrini R (2021) Opportunities in quantum reservoir computing and extreme learning machines. Adv Quantum Technol 4(8):2100027. https://doi.org/10.1002/qute.202100027
111. Brecht B, Therriault D, Weimerskirch A, Whyte W, Kumar V, Hehn T, Goudy R (2018) A security credential management system for v2x communications. IEEE Trans Intell Transp Syst 19(12):3850–3871. https://doi.org/10.1109/TITS.2018.2797529
112. Endo S, Cai Z, Benjamin SC, Yuan X (2021) Hybrid quantum-classical algorithms and quantum error mitigation. J Phys Soc Jpn 90(3):032001. https://doi.org/10.7566/JPSJ.90.032001
113. Lewis M, Soudjani S, Zuliani P (2023) Formal verification of quantum programs: theory, tools, and challenges. ACM Trans Quantum Comput 5(1):1–35. https://doi.org/10.1145/3624483
114. Chen S, Jinling H, Shi Y, Peng Y, Fang J, Zhao R, Zhao L (2017) Vehicle-to-everything (V2X) services supported by LTE-based systems and 5G. IEEE Commun Stand Mag 1(2):70–76. https://doi.org/10.1109/MCOMSTD.2017.1700015
115. Vanschoren J, van Rijn JN, Bischl B, Torgo L (2013) OpenML: networked science in machine learning. ACM SIGKDD Explorations Newsl 15(2):49–60. https://doi.org/10.1145/2641190.2641198
116. Kuznietsov A, Gyevnar B, Wang C, Peters S, Albrecht SV (2024) Explainable ai for safe and trustworthy autonomous driving: a systematic review. IEEE Trans Intell Transp Syst 25(12):19342–19364. https://doi.org/10.1109/TITS.2024.3474469

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.